

ANALYTICITY OF ALTERNATING-MAP FUNCTORS AND VECTOR BUNDLES

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ABSTRACT. A C^n functor on normed spaces turns C^n vector bundles into C^n bundles through transition cocycles. We study it for the alternating-map bifunctor Alt^k over a nontrivially normed field K .

The bifunctor is always C^∞ , so it preserves C^n bundles over arbitrary bases for $n \leq \infty$. It is analytic when $k! \neq 0$ in K and, in positive characteristic, in every degree on nonarchimedean spherically complete targets; analytic bundles then give analytic alternating-map bundles. Over bases with open K^d charts, analyticity along transition families suffices, and alternating maps preserve analytic bundles in every characteristic and degree, with arbitrary normed fibers.

The factorial criterion is sharp: in characteristic $p > 0$ and degree $k \geq p$, pullback on alternating maps is nowhere analytic for suitable Banach spaces. A shear construction turns this into analytic bundles over a Banach base whose alternating-map bundle is not analytic. For $K = \mathbb{F}_q((u))$ and $k \geq p$, the finite-coordinate hypothesis is sharp among Banach model spaces with an equivalent nonarchimedean norm.

Finally we seek a largest full subcategory on which Alt^k is analytic. Pairs with universally analytic targets can be adjoined to every such analytic domain, but for $K = \mathbb{F}_p(t)$ and $k \geq p$ no largest domain exists: two individually analytic objects can have a nonanalytic cross-action, even among nonarchimedean spaces of finite algebraic dimension. The main operator results, the analytic-bundle theorems over K^d charts and over a Banach base, and the no-largest-domain theorem are formalized in Lean with Mathlib; the remaining scope is stated separately.

1. INTRODUCTION

The passage $(E, F) \mapsto \text{Alt}^k(E; F)$ from pairs of normed spaces to bounded alternating k -linear maps is contravariant in the source and covariant in the target. We study its regularity on spaces of morphisms, both on whole hom spaces and along analytic families of morphisms; in positive characteristic these are different questions. The motivation is the construction of vector bundles of alternating maps over manifolds over an arbitrary nontrivially normed field K . This is the setting of Mathlib’s manifolds and vector bundles, with regularity indices $n \in \mathbb{N} \cup \{\infty, \omega\}$, where C^ω means analytic; compare Lang’s functors of class C^p [5, Ch. III, §4].

Informally, the construction runs as follows; the precise version is Proposition 5 in §3.1. Let M be a C^n manifold over K , and let $\mathcal{E}, \mathcal{H} \rightarrow M$ be C^n vector bundles with typical fibers E and F . Over a common trivializing cover (U_i) , their transition maps $g_{ij}: U_i \cap U_j \rightarrow \text{Aut}_K(E)$ and $h_{ij}: U_i \cap U_j \rightarrow \text{Aut}_K(F)$ are C^n , with regularity measured in the operator norm; here g_{ij}

The original counterexample, its informal proof and the original write-up were produced with Anthropic’s Claude (in Claude Code). The subsequent categorical and analytic-family results were developed with OpenAI’s Codex. The sharpness results for bundles over infinite-dimensional bases and the present reorganization were developed with Anthropic’s Claude (in Claude Code). Informal proofs were audited by independent AI agents. The original operator results were formalized in Lean with Mathlib using Codex and Autoform; Appendix J distinguishes that checked development from the later extensions. Neither the manuscript nor its formalization has yet received a human expert review.

converts coordinates in chart j to coordinates in chart i . A bundle with fibers $\text{Alt}^k(\mathcal{E}_x; \mathcal{H}_x)$ compatible with these trivializations must then have the transition maps

$$\theta_{ij}(x): m \mapsto h_{ij}(x) \circ m \circ (g_{ji}(x), \dots, g_{ji}(x)), \quad m \in \text{Alt}^k(E; F).$$

The inverse transitions $g_{ji} = g_{ij}^{-1}$ belong to the cocycle data and are already C^n , and functoriality gives the cocycle identity for free. So the only question is whether these maps are C^n in x , that is, whether the alternating-map bifunctor is regular on hom spaces, or at least along the transition families. Taking $\mathcal{E} = TM$ and $\mathcal{H} = M \times K$ gives the bundle of differential k -forms on M ; what is new is the analytic theory over fields of positive characteristic, with possibly infinite-dimensional fibers. In characteristic zero, Gouëzel proves analyticity by polarization [2] and uses it in the construction of alternating-map bundles [3]. Polarization divides by $k!$, which vanishes in K exactly when K has characteristic $p > 0$ and $k \geq p$.

Throughout, *regular* means of class C^n for some $n \in \mathbb{N} \cup \{\infty, \omega\}$. For $n \leq \infty$ there is no obstruction: over every nontrivially normed field and in every degree, with no completeness or dimension assumptions, the action of Alt^k on hom spaces is C^∞ (Proposition 7, whose argument is formalized in the draft Mathlib contribution [8]). So the construction above gives C^n alternating-map bundles over every base for every $n \leq \infty$ (Corollary 8). The dichotomy therefore occurs only at $n = \omega$, and the real question is analyticity.

We treat this question in three steps, which form the program of the paper. The first step shows that analyticity of the bifunctor, on whole hom spaces or only along the transition families, makes the alternating-map bundle analytic (Sections 3 and 4). The second step uses a shear construction to turn a failure of analyticity on hom spaces into analytic input bundles whose alternating-map bundle, with the transition maps above, is not analytic (Section 5). The third step seeks a largest analytic domain, that is, a largest class of pairs of spaces on which the bifunctor is analytic, and studies the universal targets, the targets F for which $\text{Alt}^k(-; F)$ is analytic on all hom spaces (Section 6). The Main Theorem below records the outcome of each step.

Categories and regularity. Fix a nontrivially normed field K . Let Vec_K be the category of normed K -spaces and bounded linear maps, and let Ban_K be its full subcategory of complete spaces. When K has positive characteristic, write Vec_K° for the full subcategory of nonarchimedean, spherically complete spaces. The field itself need not be complete.

For an integer $k \geq 0$, let $\text{Alt}^k(E; F)$ be the normed space of bounded alternating k -linear maps $E^k \rightarrow F$. Here *alternating* means vanishing whenever two arguments coincide, including in characteristic 2. Pullback in E and pushforward in F define the bifunctor

$$\text{Alt}^k: \text{Vec}_K^{\text{op}} \times \text{Vec}_K \longrightarrow \text{Vec}_K, \quad (E, F) \longmapsto \text{Alt}^k(E; F).$$

Its action on a pair of morphisms $u: E \rightarrow E'$ and $v: F \rightarrow F'$ is

$$\text{Alt}^k(u, v): m \longmapsto v \circ m \circ (u, \dots, u).$$

The opposite category in the first variable records pullback. Write $\mathcal{L}(E, E')$ for the hom space of bounded linear maps, equipped with the operator norm. We call the bifunctor *analytic*, or C^n for a given $n \in \mathbb{N} \cup \{\infty\}$, when every induced map on hom spaces

$$\mathcal{L}(E, E') \times \mathcal{L}(F, F') \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F'))$$

has that regularity. The product carries the maximum norm. This definition concerns the joint action on both variables, rather than separate regularity. It applies to the restrictions below as well.

The operator form. The proofs work with the action of a fixed-target functor $\text{Alt}^k(-; F)$ on hom spaces. For normed K -spaces E, E', F and a fixed degree $k \geq 0$, the pullback action is

$$(1.1) \quad \begin{aligned} Q &= Q_{E, E', F}^K: \mathcal{L}(E, E') \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)), \\ Q(f)(m) &= m \circ (f, \dots, f). \end{aligned}$$

For multilinear maps without the alternation condition, precomposition is the diagonal of a bounded multilinear map and is therefore analytic. Requiring the values to remain alternating changes the problem: polarization is unavailable when $k! = 0$ in K , and over Banach spaces this obstruction cannot be avoided (Theorem 28). For $k \geq 1$, the lift criterion (Proposition 3) shows that Q is analytic at one point if and only if it is analytic at every point, and that this holds if and only if Q has a bounded k -linear lift (Definition 1).

The translation between operators and functors uses only bounded bilinear postcomposition. Write $v_*a = v \circ a$. The map $(v, T) \mapsto v_* \circ T$ is bounded bilinear, with $\|v_* \circ T\| \leq \|v\| \|T\|$, and for $u \in \mathcal{L}(E, E')$ and $v \in \mathcal{L}(F, F')$ the joint action of the bifunctor is

$$(1.2) \quad \text{Alt}^k(u, v) = v_* \circ Q_{E, E', F}^K(u).$$

Consequently C^m regularity or analyticity of every Q gives the same regularity of the bifunctor. Conversely, restriction to $v = \text{id}_F$ recovers Q . A bounded k -linear lift P of Q gives the bounded $(k+1)$ -linear map $(u_1, \dots, u_k, v) \mapsto v_* \circ P(u_1, \dots, u_k)$, with the pushforward variable in the last slot and value $\text{Alt}^k(u, v)$ at $(u, \dots, u, v)$, so finite-power-series conclusions pass from Q to the bifunctor.

The space $\text{Alt}^k(E; F)$ is complete whenever F is complete: an operator-norm Cauchy sequence converges pointwise in F , and its boundedness, multilinearity and alternation pass to the limit. Hence the bifunctor restricts to $\text{Alt}^k: \text{Ban}_K^{\text{op}} \times \text{Ban}_K \rightarrow \text{Ban}_K$. In degree zero, $\text{Alt}^0(E; F)$ is naturally isometric to F ; pullback is constant and the joint action is linear in the pushforward variable.

The two notions used above to define Vec_K° also enter the Main Theorem below. A norm is *nonarchimedean* if $\|x + y\| \leq \max(\|x\|, \|y\|)$, and likewise for the absolute value of K . A metric space is *spherically complete* if every nonempty family of closed balls that meet pairwise has a common point; §3.5 gives the details. Every normed field of characteristic $p > 0$ is nonarchimedean (Lemma 2). Over a nonarchimedean K , if F is nonarchimedean and spherically complete then so is $\text{Alt}^k(E; F)$ for every normed E (Proposition 14); this closure property is the codomain assertion of Main Theorem (2) below.

Main results.

Main Theorem. Let K be a nontrivially normed field and let $k \geq 0$ be an integer.

- (1) The bifunctor

$$\text{Alt}^k: \text{Vec}_K^{\text{op}} \times \text{Vec}_K \longrightarrow \text{Vec}_K$$

is C^∞ . It is analytic if and only if $k! \neq 0$ in K . Both assertions hold also for its restriction

$$\text{Alt}^k: \text{Ban}_K^{\text{op}} \times \text{Ban}_K \longrightarrow \text{Ban}_K.$$

- (2) If K has positive characteristic, the restriction

$$\text{Alt}^k: \text{Vec}_K^{\text{op}} \times \text{Vec}_K^\circ \longrightarrow \text{Vec}_K^\circ$$

is analytic in every degree.

- (3) With no restriction on K or k , the bifunctor Alt^k preserves every analytic family of morphisms parametrized by an open subset of K^d . Consequently it induces a bifunctor on analytic normed vector bundles over any analytic manifold with open K^d charts, preserving operator-valued analytic bundle morphisms. No completeness or dimension hypothesis on the fibers is required.
- (4) If $k! = 0$ in K , the bundle conclusion of (3) does not extend to all Banach bases. There are an analytic manifold M modeled on a Banach space, with a single chart, and analytic normed vector bundles over M whose alternating-map bundle, formed with the induced transition maps as in (3), has a transition map that is analytic at no point of M . The two trivializations involved differ by an affine analytic family of invertible operators with affine analytic inverse.
- (5) For every prime p and every $k \geq p$, over the t -adically normed field $K = \mathbb{F}_p(t)$ there is no largest full subcategory of $\text{Vec}_K^{\text{op}} \times \text{Vec}_K$ on which Alt^k is analytic. This remains true for subcategories closed under isomorphisms and, within the corresponding ambient category, when all objects have finite algebraic dimension and nonarchimedean norms.

Items (1)–(3) contain the positive results of the first step. The analytic assertion of item (1) gives the exact range $k < p$ in characteristic $p > 0$, and every degree in characteristic zero; the positive direction is Proposition 9. Item (2) removes the restriction on the degree by restricting the target category, and no completeness of K is needed. In the analytic cases of items (1) and (2), the maps on hom spaces are given by finite power series. Over a complete field, $Q_{E,E',F}^K$ is also analytic in every degree when E or E' is finite-dimensional (Corollary 12), and a finite-dimensional target reduces the question to scalar-valued forms (Remark 46). For bundles, analytic input bundles give an analytic alternating-map bundle over every analytic base when $k! \neq 0$ in K , when K is nonarchimedean and the target fibers are nonarchimedean and spherically complete, or when the source fibers have finite continuous coordinates (Corollary 15). Over bases with open K^d charts, item (3) removes every condition on the characteristic, the degree and the fibers (Theorem 22). Its proof applies analytic reflection through a closed linear isometry to the transition families themselves, using that their parameters are finite coordinates (Theorem 19). Bases modeled on $\ell^1(I, K)$ also admit the construction when the target fibers are complete, and over a nonarchimedean field so do bases modeled on $c_0(I, K)$ when the target fibers are complete and nonarchimedean (Corollaries 26 and 24).

Item (4) is the outcome of the second step. Like the negative direction of item (1), it rests on the counterexample theorem (Theorem 28): for every K of characteristic $p > 0$ and every $k \geq p$, there are Banach spaces E, F such that $Q_{E,E',F}^K$ is analytic at no point, where F admits no equivalent nonarchimedean norm. Corollary 29 combines Theorem 28 with the factorial case, and the shear construction (Proposition 30) turns it into Corollary 31, which is item (4). This is a failure of the natural construction under analytic changes of trivialization, not a claim that the pointwise bundle admits no analytic structure, and the C^n analogues hold over every base for $n \in \mathbb{N} \cup \{\infty\}$ (Corollary 8). The base need not be an operator space: over $K = \mathbb{F}_q((u))$, with q a power of p and $k \geq p$, the open unit ball of $c_0(\mathbb{N}, K)$ with diagonal isometric transitions already fails (Corollary 34). Among Banach model spaces over this field admitting an equivalent nonarchimedean norm, finite dimension is therefore exactly the right hypothesis on the base (Corollary 35). Over the incomplete field $\mathbb{F}_p(t)$, with $k \geq p$, a base of finite algebraic dimension without continuous coordinates also fails (Remark 45).

Item (5) is the outcome of the third step. Every analytic domain (Definition 37) yields analytic alternating-map bundles over every analytic base, for input bundles whose pair of typical fibers is an object of the domain (Corollary 38). The pairs whose target F is universal,

in the sense that $\text{Alt}^k(-; F)$ is analytic on all hom spaces, form the universal-target core, and adjoining its objects to any analytic domain gives an analytic domain (Proposition 39). Nevertheless no largest analytic domain exists over $\mathbb{F}_p(t)$ for $k \geq p$ (Theorem 43). Over incomplete fields, finite algebraic dimension alone does not supply continuous coordinates; this is precisely what permits these examples. Item (5) does not decide the corresponding question over complete fields or among Banach pairs.

Related results. The characteristic-zero polarization argument of [2] works whenever $k! \neq 0$ in K ; we prove it in §3. Theorem 28 shows that the regularity range $n \leq \infty$ in the draft Mathlib contribution [8] is sharp over every field of characteristic $p > 0$ once $k \geq p$: in these examples, Mathlib’s `ContDiffAt K ω` fails at every point.

The answer is different if the target norm is required to be nonarchimedean. If K is nonarchimedean and F is nonarchimedean and spherically complete, then Q is analytic for all normed E, E' (Main Theorem). The proof, outlined in §3.5 and given in Appendix B, constructs a contracting retraction from multilinear maps to alternating maps by spherical completeness and extension of bounded linear maps. This positive result is independent of the counterexample theorem and Corollary 29. A special case appears in [7, Remark 16]. Over a discretely valued base, the Main Theorem in fact rules out a Banach counterexample whenever F has an equivalent nonarchimedean norm (Proposition 47).

The earlier characteristic-2 construction [7] uses a nonarchimedean target and gives non-analyticity at 0 for $E = E' = \ell^\infty$ and $k = 2$ over every complete nonarchimedean field of characteristic 2 that is not spherically complete. Its formal hypothesis, a descending sequence of closed balls with empty intersection, is equivalent to failure of spherical completeness (Remark 77). The higher-characteristic cases were left open there. The present construction works over every nontrivially normed field of positive characteristic, including spherically complete fields, by allowing targets with no equivalent nonarchimedean norm. The remaining questions for nonarchimedean targets are discussed in §7.

Organization and verification scope. After the motivation above, the paper carries out the three steps of the program; Section 2 first fixes conventions and states the lift criterion. The first step, from analytic functors to analytic bundles, occupies Section 3, on whole hom spaces, and Section 4, along analytic families. The second, from a nonanalytic functor to a nonanalytic bundle through the shear, is Section 5. The third, the search for the largest analytic domain, is Section 6. Section 7 collects open problems. The long technical proofs are deferred to appendices, in the order of the main text, so that it can be read without them: Appendices A–C for the lift criterion and the first step, Appendices D–G for Theorem 28, and Appendix H for Theorem 43. Appendix I collects further complements on nonarchimedean targets.

The counterexample theorem and the factorial classification (Theorem 28 and Corollary 29), including their supporting finite-field and functional-analytic arguments, are formalized in Lean 4 with Mathlib. The checked development also contains the operator form of Main Theorem (2), the discrete-target result and the sorted-basis lift. The analytic-family, bundle and no-largest-domain results are later extensions. Their mathematical proofs are included here; Appendix J and the repository’s theorem map give their separate formalization status.

2. ANALYTIC PRELIMINARIES

The lift criterion reduces the analytic question to a bounded multilinear identity. Its proof uses uniqueness on scalar lines and applies in every characteristic, without completeness assumptions.

We first fix the conventions used throughout and then state the lift criterion, Proposition 3. The one-variable uniqueness of power-series coefficients on which it rests is Lemma 48, and both proofs are given in Appendix A.

2.1. Conventions. We write $\mathbb{N} = \{0, 1, 2, \dots\}$. Throughout, K denotes the base field and k the degree. A *normed field* K carries a multiplicative absolute value $|\cdot|$ satisfying the triangle inequality. It is *nontrivially normed* if some $c \in K$ has $0 < |c| < 1$, and *nonarchimedean* if $|x + y| \leq \max(|x|, |y|)$. A *normed K -space* X satisfies $\|\lambda x\| = |\lambda| \|x\|$; it is *Banach* if it is complete.

We write $\mathcal{L}(X, Y)$ for the bounded linear maps, $\text{Mult}^k(X; Y)$ for the bounded k -linear maps $X^k \rightarrow Y$, and $\text{Alt}^k(X; Y) \subseteq \text{Mult}^k(X; Y)$ for the closed subspace of alternating maps, all with the operator norm. A subscript K records the field when needed. An alternating map m is antisymmetric: expanding $0 = m(\dots, y + z, \dots, y + z, \dots)$ gives $m(\dots, y, \dots, z, \dots) = -m(\dots, z, \dots, y, \dots)$, and hence $m(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \text{sgn}(\sigma) m(x_1, \dots, x_k)$ for $\sigma \in S_k$. In characteristic 2 the converse fails.

We index the k arguments by $\{1, \dots, k\}$ throughout. Reindexing by any finite set with k elements is an isometric isomorphism of alternating-map spaces and intertwines precomposition. Thus the results also apply to the arbitrary finite index types used in Lean, with the same counterexample spaces. For each fixed degree we write

$$Q = Q_{E, E', F}^K: \mathcal{L}(E, E') \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)),$$

where $Q(f)(m)(x_1, \dots, x_k) = m(fx_1, \dots, fx_k)$.

Definition 1. A *bounded k -linear lift* of Q is a bounded k -linear map $P: \mathcal{L}(E, E')^k \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$ with $P(f, \dots, f) = Q(f)$ for all $f \in \mathcal{L}(E, E')$.

A map $g: X \rightarrow Z$ between normed spaces is *analytic at x_0* if there are bounded n -linear maps $p_n: X^n \rightarrow Z$ ($n \geq 0$) and a radius $R > 0$ with $\sup_n \|p_n\| R^n < \infty$ such that $g(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$, a convergent series, whenever $\|h\| < R$. Up to shrinking R this is Mathlib's `AnalyticAt` [6], and in Mathlib `ContDiffAt K ω` at a point implies `AnalyticAt` there (`ContDiffAt.analyticAt`).

Lemma 2. A normed field of characteristic $p > 0$ is nonarchimedean.

Proof. Frobenius and the triangle inequality give, for every $m \geq 0$,

$$|x + y|^{p^m} = |x^{p^m} + y^{p^m}| \leq 2 \max(|x|, |y|)^{p^m}.$$

Taking p^m -th roots and letting $m \rightarrow \infty$ proves the assertion. $\square$

2.2. The lift criterion. A power series for Q at any base point determines a lift through its degree- k term. The only uniqueness statement needed is one-dimensional, so the argument does not use a polarization formula or division by $k!$.

Proposition 3 (lift criterion). *Let K be any nontrivially normed field, E, E', F normed K -spaces and $k \geq 1$.*

- (1) *If Q has a bounded k -linear lift P , then at every $f_0 \in \mathcal{L}(E, E')$ the map Q is given by a finite power series of infinite radius; in particular Q is analytic at every point.*
- (2) *If Q is analytic at some point f_0 , with power series $(p_n)_n$ there, then p_k is a bounded k -linear lift of Q .*
- (3) *Hence the following are equivalent: Q has a bounded k -linear lift; Q is analytic at every point; Q is analytic at some point. If they fail, then Mathlib's `ContDiffAt K ω Q f_0` fails at every f_0 .*

The proof is given in Appendix A.

Remark 4. The degree- k term of *every* power series representing Q is a lift, regardless of the base point. This does not assert uniqueness of the multilinear lift: in positive characteristic, different multilinear maps can have the same diagonal.

3. FROM REGULAR FUNCTORS TO REGULAR BUNDLES

This section works on whole hom spaces. Regularity of a functor on hom spaces passes to the vector bundles it induces (Proposition 5). The alternating functor is C^∞ in every characteristic (Proposition 7), because smoothness descends from all multilinear maps through a closed isometric inclusion. Three constructions give bounded multilinear lifts of $Q_{E,E',F}^K$, and hence analyticity by Proposition 3: division by $k!$ when it is nonzero in K (Proposition 9); determinant formulas when E or E' has a finite basis with continuous coordinate functionals (Proposition 10); and, over a nonarchimedean field, a contractive retraction onto alternating maps when F is nonarchimedean and spherically complete (§3.5 and Main Theorem (2)). Corollary 15 collects the consequences for alternating-map bundles.

3.1. Bundles from regular functors. Fix $n \in \mathbb{N} \cup \{\infty, \omega\}$, where C^ω means analytic, and let M be a C^n manifold over K , modeled on any normed K -space. A C^n vector bundle $\mathcal{V} \rightarrow M$ with typical fiber V is given by an open cover (U_i) of M and C^n transition maps $g_{ij}: U_i \cap U_j \rightarrow \text{Aut}_K(V) \subseteq \text{End}_K(V) = \mathcal{L}(V, V)$. Here $\text{Aut}_K(V)$ consists of bounded linear isomorphisms with bounded inverse, and regularity is measured in the operator norm. The map g_{ij} converts coordinates in chart j to coordinates in chart i , and $g_{ii}(x) = \text{id}_V$ and $g_{ij}(x) \circ g_{j\ell}(x) = g_{i\ell}(x)$ on the relevant overlaps. Conversely, such a cocycle glues the products $U_i \times V$ into a C^n vector bundle: the coordinate changes $(x, v) \mapsto (x, g_{ij}(x)v)$ are C^n because operator evaluation is bounded bilinear. In particular $g_{ji} = g_{ij}^{-1}$, and both directions are C^n by the cocycle data.

A multifunctor of mixed variance in r variables is a functor $\mathcal{F}: \mathbf{C}_1 \times \cdots \times \mathbf{C}_r \rightarrow \text{Vec}_K$, where each $\mathbf{C}_a$ is Vec_K or Vec_K^{op} . As for Alt^k , it is C^n when its induced maps on products of hom spaces, with the maximum norm, are jointly C^n . For transition maps $\mathbf{g}_{ij} = (g_{ij}^1, \dots, g_{ij}^r)$ we write $\mathcal{F}(\mathbf{g}_{ij}(x))$ for $\mathcal{F}$ applied to $g_{ij}^a(x)$ in each covariant variable and to $g_{ji}^a(x)$ in each contravariant variable.

Proposition 5 (bundles from regular functors). *Let $n \in \mathbb{N} \cup \{\infty, \omega\}$, let M be a C^n manifold over K , and let $\mathcal{F}$ be a multifunctor of mixed variance in r variables. Let $\mathcal{V}_1, \dots, \mathcal{V}_r$ be C^n vector bundles over M with a common trivializing cover, typical fibers V_a and transition maps g_{ij}^a .*

- (1) *Suppose that $\mathcal{F}$ is C^n . Then the maps $\theta_{ij}(x) = \mathcal{F}(\mathbf{g}_{ij}(x))$ form a C^n cocycle with values in $\text{End}_K(\mathcal{F}(V_1, \dots, V_r))$. It defines a C^n vector bundle $\mathcal{F}(\mathcal{V}_1, \dots, \mathcal{V}_r)$ with fiber $\mathcal{F}((\mathcal{V}_1)_x, \dots, (\mathcal{V}_r)_x)$ over x . Up to canonical isomorphism this bundle is independent of the cover and the trivializations, and it is functorial, with the variance of $\mathcal{F}$, in C^n bundle morphisms.*
- (2) *More generally, the same conclusions hold without assuming that $\mathcal{F}$ is C^n , provided the maps $x \mapsto \mathcal{F}(\mathbf{g}_{ij}(x))$ are C^n , and so are the corresponding composites of $\mathcal{F}$ with the changes of trivialization and with the local expressions of bundle morphisms.*

Proof. Functoriality gives $\theta_{ii}(x) = \text{id}$ and $\theta_{ij}(x) \circ \theta_{j\ell}(x) = \mathcal{F}(\mathbf{g}_{ij}(x) \circ \mathbf{g}_{j\ell}(x)) = \theta_{i\ell}(x)$, the composite in a contravariant variable being $g_{\ell j}^a(x) \circ g_{ji}^a(x) = g_{\ell i}^a(x)$. In case (1), θ_{ij} is C^n as the composite of $x \mapsto \mathbf{g}_{ij}(x)$ with the C^n map induced by $\mathcal{F}$ on hom spaces; in case (2), this

is the hypothesis. The inverse transitions used in contravariant variables are already C^n by the cocycle data, so no separate regularity theorem for inversion is needed. Applying $\mathcal{F}$ in the same way to the C^n operator-valued local expressions of changes of trivialization and of bundle morphisms gives C^n local expressions, by composition in case (1) and by hypothesis in case (2). They are compatible with the cocycles θ_{ij} by functoriality. This identifies the bundles obtained from different choices and gives the functoriality in bundle morphisms. $\square$

In particular, a multifunctor may be applied repeatedly to the same bundle.

For $\mathcal{F} = \text{Alt}^k$, let $\mathcal{E}, \mathcal{H} \rightarrow M$ have typical fibers E, F and transition maps g_{ij}, h_{ij} , so that $\mathbf{g}_{ij} = (g_{ij}, h_{ij})$ with $\mathcal{E}$ in the contravariant variable. The cocycle of Proposition 5 for $\text{Alt}^k(\mathcal{E}; \mathcal{H})$ is

$$\theta_{ij}(x): m \mapsto h_{ij}(x) \circ m \circ (g_{ji}(x), \dots, g_{ji}(x)), \quad m \in \text{Alt}^k(E; F).$$

By (1.2), $\theta_{ij}(x)$ is obtained from $Q_{E,E,F}^K(g_{ji}(x))$ by postcomposition with $h_{ij}(x)$, which is bounded bilinear in the pair. These transitions are therefore C^n as soon as $Q_{E,E,F}^K$ is C^n , and bundle morphisms between bundles with different typical fibers require the same of $Q_{E,E',F}^K$. Part (2) of the proposition is the version used in Corollaries 15 and 38, which need analyticity only on the hom spaces between fibers satisfying their hypotheses, and in Section 4, where analyticity is established only along transition families. The rest of this section supplies the regularity of Q on whole hom spaces.

3.2. Smoothness in every characteristic. We first record why finite-order differentiability descends through a closed isometric inclusion. The argument also explains why smoothness survives when a multilinear analytic lift does not.

Lemma 6 (smoothness through a closed isometric inclusion). *Let U, V, W be normed spaces over a nontrivially normed field K , and let $j: V \rightarrow W$ be a linear isometry with closed range. For a map $g: U \rightarrow V$ and $n \in \mathbb{N} \cup \{\infty\}$, the map g is C^n if and only if $j \circ g$ is C^n .*

Proof. One direction follows by composition with a continuous linear map. For the converse, continuity follows from the isometry. Suppose $j \circ g$ has Fréchet derivative $A: U \rightarrow W$ at x . For each $h \in U$,

$$Ah = \lim_{t \rightarrow 0, t \neq 0} \frac{j(g(x + th)) - j(g(x))}{t}.$$

The quotients lie in $j(V)$, and nontriviality of the norm supplies nonzero scalars tending to zero. Since $j(V)$ is closed, $Ah \in j(V)$. Thus $A = j \circ B$ for a unique linear map $B: U \rightarrow V$. The identity $\|Bh\| = \|Ah\|$ makes B bounded, and the isometry transfers the Fréchet remainder estimate from A to B . Consequently g is differentiable and

$$D(j \circ g) = j_* \circ Dg, \quad j_*: \mathcal{L}(U, V) \longrightarrow \mathcal{L}(U, W), \quad T \longmapsto j \circ T.$$

The map j_* is again a linear isometry with closed range. Indeed, an operator belongs to its range exactly when all its values lie in $j(V)$; this is a closed condition, being an intersection of inverse images under continuous evaluation maps. Applying the same argument to j_* proves the assertion for C^{n+1} from the assertion for C^n . Induction proves every finite order, and hence C^∞ . $\square$

Proposition 7 (smoothness of precomposition). *For every nontrivially normed field K , all normed K -spaces E, E', F , and every integer $k \geq 0$, the map $Q_{E,E',F}^K$ is C^n for every $n \in \mathbb{N} \cup \{\infty\}$. In particular, precomposition is always C^∞ , without completeness, finite-dimensionality, or characteristic assumptions.*

Proof. The inclusion $\text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$ is isometric and has closed range: alternation is the intersection of the closed conditions $m(x_1, \dots, x_k) = 0$ for tuples with two equal entries. Postcomposition therefore gives a linear isometry with closed range

$$\mathcal{I}: \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)).$$

In the larger target, precomposition is the diagonal of the bounded k -linear map

$$A(f_1, \dots, f_k)(m)(x_1, \dots, x_k) = m(f_1 x_1, \dots, f_k x_k),$$

whose norm is at most 1. Hence $\mathcal{I} \circ Q$ is a continuous polynomial and is C^∞ . Lemma 6 gives the result. The degree-zero case is again constant. $\square$

This is the argument formalized in [8]. It uses closedness to recover derivatives, rather than a projection onto alternating maps. The induction proves all finite orders but does not recover analytic coefficients in positive characteristic. Theorem 28 shows that the resulting C^∞ conclusion cannot in general be strengthened to C^ω .

Combined with the cocycle construction, Proposition 7 settles every finite order of differentiability and the case $n = \infty$.

Corollary 8 (smooth alternating-map bundles). *Let K be a nontrivially normed field, let $k \geq 0$ be an integer, let $n \in \mathbb{N} \cup \{\infty\}$, and let M be a C^n manifold over K modeled on any normed space. Fiberwise alternating maps define a bifunctor*

$$(\mathcal{E}, \mathcal{H}) \longmapsto \text{Alt}^k(\mathcal{E}; \mathcal{H}), \quad \text{Alt}^k(\mathcal{E}; \mathcal{H})_x = \text{Alt}^k(\mathcal{E}_x; \mathcal{H}_x),$$

contravariant in the first variable, on C^n normed vector bundles over M with arbitrary normed typical fibers and C^n bundle morphisms. No completeness, dimension or characteristic hypothesis is needed.

Proof. By Proposition 7, every $Q_{E, E', F}^K$ is C^n . Since $(v, T) \mapsto v_* \circ T$ is bounded bilinear, (1.2) shows that the joint action of Alt^k on hom spaces is C^n . Proposition 5(1) now applies. $\square$

The remaining case is $n = \omega$, which occupies the rest of the paper.

3.3. Invertible factorial: in particular, $p > k$.

Proposition 9. *Let K be a nontrivially normed field and let E, E', F be normed K -spaces. If $k! \neq 0$ in K , then Q is given at every point by a finite power series of infinite radius. In particular, it is analytic whenever $\text{char } K = 0$, or whenever $\text{char } K = p > k$. No completeness or dimension hypothesis is needed.*

Proof. For $k = 0$, precomposition is constant. Suppose $k \geq 1$ and define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \frac{1}{k!} \sum_{\sigma \in S_k} m(f_{\sigma(1)} x_1, \dots, f_{\sigma(k)} x_k).$$

This is multilinear in the f_i and in the x_j , linear in m , and satisfies

$$\|P(f_1, \dots, f_k)(m)(x_1, \dots, x_k)\| \leq |(k!)^{-1}| k! \|m\| \prod_i \|f_i\| \prod_j \|x_j\|,$$

where the second $k!$ on the right is a real number counting the terms. To check alternation, suppose $x_a = x_b$ and let τ exchange a and b . The summands indexed by σ and $\sigma \circ \tau$ cancel by antisymmetry of m . This pairing is a fixed-point-free involution, so it proves vanishing also in characteristic 2. Thus P takes values in the required space of alternating maps and is a bounded k -linear lift: on the diagonal, all $k!$ summands equal $Q(f)(m)(x)$. Proposition 3 gives the claimed power series.

In a field of characteristic $p > 0$, the scalar $k!$ is nonzero precisely when $k < p$. This gives the stated positive-characteristic case. $\square$

3.4. Finite-coordinate operator domains and codomains. The finite-dimensional argument uses continuous coordinate functionals. We state this hypothesis explicitly so that the result also applies over incomplete fields whenever the chosen finite-dimensional space has its usual coordinate topology.

Proposition 10 (finite coordinate bases). *Let K be a nontrivially normed field and let E, E', F be normed K -spaces. If either E or E' has a finite algebraic basis whose coordinate functionals are continuous, then $Q_{E, E', F}^K$ has a bounded k -linear lift in every degree $k \geq 1$. In particular, it is continuously polynomial, and hence analytic, at every point. No completeness or nonarchimedean hypothesis on F is required.*

The proof is given in Appendix A.

Lemma 11. *Let $\mathbb{K}$ be a complete nontrivially normed field and U a finite-dimensional normed $\mathbb{K}$ -space. Then every linear functional on U is bounded, and U is complete.*

Proof. We prove both assertions by induction on $n = \dim U$, the case $n = 0$ being immediate. Suppose $n \geq 1$ and let $g \neq 0$ be a linear functional on U . Its kernel H has dimension $n - 1$, so the induction hypothesis makes H complete and therefore closed in U . Choose v with $g(v) = 1$. Since $v \notin H$, $\delta := \inf_{h \in H} \|v - h\| > 0$. Write $x = g(x)v + h$ with $h \in H$. If $g(x) \neq 0$, then

$$\|x\| = |g(x)| \|v + g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus $|g(x)| \leq \delta^{-1}\|x\|$ for every x , proving boundedness.

Choose a basis of U . Its coordinate functionals are now bounded, so the coordinate map $U \rightarrow \mathbb{K}^n$, with the max norm on $\mathbb{K}^n$, is bounded. The inverse map $(c_i) \mapsto \sum_i c_i e_i$ is bounded by $\sum_i \|e_i\|$. Hence U is bi-Lipschitz equivalent to the complete space $\mathbb{K}^n$, and is itself complete. $\square$

Corollary 12. *Let K be a complete nontrivially normed field, and let E, E', F be normed K -spaces. If either E or E' is finite-dimensional over K , then $Q_{E, E', F}^K$ is analytic at every point in every degree. No completeness assumption on E, E' or F is needed in the statement.*

Proof. Lemma 11 gives continuous coordinates on whichever of E, E' is finite-dimensional. Proposition 10 then applies. $\square$

Remark 13 (the coordinate hypothesis over an incomplete field). Finite algebraic dimension alone does not imply continuity of coordinate functionals over an incomplete field. For example, choose $a \in \widehat{K} \setminus K$ and give $H := K + Ka \subseteq \widehat{K}$ the induced norm. The vectors $1, a$ form a K -basis. If $a_j \in K$ tends to a , then $a - a_j \rightarrow 0$ in H , whereas its a -coordinate is always 1. Thus that coordinate functional is discontinuous. This prevents dropping the coordinate hypothesis from the proof of Proposition 10; it does not by itself give a counterexample to analyticity of Q for such spaces. Over an incomplete field the proposition still applies, for example, to finite-dimensional spaces whose norm is equivalent to a coordinate maximum norm.

Finite-dimensionality of the target F leads to a different question. Remark 46 reduces it to scalar-valued forms and gives a positive result over a spherically complete nonarchimedean base.

3.5. Spherically complete targets. This subsection and Appendix B prove part (2) of the Main Theorem. Spherical completeness allows us to project bounded multilinear maps onto alternating maps without increasing the norm. Applying this projection to the ordinary multilinear precomposition formula produces an analytic lift in every characteristic and degree. The argument is independent of the counterexample construction. The retraction strategy reflects the classical role of spherically complete spaces as injective objects in nonarchimedean Banach-space theory; see Schikhof [14, §2.8] for this background over complete fields. A special case is stated in [7, Remark 16]; the proof here follows the extension and retraction arguments in the Lean development. Related Lean formalizations of compatible extension and spherical completion in the nonarchimedean setting appear in Yuan [15, Prop. 6.6 and Thm. 7.3].

Nonarchimedean norms and spherical completeness are as defined in the introduction. We write $\overline{B}(a, r) = \{x : \|x - a\| \leq r\}$, with $r \geq 0$. In a nonarchimedean space, two such balls meet if and only if the distance between their centres is at most the larger radius; in that case the ball of smaller radius is contained in the other.

Recall from Section 2 that $\text{Mult}^k(X; Y)$ and $\text{Alt}^k(X; Y)$ carry the operator norms determined by

$$\|m(x_1, \dots, x_k)\| \leq \|m\| \prod_{i=1}^k \|x_i\|.$$

When the target norm is nonarchimedean, both operator spaces are nonarchimedean, regardless of the norm on X .

Throughout this subsection, K is a nonarchimedean nontrivially normed field and $k \geq 0$ is an integer. Positive characteristic implies the nonarchimedean hypothesis by Lemma 2. We prove the following operator statement, which also records every smoothness order. For normed K -spaces E, E' and a nonarchimedean, spherically complete F ,

$$Q_{E, E', F}^K \text{ is } C^n \quad (n \in \mathbb{N} \cup \{\infty, \omega\}).$$

Together with spherical completeness of $\text{Alt}^k(E; F)$ and (1.2), this proves Main Theorem (2).

No completeness assumption on K, E or E' is needed, and no nonarchimedean hypothesis is imposed on E or E' . For $k \geq 1$, put $A = \text{Alt}^k(E; F)$ and $M = \text{Mult}^k(E; F)$. The desired projection is a bounded linear map $\rho: M \rightarrow A$ of norm at most one that fixes every alternating map. Once it is available, the formula

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k))$$

is bounded and multilinear in the operators f_i , and has diagonal Q . This is the required analytic lift: expansion around any operator gives a finite power series. The construction of ρ , outlined after Proposition 14 and carried out in Appendix B, rests on the spherical completeness of A .

Proposition 14. *For a nonarchimedean nontrivially normed field K , a normed K -space E and a nonarchimedean, spherically complete F , the space $\text{Alt}^k(E; F)$ is a nonarchimedean, spherically complete normed space for every $k \geq 0$.*

The proof is given in Appendix B.

Proposition 14 is the codomain assertion of Main Theorem (2). Appendix B first proves an extension lemma for spherically complete targets that keeps a prospective extension in several operator balls at once (Lemma 49). It then deduces Proposition 14 by identifying $\text{Alt}^k(E; F)$ isometrically with the bounded linear maps into F on a free seminormed space that vanish on the multilinearity and equal-input relations. A second application of the lemma, now with target A , extends the inverse of the isometric inclusion $A \rightarrow M$ to a contraction $\rho: M \rightarrow A$

that fixes every alternating map; inserted into the lift formula above, it gives a bounded k -linear lift of Q of norm at most one, which completes the proof of Main Theorem (2).

3.6. Consequences for alternating-map bundles. Each of the three mechanisms gives analytic alternating-map bundles over an arbitrary analytic base, provided the fibers satisfy the corresponding hypothesis.

Corollary 15 (analytic alternating-map bundles). *Let K be a nontrivially normed field, let $k \geq 0$, and let M be an analytic manifold over K modeled on an arbitrary normed space. Let $\mathcal{E}, \mathcal{H} \rightarrow M$ be analytic normed vector bundles with typical fibers E, F . Then $\text{Alt}^k(\mathcal{E}; \mathcal{H})$, with fiber $\text{Alt}^k(\mathcal{E}_x; \mathcal{H}_x)$ over x , is an analytic normed vector bundle, functorially in analytic bundle morphisms between bundles satisfying the same hypothesis, in each of the following cases:*

- (a) $k! \neq 0$ in K ;
- (b) K is nonarchimedean, and F is nonarchimedean and spherically complete, or merely admits an equivalent norm with these properties;
- (c) E has a finite algebraic basis with continuous coordinate functionals; for instance, K is complete and E is finite-dimensional.

Proof. For typical source fibers E_1, E_2 and a typical target fiber F_1 of bundles satisfying the hypothesis, the map Q_{E_1, E_2, F_1}^K is analytic: in case (a) by Proposition 9, and in case (c) by Proposition 10 and Lemma 11. In case (b) we may assume $k! = 0$ in K , so that K has positive characteristic. Main Theorem (2) then applies after an equivalent change of norm on F_1 , which alters the norms of the alternating-map and operator spaces only by fixed factors and so preserves analyticity. By (1.2), Alt^k is analytic on the relevant hom spaces, and its composites with the transition maps, the changes of trivialization and the local expressions of bundle morphisms are analytic. Proposition 5(2) therefore applies (part (1) is not available in cases (b) and (c), where Alt^k need not be analytic on all hom spaces). $\square$

Cases (a) and (b) concern the target fibers alone. Call a normed space F a *universal target* if $Q_{E, D, F}^K$ is analytic for all normed spaces E, D , and write $\mathsf{T}_k(K)$ for the class of universal targets (formally Definition 37 in Section 6). Cases (a) and (b) say that $\mathsf{T}_k(K)$ contains every normed space when $k! \neq 0$ in K , and every space with an equivalent nonarchimedean, spherically complete norm. Their proof uses only this membership, so the corollary holds verbatim whenever the typical target fibers lie in $\mathsf{T}_k(K)$. When $k! = 0$ in K , the analytic conclusion over a general base needs some hypothesis on the fibers. Section 4 removes it for bases with finite continuous coordinates, and Section 5 shows that it cannot be removed over general Banach bases.

4. ANALYTIC FAMILIES AND BUNDLES OVER FINITE-COORDINATE BASES

Proposition 5(2) requires analyticity of the alternating functor only along the transition families of the input bundles, not on whole hom spaces. This section shows that the weaker requirement holds for every analytic family of morphisms parametrized by an open subset of K^d , in every characteristic and every degree and with arbitrary normed fibers (Corollary 21). The alternating-map bundle of analytic input bundles is therefore analytic over every base with finite continuous coordinates (Theorem 22), which proves Main Theorem (3). For infinite-dimensional parameter spaces we prove the same conclusion under hypotheses on the final target fiber, the codomain of the value-space component of a family. Over a nonarchimedean field, families parametrized by open subsets of $c_0(I, K)$ are covered when this fiber is complete and nonarchimedean (Corollary 24). Over any K , families parametrized by open subsets of

$\ell^1(I, K)$ with the sum norm are covered when this fiber is complete (Corollary 26). The longer proofs for these two classes are in Appendix C.

Regularity on an entire operator space is stronger than regularity along a prescribed family of operators. This distinction gives a useful replacement for restricting the objects of the alternating-map bifunctor. We retain every object and every individual morphism, and ask which analytic families the bifunctor preserves. The resulting family structure is canonical. Its usefulness comes from the concrete parameter classes for which the additional condition is automatic.

4.1. The canonical family structure. For objects $X = (E, F)$ and $Y = (E', F')$ of $\text{Vec}_K^{\text{op}} \times \text{Vec}_K$, write

$$V_X = \text{Alt}^k(E; F), \quad H_{XY} = \mathcal{L}(E', E) \times \mathcal{L}(F, F'),$$

and denote the action on morphisms by

$$A_{XY}: H_{XY} \longrightarrow \mathcal{L}(V_X, V_Y), \quad A_{XY}(u, v)m = v \circ m \circ (u, \dots, u).$$

The product defining H_{XY} has the maximum norm. These conventions make the contravariance in E explicit: the first component of a morphism $X \rightarrow Y$ is a map $E' \rightarrow E$.

Definition 16. Let U be open in a normed K -space. A map $\gamma: U \rightarrow H_{XY}$ is *admissible* if both γ and $A_{XY} \circ \gamma$ are analytic. Families on analytic manifolds are admissible when their expressions in parameter charts are admissible.

Proposition 17. *Admissible families contain constants and are closed under analytic reparameterization, local gluing, and categorical composition. Composition is allowed to depend on either a common parameter or independent parameters. They form the largest collection of ordinary analytic families for which the action of Alt^k is analytic.*

Proof. Constants, reparameterization, and gluing follow from the corresponding properties of analytic maps. For composable families $\gamma: U \rightarrow H_{XY}$ and $\eta: U \rightarrow H_{YZ}$, composition in H is bounded bilinear, and functoriality gives

$$A_{XZ}(\eta \circ \gamma) = A_{YZ}(\eta) \circ A_{XY}(\gamma).$$

Composition of bounded operators is bounded bilinear as well, so both required maps are analytic. For independent parameter spaces, pull the two families back to their product and apply the same argument. Finally, any ordinary analytic family preserved analytically by Alt^k satisfies exactly the two conditions of Definition 16, and hence belongs to this collection. $\square$

Equivalently, identify the set H_{XY} with the graph

$$\Gamma_{A_{XY}} = \{(h, A_{XY}(h)) : h \in H_{XY}\} \subseteq H_{XY} \times \mathcal{L}(V_X, V_Y)$$

and declare a family analytic when its map to the ambient product is analytic. No assertion that this graph is an analytic submanifold is involved. In the language of plots, this is a category whose hom sets are analytic plot spaces: the parameter domains are open subsets of normed K -spaces, the reparameterizations are analytic maps, and covers are open covers. The smooth concrete-sheaf framework of Baez and Hoffnung [1] motivates this formulation; the analytic family structure used here is specified directly by Definition 16 and Proposition 17.

This structure need not preserve the vector-space operations on hom sets. Suppose $k > 0$ and $u: U \rightarrow \mathcal{L}(E', E)$ is analytic but its pullback action is not. With $F' = F$, both $(u, 0)$ and $(0, \text{id}_F)$ are admissible because their induced actions vanish. Their sum (u, id_F) is not admissible. Thus addition of admissible families can fail. Multiplication by an analytic scalar family does preserve admissibility, since $A_{XY}(\lambda u, \lambda v) = \lambda^{k+1} A_{XY}(u, v)$.

4.2. Reflection through a closed subspace. The range condition behind admissibility can be separated from the operator construction. Set

$$W = \mathcal{L}(V_X, \text{Alt}^k(E'; F')), \quad Z = \mathcal{L}(V_X, \text{Mult}^k(E'; F')),$$

and let $j: W \hookrightarrow Z$ be the map induced by forgetting alternation. It is a linear isometry with closed image. Indeed, an element of Z belongs to its image precisely when every value vanishes on every tuple with two equal entries. These are closed linear evaluation conditions; the operator norms before and after forgetting alternation agree. This argument does not require completeness.

Although A_{XY} can fail to be analytic, its composite with j is always a bounded homogeneous polynomial of degree $k + 1$. A multilinear representative is obtained by using k independent input operators and one independent output operator:

$$(4.1) \quad B((u_1, v_1), \dots, (u_{k+1}, v_{k+1}))(m)(x_1, \dots, x_k) = v_{k+1}(m(u_1 x_1, \dots, u_k x_k)).$$

Its norm is at most one, and its diagonal is jA_{XY} . For pullback alone, the corresponding degree is k . Thus an analytic family always gives an analytic ambient map $jA_{XY}\gamma$. The remaining question is whether this map is analytic into W .

We first record how closedness detects homogeneous coefficients. The conclusion concerns their diagonals, not the values of a chosen multilinear representative on arbitrary tuples.

Lemma 18 (closedness of homogeneous terms). *Let W be a closed linear subspace of a normed space Z , and suppose*

$$f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$$

is an analytic expansion with values in Z . If f takes values in W near x_0 , then $p_n(x, \dots, x) \in W$ for every n and every parameter vector x .

The proof is given in Appendix C.

Theorem 19 (finite-coordinate analytic reflection). *Let $j: W \hookrightarrow Z$ be a linear isometry with closed image, let $U \subseteq K^d$ be open, and let $f: U \rightarrow W$. If jf is analytic, then f is analytic. The spaces W, Z and the field K need not be complete. The same statement holds when the parameter space is continuously linearly isomorphic to K^d .*

The proof is given in Appendix C.

Remark 20 (Preservation of the coefficient bound). In the proof of Theorem 19 (§C.1), the Taylor coefficients p_n of jf are regrouped into monomial coefficients $c_\alpha \in W$ and W -valued multilinear maps q_n with the same diagonals. If Z has a nonarchimedean norm, both finite sums in that construction satisfy maximum estimates. Thus $\|c_\alpha\| \leq \|p_n\|$ and $\|q_n\| \leq \|p_n\|$ for maximum-norm coordinates on K^d . The reflected expansion has at least the radius of the ambient expansion, without completeness of W or Z . The c_0 theorem below extends this same estimate to infinitely many coordinates; there completeness supplies the infinite coefficient sums.

Corollary 21. *Every analytic morphism family parametrized by an open subset of K^d is admissible for Alt^k , in every characteristic and every degree, with arbitrary normed fibers.*

Proof. Apply Theorem 19 to the closed isometric inclusion j above and the analytic map $jA_{XY}\gamma$, using (4.1). $\square$

Over an incomplete field, finite algebraic dimension alone does not imply the existence of continuous coordinates. The coordinate hypothesis in Theorem 19 is consequently essential to its stated scope. When K is complete, every finite-dimensional normed parameter space has such coordinates. In particular, standard finite-dimensional analytic probes cannot detect the failure of analyticity of the full operator action. Arbitrary normed parameter spaces, including the operator spaces themselves, are needed to recover the regularity notion used in the Main Theorem.

4.3. Bundles over finite-coordinate bases. The finite-coordinate theorem gives a bundle construction in every characteristic, even when the fibers are infinite-dimensional. For certain infinite-dimensional bases, summability gives further positive results. These are conditions on the parameter space of a family, and should be distinguished from the conditions on source or target fibers in the earlier positive theorems.

Here an analytic normed vector bundle has operator-valued analytic transition maps in its local trivializations. Bundle morphisms are required to have operator-valued analytic expressions as well. For bundles $\mathcal{E}, \mathcal{H}$ over the same analytic manifold M , with typical fibers E, F , take a common trivializing cover with transition maps g_{ij} and h_{ij} . As in §3.1, the candidate transition maps on alternating forms are

$$(4.2) \quad \theta_{ij}(x): m \mapsto h_{ij}(x) \circ m \circ (g_{ji}(x), \dots, g_{ji}(x)), \quad m \in \text{Alt}^k(E; F).$$

The inverse $g_{ji} = g_{ij}^{-1}$ is an analytic operator family by the bundle atlas. Functoriality gives the cocycle identity for (4.2).

Theorem 22 (alternating-map bundles). *Let M be an analytic K -manifold modeled on K^d , or on a normed space with continuous linear coordinates in K^d . For every $k \geq 0$, fiberwise alternating maps define a bifunctor*

$$\text{Alt}^k: \text{VB}_K^\omega(M)^{\text{op}} \times \text{VB}_K^\omega(M) \longrightarrow \text{VB}_K^\omega(M),$$

where $\text{VB}_K^\omega(M)$ denotes analytic normed vector bundles over M with the operator-valued analytic bundle morphisms just specified. There is no restriction on the characteristic, the degree, or the dimensions and completeness of the fibers.

Proof. In each base chart, the pair (g_{ji}, h_{ij}) is an analytic family with a finite-coordinate parameter space. Corollary 21 makes (4.2) analytic. The cocycle identity therefore glues the local products with fiber $\text{Alt}^k(E; F)$ to an analytic normed vector bundle. The same argument applies to a change of trivialization and to the local expressions of a pair of bundle morphisms, with the first morphism directed contravariantly. The identities and composition laws hold fiberwise by functoriality of alternating maps, giving the stated bifunctor. $\square$

Together with the unconditional C^∞ result, this supplies the alternating-map bundle construction for every smoothness order over such finite-coordinate bases. The analytic conclusion does not assert that the action on arbitrary operator spaces is analytic; it uses the base charts as parameter spaces. Over a complete field, the hypothesis includes every finite-dimensional normed model space.

Finite dimension enters the proof only through Theorem 19: the ambient transition $j \circ \theta_{ij}$ of (4.2) is analytic over every base, by (4.1), and the finitely many monomial coefficients give W -valued representatives with the exponential bound (C.1). No reflection step is needed, over any base, when the joint action itself has a bounded multilinear lift: when $k! \neq 0$ in K , when the target fibers lie in $\mathcal{T}_k(K)$, or when the source fibers have finite continuous coordinates. The hypothesis on the base cannot simply be removed: Corollary 31 in Section 5 gives analytic bundles over a Banach base whose alternating-map bundle is not analytic.

4.4. Nonarchimedean c_0 parameters. For an arbitrary index set I , write $c_0(I, K)$ for the space of scalar families tending to zero outside finite sets, with its supremum norm. Completeness is imposed on the output of the coefficient construction, not on the field or the parameter space.

Theorem 23. *Let K be nonarchimedean, let $j: W \hookrightarrow Z$ be a linear isometry, and assume that W is complete and Z has a nonarchimedean norm. If $U \subseteq c_0(I, K)$ is open and $jf: U \rightarrow Z$ is analytic for a map $f: U \rightarrow W$, then f is analytic.*

The proof is given in Appendix C.

Corollary 24. *Over a nonarchimedean field, every analytic morphism family parametrized by an open subset of $c_0(I, K)$ is admissible for Alt^k when its final value-space target F' is complete and nonarchimedean. The same holds after a bounded linear isomorphism of parameter spaces.*

Proof. When F' is complete and nonarchimedean, so are $\text{Alt}^k(E'; F')$ and the operator space W ; the ambient operator space Z is nonarchimedean as well. Apply Theorem 23 to $A_{XY}\gamma$. $\square$

4.5. Ordinary ℓ^1 parameters. For this subsection, $\ell^1(I, K)$ has the ordinary sum norm $\|x\|_1 = \sum_{i \in I} |x_i|$, even when K is nonarchimedean. The proof of Theorem 25 below uses the fixed degree of the outer polynomial. It does not require a uniform lifting bound for all Taylor degrees of an arbitrary ambient analytic map.

Theorem 25 (fixed-degree reflection along ℓ^1 families). *Let W be a closed subspace of a Banach space Z , let H be normed, and suppose that $P: H \rightarrow W$ is the diagonal, in Z , of a bounded d -linear map $B: H^d \rightarrow Z$. For every analytic map $\gamma: U \rightarrow H$, where U is open in $\ell^1(I, K)$, the map $P\gamma: U \rightarrow W$ is analytic. The field K need not be complete.*

The proof is given in Appendix C.

Corollary 26. *Every analytic morphism family parametrized by an open subset of $\ell^1(I, K)$ is admissible for Alt^k when the final value-space target F' is complete. The same holds when the parameter space is a bounded linear retract of an $\ell^1(I, K)$ space. No completeness is required of the other fibers.*

Proof. The ambient space $Z = \mathcal{L}(V_X, \text{Mult}^k(E'; F'))$ is Banach, and W is closed in Z . Apply Theorem 25 to the degree- $(k+1)$ polynomial (4.1). For pullback alone one may instead use degree k . If $i: P \rightarrow \ell^1(I, K)$ and $r: \ell^1(I, K) \rightarrow P$ are bounded linear with $ri = \text{id}_P$, apply the result to γr on the open set $r^{-1}(U)$, and then restrict along i . $\square$

Corollary 27 (Failure of descent through quotient parameters). *Let K be a complete nontrivially normed field of characteristic $p > 0$, and let $k \geq p$. There are Banach spaces E, F , an index set J , and a bounded open linear surjection*

$$q: \ell^1(J, K) \longrightarrow H := \mathcal{L}(E, E)$$

such that $Q_{E,E,F}^K \circ q$ is analytic, whereas $Q_{E,E,F}^K$ is analytic at no point. The map q has no local right inverse on a nonempty open subset of H that is Fréchet differentiable at even one point of that subset.

The proof is given in Appendix C.

Thus an analytic family can become admissible after a bounded linear quotient reparameterization even though it was not admissible originally. The open-cover locality of admissible families does not imply descent through such quotient maps. In particular, the bad parameter space H cannot be a bounded linear retract of an ordinary ℓ^1 space.

The bundle argument of Theorem 22 now gives two further forms of the construction: over bases modeled on $c_0(I, K)$ when the output fibers are complete and nonarchimedean, and over

bases modeled on $\ell^1(I, K)$ or its bounded linear retracts when the output fibers are complete. For a statement about an entire category of bundles, one can restrict all fibers respectively to nonarchimedean Banach spaces or to Banach spaces; alternating-map fibers remain in the same category.

On arbitrary analytic bases one can instead specify admissible transition families and require admissibility of changes of atlas and bundle morphisms. Proposition 17 ensures compatibility with composition. When the alternating-map spaces at both endpoints are Banach (in particular, when the value fibers are complete), an admissible family of isomorphisms with analytic underlying inverse family has an admissible inverse family. Its alternating action is the inverse of the original action. Locally identify the two endpoint spaces by one fixed isomorphism; inversion is then analytic by the convergent Neumann series in the Banach endomorphism algebra. The shear example of Corollary 31 shows why arbitrary analytic changes of atlas cannot simply be substituted for admissible ones in this formulation.

4.6. Universal analytic reflection. The parameter theorems (Theorems 19, 23 and 25) suggest an intrinsic question: when does every analytic map into a Banach space remain analytic into each closed subspace containing its values? Appendix C first isolates the exact local coefficient problem for a general parameter space (Proposition 50). Over a complete field, §C.4 then characterizes this universal reflection property of a Banach space P (Definition 53) by projections R_n from the completed projective tensor powers $T_n(P)$ onto their diagonal subspaces $\Delta_n(P)$, the closed spans of the pure powers $x^{\otimes n}$, with exponentially bounded norms (Theorem 54).

For K^d with the maximum norm, the coordinate construction gives $\|R_n\| \leq d^n$. For the ordinary sum-norm space $\ell^1(I, K)$ over a complete nonarchimedean field, with I infinite, the infimum of the norms of such projections is exactly $n!$ (Proposition 55). So $\ell^1(I, K)$ lacks universal analytic reflection, although every analytic morphism family parametrized by an open subset of it is admissible for Alt^k when its final value-space target is complete (Corollary 26). The alternating-family property is strictly weaker because the ambient alternating action has one fixed polynomial degree, so factoring through ℓ^1 costs a single factorial bound rather than one for each Taylor degree. Over a general Banach base no such reflection need be available: Section 5 gives, in characteristic $p > 0$ and degree $k \geq p$, analytic bundles over a Banach base whose alternating-map bundle, with the induced transitions, is not analytic.

5. NONANALYTIC FUNCTORS GIVE NONANALYTIC BUNDLES

The analytic results of Sections 3 and 4 each use an extra hypothesis, such as $k! \neq 0$ in K , nonarchimedean spherically complete target fibers, source fibers with finite continuous coordinates, a base with finite continuous coordinates, or a c_0 or ℓ^1 base with complete target fibers (nonarchimedean in the c_0 case). This section carries out the second step of the program described in the introduction. When $k! = 0$ in K and no such hypothesis is imposed, analyticity fails both for the alternating functor and for the alternating-map bundles it induces.

When $k! = 0$ in K , there are Banach spaces E and F for which the pullback action $Q_{E,E,F}^K$ is analytic at no point (Theorem 28). Hence Q is analytic for all Banach spaces exactly when $k! \neq 0$ in K (Corollary 29). The shear of Proposition 30 turns a point of nonanalyticity of Q into a point of nonanalyticity of the pullback along an affine analytic family of automorphisms. Applied to Theorem 28, it gives an alternating-map bundle over a Banach base whose transition between the charts induced by two analytic trivializations is analytic at no point (Corollary 31). This proves Main Theorem (4).

Two refinements concern the base. In §5.4, over $K = \mathbb{F}_q((u))$, the operator base is replaced by the open unit ball of $c_0(\mathbb{N}, K)$ and the shear by diagonal isometric transitions. Among Banach model spaces admitting an equivalent nonarchimedean norm over this field, finite dimension is then exactly the right hypothesis on the base (Corollary 35). Over the incomplete field $\mathbb{F}_p(t)$, the alternating-map bundle of analytic input bundles can also fail to be analytic over a base of finite algebraic dimension without continuous coordinates. This uses the objects of Section 6 and is recorded there as Remark 45.

5.1. A nowhere analytic alternating functor. As noted in the introduction, the usual polarization argument divides by $k!$ and does not apply when that scalar vanishes. The following theorem shows that this obstruction is unavoidable if analyticity is required for all Banach spaces over K .

Theorem 28 (Counterexample theorem). *Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \geq p$. There are Banach spaces E and F over K such that the degree- k map*

$$Q: \mathcal{L}(E, E) \rightarrow \mathcal{L}(\text{Alt}^k(E; F), \text{Alt}^k(E; F)), \quad Q(f)(m) = m \circ (f, \dots, f),$$

is analytic at no point of $\mathcal{L}(E, E)$. Equivalently, there is no bounded k -linear map P on $\mathcal{L}(E, E)$ with $P(f, \dots, f) = Q(f)$ for all f . The Banach space F admits no equivalent nonarchimedean norm. The same E, F work for any reindexing of the k arguments.

Corollary 29. *Let K be any nontrivially normed field and $k \geq 0$ any integer. Then $Q_{E, E', F}^K$ is analytic for all K -Banach spaces E, E', F if and only if $k! \neq 0$ in K . More precisely:*

- (1) *if $k! \neq 0$ in K , then $Q_{E, E', F}^K$ is analytic, indeed given at every point by a finite power series, for all normed spaces E, E', F , complete or not;*
- (2) *if $k! = 0$ in K , then there are Banach spaces $E = E'$ and F , with F admitting no equivalent nonarchimedean norm, such that $Q_{E, E', F}^K$ is analytic at no point.*

Proof. If $k! \neq 0$ in K , Proposition 9 gives a finite power series at every point for arbitrary normed E, E', F , including degree zero. If $k! = 0$ in K , the field has positive prime characteristic $p \leq k$. Theorem 28 gives the required Banach counterexample. Together these prove the classification and both strengthened conclusions. $\square$

Thus the factorial condition is exact, even when the source and target of the linear maps coincide. The failure is also local in the strongest sense: in the examples, analyticity fails at every base point.

Proof of Main Theorem (1). The map $(v, T) \mapsto v_* \circ T$ in (1.2) is bounded bilinear, and every $Q_{E, E', F}^K$ is C^∞ by Proposition 7. Hence every joint action, and thus the bifunctor, is C^∞ . Suppose $k! \neq 0$ in K . In degree zero, Q is constant and the joint action is a bounded linear function of v alone. For $k \geq 1$, Corollary 29 and Proposition 3 give a bounded k -linear lift P of Q . Placing the pushforward variable in the last slot,

$$((u_1, v_1), \dots, (u_{k+1}, v_{k+1})) \longmapsto (v_{k+1})_* \circ P(u_1, \dots, u_k)$$

is a bounded $(k+1)$ -linear lift of the joint action. Expanding it as in Proposition 3(1) gives a finite power series at every point. Suppose $k! = 0$ in K , and let $E = E'$ and F be the Banach spaces of Corollary 29(2). By (1.2), composing the joint action with the affine map $u \mapsto (u, \text{id}_F)$ recovers $Q_{E, E', F}^K$. Hence the joint action is analytic at no point of the form (u, id_F) . For the restriction to Ban_K , recall from the introduction that $\text{Alt}^k(E; F)$ is complete whenever F is complete. The restricted bifunctor has the same maps on hom spaces, and the counterexample spaces are Banach, so both assertions persist. $\square$

It remains to prove Theorem 28. The shear construction below uses only the statement of the theorem, so it comes first. The Banach spaces behind the theorem and an outline of its proof follow in §5.3, and a detailed roadmap of the proof opens Appendix D.

5.2. The shear construction. Restricting to changes of frame does not remove the obstruction on unrestricted parameter spaces. The following elementary construction transfers any bad pullback map to a family of automorphisms.

Proposition 30 (invertible shear family). *Let E, D, F be normed K -spaces, and suppose*

$$Q: \mathcal{L}(E, D) \longrightarrow \mathcal{L}(\text{Alt}^k(D; F), \text{Alt}^k(E; F))$$

is not analytic at u_0 . Put $H = D \oplus E$, with the maximum norm, and define

$$g(u) = \begin{pmatrix} \text{id}_D & u \\ 0 & \text{id}_E \end{pmatrix}.$$

Then g and g^{-1} are affine analytic operator families, but $u \mapsto g(u)^$ on $\text{Alt}^k(H; F)$ is not analytic at u_0 . In particular, the counterexample can occur at $u_0 = 0$, where $g(0) = \text{id}_H$.*

Proof. The inverse is $g(-u)$. Let

$$J: \text{Alt}^k(D; F) \rightarrow \text{Alt}^k(H; F), \quad Jm = m \circ \pi_D^{\times k},$$

and let $R: \text{Alt}^k(H; F) \rightarrow \text{Alt}^k(E; F)$ be restriction along $i_E: E \rightarrow H$. Both maps are bounded linear, and $\pi_D g(u) i_E = u$, so

$$(5.1) \quad R \circ g(u)^* \circ J = Q(u).$$

If $g(u)^*$ were analytic at u_0 , bounded linear composition with R and J would make Q analytic there. For the last assertion, take the counterexample of Theorem 28, which is nonanalytic at zero. $\square$

Theorem 28 makes E, D, F , and hence the parameter space $\mathcal{L}(E, D)$, Banach. Thus even an analytic family of invertible operators, with analytic inverse, can induce a nonanalytic family on alternating maps over a Banach base. Equation (5.1) also makes the obstruction precise: it is already visible after bounded linear extension and restriction of forms.

Corollary 31 (failure over a Banach base). *Let K be a nontrivially normed field of characteristic $p > 0$, let $k \geq p$, and let E, F be Banach spaces as in Theorem 28. Put $M = \mathcal{L}(E, E)$, an analytic manifold with one chart, and $H = E \times E$. Let $\mathcal{E} \rightarrow M$ be the trivial bundle with fiber H , equipped with the two global trivializations*

$$\tau_0(u, z) = (u, z), \quad \tau_1(u, z) = (u, g(u)z),$$

where g is the shear of Proposition 30, and let $M \times F$ be the trivial bundle with fiber F . Then $\mathcal{E}$ and $M \times F$ are analytic normed vector bundles, but the alternating-map bundle $\text{Alt}^k(\mathcal{E}; M \times F)$ with the transitions (4.2) is not analytic: its transition between the charts induced by τ_0 and τ_1 is analytic at no point of M . Equivalently, $u \mapsto g(u)$ is an automorphism of $M \times H$ in $\text{VB}_K^{\omega}(M)$ whose fiberwise alternating action on $M \times \text{Alt}^k(H; F)$ is not an analytic bundle morphism. Thus Theorem 22 fails over this Banach base, both for objects and for morphisms.

Proof. The transition of $\mathcal{E}$ from τ_0 to τ_1 is the affine analytic family g , with affine analytic inverse $u \mapsto g(-u)$. By (4.2), the induced transitions of the alternating bundle are pullbacks by $g(\pm u)$, with the identity on F . Theorem 28 makes $Q_{E,E,F}^K$ analytic at no point, so Proposition 30 shows that $u \mapsto g(u)^*$ is analytic at no point. The same holds for $u \mapsto g(-u)^*$ after the linear reparameterization $u \mapsto -u$. The morphism statement is the same computation in the global product chart of $M \times \text{Alt}^k(H; F)$. $\square$

The proof uses Theorem 28 only through Proposition 30. Hence, whenever $Q_{E,D,F}^K$ fails to be analytic at some u_0 , the same construction over $M = \mathcal{L}(E, D)$, with fiber $D \oplus E$ and target bundle $M \times F$, gives analytic input bundles whose alternating-map bundle has a transition that is not analytic at u_0 . Every failure of analyticity of the functor therefore appears as a failure for bundles. The corollary concerns the compatibility of the natural construction with the given analytic atlas, or equivalently its functoriality. It does not assert that the trivial pointwise alternating-map bundle admits no analytic structure, and the C^n forms of the construction hold over every base for $n \leq \infty$ (Corollary 8). The Lean formalization proves the corresponding contradiction for Mathlib's existing alternating-bundle topology (Appendix J).

5.3. The Laurent series counterexample. We describe the Banach spaces behind Theorem 28 and outline its proof. Appendix E gives the details of the construction. Let κ be a finite field of characteristic p , let $k \geq 1$ and $r \in (0, 1)$, and let $K_1 := \kappa((X))$ carry the absolute value $|a| = r^{\text{ord } a}$, so that $|X| = r$. Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the supremum norm. The projective exterior norm on $\Lambda_{K_1}^k E_1$ is

$$\|\omega\|_\pi := \inf \left\{ \sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty : \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k} \right\},$$

the infimum over finite decompositions of an ordinary sum rather than a maximum. Let B be the completion. For $k \geq 2$, sums of unit wedges on disjoint blocks of coordinates have norms in B growing linearly with the number of blocks, so B admits no equivalent nonarchimedean norm (Proposition 88). The universal wedge map $W_B \in \text{Alt}_{K_1}^k(E_1; B)$ is $W_B(x_1, \dots, x_k) = x_1 \wedge \cdots \wedge x_k$, and for $a \in E_1$ we write $D_a \in \mathcal{L}(E_1, E_1)$ for multiplication by a , $D_a x = ax$.

By Proposition 3, it suffices to exclude a bounded k -linear lift of Q . Let $E_0 = \kappa^\mathbb{N} \subseteq E_1$ be the κ -subspace of sequences with entries in κ . When $k \geq p$, the multiplier theorem over a finite field (Theorem 65, Appendix D) rules out a multilinear lift of coordinatewise multiplication on E_0 that satisfies a polarized diagonal identity and whose values have exterior support of uniformly bounded dimension. To prove the following theorem, we evaluate a hypothetical bounded k -linear lift of $Q_{E_1, E_1, B}^{K_1}$ on multiplication operators $D_{a_1}, \dots, D_{a_k}$ and at W_B , with all a_i and all arguments in E_0 , and apply the κ -linear constant-coefficient map $\eta: B \rightarrow \Lambda_\kappa^k E_0$. Its support dimension is bounded on bounded subsets of B (Proposition 73). This converts the norm bound on the lift into an algebraic obstruction: the result is a finite-field lift of coordinatewise multiplication of the kind excluded by Theorem 65 when $k \geq p$ (Lemma 74).

Theorem 32. *Let κ be a finite field of characteristic p , let $k \geq p$ and $r \in (0, 1)$, and let $K_1 = \kappa((X))$ with $|X| = r$. Let $E_1 = \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda_{K_1}^k E_1$ under the projective exterior norm. Then $Q_{E_1, E_1, B}^{K_1}$ admits no bounded k -linear lift over K_1 . Consequently it is analytic at no point of $\mathcal{L}(E_1, E_1)$.*

The proof is given in Appendix E. Theorem 28 follows by ascent, with $\kappa = \mathbb{F}_p$. For any $t \in K$ with $0 < |t| < 1$, the closure of $\mathbb{F}_p(t)$ in the completion of K is such a Laurent series field. Ingleson's Hahn–Banach theorem and projective base change pass the obstruction to the completion, and density passes it to K itself (Appendix F). The linear growth of sums of unit wedges on disjoint blocks persists after base change, so the resulting target F , like B , admits no equivalent nonarchimedean norm (Proposition 88). Appendix G assembles the proof.

5.4. Multiplier families and c_0 bases. The proof of Theorem 32 uses the hypothetical lift only on coordinatewise multiplication operators and only at the single form W_B . Moreover,

Theorem 65(1) needs only finitely supported test vectors with entries in $\{0, \pm 1\}$. Both observations give a much smaller bad parameter space. Throughout this subsection κ , k , r , K_1 , E_1 , B and W_B are as in Theorem 32, $c_0(\mathbb{N}, K_1) \subseteq E_1$ carries the supremum norm, and $D_a \in \mathcal{L}(E_1, E_1)$ denotes multiplication by a .

Proposition 33 (multipliers on c_0). *The homogeneous polynomial*

$\sigma: c_0(\mathbb{N}, K_1) \rightarrow \text{Alt}_{K_1}^k(E_1; B)$, $\sigma(a)(x_1, \dots, x_k) = W_B(ax_1, \dots, ax_k) = (ax_1) \wedge \dots \wedge (ax_k)$,
has no bounded k -linear lift with values in $\text{Alt}_{K_1}^k(E_1; B)$. Consequently, for every $u_0 \in \mathcal{L}(E_1, E_1)$, the map $a \mapsto Q_{E_1, E_1, B}^{K_1}(u_0 + D_a)(W_B)$ is analytic at no point of $c_0(\mathbb{N}, K_1)$. In particular $a \mapsto Q_{E_1, E_1, B}^{K_1}(u_0 + D_a)$ is analytic at no point.

The proof is given in Appendix E.

Corollary 34 (diagonal transitions over a c_0 base). *Let U be the open unit ball of $c_0(\mathbb{N}, K_1)$, an analytic manifold with one chart. Let $\mathcal{V} \rightarrow U$ be the trivial bundle with fiber E_1 and the two global trivializations $\tau_0(a, z) = (a, z)$ and $\tau_1(a, z) = (a, D_{1+a}z)$, and let $\mathcal{B} = U \times B$. Then $\mathcal{V}$ is an analytic normed vector bundle whose transitions are isometric diagonal automorphisms of E_1 . The alternating-map bundle $\text{Alt}^k(\mathcal{V}; \mathcal{B})$ is not analytic: the section that is constantly W_B in the chart induced by τ_1 is, in the chart induced by τ_0 , the map $a \mapsto W_B \circ (D_{1+a}, \dots, D_{1+a})$, which is analytic at no point of U .*

Proof. For $a \in U$ every coordinate satisfies $|1 + a_i| = 1$, so D_{1+a} is an isometry of E_1 . Its inverse $D_{(1+a)^{-1}} = \sum_{n \geq 0} (-D_a)^n$ is a power series of radius one in a , since $D_a^n = D_{a^n}$ has norm at most $\|a\|^n$. Hence both transitions are analytic. If $\omega_a \in \text{Alt}^k(\mathcal{V}_a; B)$ has coordinates $m_0(a)$ and $m_1(a)$ in the two induced charts, then $m_0(a) = m_1(a) \circ (D_{1+a}, \dots, D_{1+a})$. With $m_1 \equiv W_B$, this is the map of Proposition 33 with $u_0 = \text{id}$, which is analytic at no point. $\square$

Thus the analyticity of a section of the alternating-map bundle depends on the analytic frame used to read it, at every point of the base. This shows that the nonarchimedean hypothesis on the final value space F' in Corollary 24 cannot be removed: B is complete but admits no equivalent nonarchimedean norm (Proposition 88). It also contrasts with Corollary 26. On $\ell^1(\mathbb{N}, K_1)$, with its ordinary sum norm, the same multiplier family is admissible; its sorted-word lift (Lemma 51) has norm at most $k!$. The test vectors above are finitely supported in both spaces, so only the norm on the parameter space distinguishes them. This is a concrete analogue of Corollary 27, with the bounded inclusion $\ell^1(\mathbb{N}, K_1) \rightarrow c_0(\mathbb{N}, K_1)$ in place of a quotient map.

Corollary 35 (finite dimension is sharp for nonarchimedean bases). *Let $K = \mathbb{F}_q((u))$, where q is a power of p , with a u -adic absolute value, and let $k \geq p$. Let P be a K -Banach space admitting an equivalent nonarchimedean norm. Then the conclusion of Theorem 22 holds over every analytic manifold modeled on P , with arbitrary normed fibers, if and only if P is finite-dimensional.*

Proof. If $\dim P < \infty$, Lemma 11 supplies continuous coordinates and Theorem 22 applies. Conversely, suppose $\dim P = \infty$. Remark 90 allows us to apply Theorem 32 with $\kappa = \mathbb{F}_q$ and $K_1 = K$. By Serre's theorem cited in Remark 95, P is topologically isomorphic to $c_0(I, K)$ for an infinite set I . Choosing a countable subset of I gives bounded linear maps $\iota: c_0(\mathbb{N}, K) \rightarrow P$ and $\pi: P \rightarrow c_0(\mathbb{N}, K)$ with $\pi\iota = \text{id}$. On the open set $V = \{x \in P : \|\pi x\| < 1\}$, take the bundle with fiber E_1 and transition $D_{1+\pi(x)}$, as in Corollary 34, together with the trivial bundle with fiber B . If the induced transition were analytic at $x_0 \in V$, restriction to the affine subspace $x_0 + \iota(c_0(\mathbb{N}, K))$ would make $b \mapsto Q_{E_1, E_1, B}^K(D_{1+\pi(x_0)} + D_b)$ analytic at zero. This contradicts Proposition 33 with $u_0 = D_{1+\pi(x_0)}$. $\square$

The ordinary sum norm on ℓ^1 is not equivalent to a nonarchimedean norm, so Corollary 26 does not conflict with this dichotomy. Over $\mathbb{F}_q((u))$, every counterexample must also use a target fiber outside $\mathsf{T}_k(K)$; by Proposition 47, a Banach target fiber must then admit no equivalent nonarchimedean norm. Thus the obstruction does not occur among bundles all of whose target fibers are nonarchimedean Banach spaces.

Remark 36 (pullback of differential forms). The same computation shows that pullback along analytic maps does not preserve analytic B -valued forms on infinite-dimensional bases. Let $f: E_1 \times E_1 \rightarrow E_1$ be the bounded bilinear map $f(x, y) = xy$, and let $\omega_0 \equiv W_B$ be the constant B -valued k -form on E_1 . At (x, y) , the restriction of $f^*\omega_0 = W_B \circ (Df, \dots, Df)$ to vectors in $\{0\} \times E_1$ is $\sigma(x)$, because $Df(x, y)(0, h) = xh$. By Proposition 33 and the lift criterion at points of E_1 , the form $f^*\omega_0$ is analytic at no point of $E_1 \times E_1$. In the same way, the chart change $(x, y) \mapsto ((1+y)x, y)$ on $\{(x, y) : \|y\| < 1\}$ has derivative $\begin{pmatrix} D_{1+y} & D_x \\ 0 & \text{id} \end{pmatrix}$, which is the shear with $u = D_x$ along $y = 0$. Hence the bundle of B -valued differential k -forms on this open subset of $E_1 \times E_1$ is not analytic for the atlas induced by the analytic structure. For scalar-valued forms the target K is spherically complete when $K = \mathbb{F}_q((u))$, so Main Theorem (2) makes the bundle of differential k -forms analytic over every base.

6. THE LARGEST ANALYTIC DOMAIN

The analytic cases of Section 3 in degrees with $k! = 0$ in K rest on hypotheses on objects: finite continuous coordinates on a source, or a spherically complete nonarchimedean target. This suggests restricting the objects on which the alternating-map bifunctor acts. There is, however, an obstruction to a single best restriction: two objects can each have analytic endomorphism action while the action on morphisms between them is not analytic.

The bundle problem suggests the same restriction of objects. By Section 5, when $k! = 0$ in K the analytic construction of alternating-map bundles cannot work for all fibers over all bases. Section 4 restricted the base. Here the base stays arbitrary, and we ask for the largest class of fiber pairs on which Alt^k is analytic.

We define analytic domains and record their bundles (Corollary 38). The targets F for which $Q_{E,D,F}^K$ is analytic for all normed E, D form a canonical core compatible with every analytic domain (Proposition 39). Within any set of objects, every analytic domain extends to a maximal one (Proposition 40), and pairs split in the sense of Definition 41 form a further analytic domain (Lemma 42). A largest analytic domain need not exist: for every prime p and every $k \geq p$, over $K = \mathbb{F}_p(t)$ with its t -adic absolute value, there is no largest full analytic domain (Theorem 43, with the construction in Appendix H). The section ends with what is known about universal targets (§6.3).

Definition 37. A full subcategory $\mathcal{D} \subseteq \text{Vec}_K^{\text{op}} \times \text{Vec}_K$ is an *analytic domain* for Alt^k if the restriction of Alt^k to $\mathcal{D}$ is analytic on every hom space. Let $\mathsf{T}_k(K)$ be the full subcategory of normed spaces F such that $Q_{E,D,F}^K$ is analytic for every pair of normed spaces E, D .

Corollary 38 (bundles with fibers in an analytic domain). *Let $\mathcal{D}$ be an analytic domain for Alt^k , and let M be an analytic manifold over K modeled on an arbitrary normed space. Consider the category whose objects are pairs $(\mathcal{E}, \mathcal{H})$ of analytic normed vector bundles over M whose typical fiber pair (E, F) is an object of $\mathcal{D}$. A morphism $(\mathcal{E}, \mathcal{H}) \rightarrow (\mathcal{E}', \mathcal{H}')$ is a pair of analytic bundle morphisms $\varphi: \mathcal{E}' \rightarrow \mathcal{E}$ and $\psi: \mathcal{H} \rightarrow \mathcal{H}'$; since $\mathcal{D}$ is full, its local expressions are analytic families of morphisms of $\mathcal{D}$ from (E, F) to (E', F') . On this category, fiberwise alternating maps define a functor $(\mathcal{E}, \mathcal{H}) \mapsto \text{Alt}^k(\mathcal{E}; \mathcal{H})$ with values in analytic normed vector bundles over M .*

Proof. Over a common trivializing cover, the transition maps g_{ij} of $\mathcal{E}$ and h_{ij} of $\mathcal{H}$ give an analytic family $x \mapsto (g_{ji}(x), h_{ij}(x))$ of endomorphisms of (E, F) in $\mathcal{D}$. Changes of trivialization and local expressions of morphisms are analytic families in hom spaces of $\mathcal{D}$ in the same way. The action of Alt^k on these hom spaces is analytic, so its composites with all these families are analytic. This is the hypothesis of Proposition 5(2), the form of part (1) that requires regularity only along the families that occur, and it gives the bundle $\text{Alt}^k(\mathcal{E}; \mathcal{H})$ together with its functoriality. $\square$

Corollary 15 is the special case of Corollary 38 for two analytic domains. The first is the core $\text{Vec}_K^{\text{op}} \times \mathbb{T}_k(K)$ of Proposition 39 below, whose target class contains the target fibers of the factorial and spherically complete cases. The second consists of the pairs whose source has a finite algebraic basis with continuous coordinate functionals; it is an analytic domain by Proposition 10 and (1.2). A largest analytic domain, if it existed, would carry the largest such bundle construction valid over every analytic base. This motivates the question studied next.

6.1. A core compatible with every analytic domain. The definition of an analytic domain includes morphisms between distinct objects. Requiring only that each endomorphism action be analytic is strictly weaker, as Theorem 43 will show.

Proposition 39 (The universal-target core). *The full subcategory $\text{Vec}_K^{\text{op}} \times \mathbb{T}_k(K)$ is an analytic domain. It is the largest analytic domain of the form $\text{Vec}_K^{\text{op}} \times \mathcal{T}$. Moreover, adjoining all its objects to any analytic domain again gives an analytic domain. The category $\mathbb{T}_k(K)$ is closed under bounded linear isomorphisms, bounded linear retracts, and finite products.*

Proof. Consider a morphism from (E, F) to (D, G) , represented by $u: D \rightarrow E$ and $v: F \rightarrow G$. Writing v_* for postcomposition, its action has the two factorizations

$$(6.1) \quad \text{Alt}^k(u, v) = v_* \circ Q_{D,E,F}^K(u) = Q_{D,E,G}^K(u) \circ v_*.$$

Operator composition is bounded bilinear. The first expression is therefore jointly analytic if $F \in \mathbb{T}_k(K)$, and the second is jointly analytic if $G \in \mathbb{T}_k(K)$. Thus a universally analytic target at either endpoint suffices. This proves both analyticity of the core and its compatibility with every analytic domain. Conversely, analyticity on a product with unrestricted first variable, restricted to $v = \text{id}_F$, is exactly the defining property of $\mathbb{T}_k(K)$.

Suppose $i: F \rightarrow G$ and $r: G \rightarrow F$ are bounded linear maps with $ri = \text{id}_F$. Then

$$Q_{E,D,F}^K(u) = r_* \circ Q_{E,D,G}^K(u) \circ i_*.$$

Consequently the property passes from G to its retract F , and in particular is invariant under bounded linear isomorphism. For a finite product of targets, alternating maps identify boundedly with the product of the corresponding alternating-map spaces; under this identification, pullback acts componentwise. This proves closure under finite products. $\square$

Proposition 40 (Compatibility and maximal domains). *Fix a set of objects in the ambient product category. Form a graph whose vertices are those objects with analytic endomorphism action, joining two distinct vertices when both directed morphism-space actions are analytic. Full analytic domains are exactly the cliques in this graph. Every analytic domain extends to one maximal under inclusion, and every maximal domain contains every object of the universal-target core that belongs to the chosen set.*

Proof. The clique condition states exactly that every hom-space action in the full subcategory is analytic. The union of a chain of cliques is again a clique: any two of its vertices occur together in one member of the chain. Zorn's lemma gives a maximal extension. Proposition 39 permits adjoining every core object, so maximality forces each such object to be present. $\square$

This description distinguishes existence of maximal restrictions from existence of a largest one. The counterexample below exhibits two vertices that are not joined. Thus no largest clique exists in any chosen ambient set containing both constructed objects. The set-of-objects qualification makes the use of Zorn's lemma explicit; no choice principle for a proper class of objects is being asserted.

The core is canonical, but its definition is a regularity condition rather than an intrinsic classification of targets. Main Theorem (2) supplies many of its objects. A different sufficient condition depends on the source and target together.

Definition 41. A pair (E, F) is *split in degree k* if the isometric inclusion

$$j_{E,F}: \text{Alt}^k(E; F) \longrightarrow \text{Mult}^k(E; F)$$

has a bounded linear retraction.

Lemma 42. *Every morphism-space action whose destination is a split pair is analytic. In particular, the full subcategory of split pairs is an analytic domain.*

Proof. Let (E, F) be split, with retraction R , and let the other object be (D, G) . For $u: E \rightarrow D$ and $v: G \rightarrow F$, the action is

$$m \longmapsto R(v \circ m \circ (u, \dots, u)).$$

Before applying R , replace the k copies of u by independent variables. This gives a bounded $(k+1)$ -linear lift of the joint action, with values in $\mathcal{L}(\text{Alt}^k(D; G), \text{Mult}^k(E; F))$. Postcomposition with R gives a lift with the required alternating codomain. No compatibility between the retractions chosen for different objects is needed. $\square$

6.2. Two individually analytic objects with nonanalytic cross-action.

Theorem 43 (No largest full analytic domain). *Let p be prime, let $k \geq p$, and equip $K = \mathbb{F}_p(t)$ with its t -adic absolute value. There is no largest full analytic domain for*

$$\text{Alt}^k: \text{Vec}_K^{\text{op}} \times \text{Vec}_K \longrightarrow \text{Vec}_K.$$

The assertion remains true if the domains must be closed under isomorphisms, and if their objects are restricted to pairs of finite-dimensional, nonarchimedean normed K -spaces. In the latter case, isomorphism closure is understood inside that restricted ambient category.

More precisely, there are objects $X_k = (E_k, G)$ and $Y_k = (D_k, G)$ whose singleton full subcategories are analytic, such that Y_k is split and the action on morphisms from Y_k to X_k is not analytic.

Here finite-dimensional means finite *algebraic* dimension over the incomplete field K . These spaces need not admit continuous coordinates over K . The theorem does not give a counterexample over a complete field, or a counterexample with Banach objects.

The proof of Theorem 43 is given in Appendix H. We outline it and fix the objects used in the remarks below. Let $L = \mathbb{F}_p((t))$ be the completion of K , and let $A = L[\epsilon]/(\epsilon^p)$ carry the maximum norm in the basis $e_i = \epsilon^i$ ($0 \leq i < p$). Choose $a = \sum_i a_i e_i$ with $a_0, \dots, a_{p-1} \in L$ such that these scalars and finitely many auxiliary scalars $\tau_w \in L$ are algebraically independent over K . Then $E = K^p + Ka \subseteq A$ is a K -space of dimension $p+1$ with completion A . By Lemma 91 it is rigid: its bounded endomorphisms are the scalars in K , and it has no nonzero bounded linear functional. Let $D \subseteq A$ be spanned by K^p , the vectors $\epsilon^i a$ and a^2 , and the generators $\tau_w w$, where w runs over the vectors e_i and $e_i + e_j$ with $i < j$. Let $G \subseteq L$ be spanned by the determinants, in the basis (e_i) , of p -tuples from D ; the completions of D and G are A and L . By rigidity, the self-action of $X = (E, G)$ is $(s, v) \mapsto s^p v_*$, so its singleton is analytic. The

generators $\tau_w w$ make every bounded p -linear map $D^p \rightarrow G$ alternating, so $Y = (D, G)$ is split (Lemma 92) and its singleton is analytic by Lemma 42.

Multiplication by $x \in E$ in A gives a bounded linear family $E \rightarrow \mathcal{L}(E, D)$, $x \mapsto M_x$, with $\det M_x = x_0^p$, where x_0 is the e_0 -coefficient of x . Identify $\text{Alt}^p(E; G)$ with $C \det$, where $C = \{\lambda \in G : a_i \lambda \in G \text{ for every } i\}$, as in (H.2). Through this identification, the cross-action from Y to X along $x \mapsto M_x$, with target morphism id_G and evaluated at $\det$, is the coordinate map $x \mapsto x_0^p$ of (H.3). This map has no bounded p -linear lift, because a coefficient is missing: a lift would place in C an element whose specialization at $\tau_w = 0$ has a nonzero homogeneous component of degree one in the a_i , and Lemma 93 shows that the specialized coefficient space has no such component. Analyticity at zero would produce a lift by the homogeneous coefficient argument of Proposition 3, so the cross-action is not analytic.

For $k = p + r$ we pad by direct sums: $E_k = E \oplus K^r$ and $D_k = D \oplus K^r$, the bounded affine family $u(x) = M_x \oplus I_r$ in $\mathcal{L}(E_k, D_k)$, and $\det_k$ in place of $\det$. This gives $X_k = (E_k, G)$ and $Y_k = (D_k, G)$ with the same properties. Theorem 19 does not apply to the family u : its parameter space E has finite algebraic dimension, but no nonzero bounded linear functional and hence no continuous coordinates.

Remark 44. The theorem also shows that X_k cannot be adjoined to the entire split-pair category, since that category contains Y_k . It does not establish that the split-pair category has no other strict analytic extension. Nor does it resolve the largest-domain question for complete fields or Banach pairs. Completing this particular construction removes the obstruction: the sources become finite-dimensional L -spaces, the target becomes L , and the coordinate polynomial acquires a bounded multilinear lift.

Remark 45 (finite algebraic dimension does not suffice for bundles). The same objects show that the coordinate hypothesis in Theorem 22 cannot be weakened to finite algebraic dimension over an incomplete field. Take $M = E$, of dimension $p + 1$ over $K = \mathbb{F}_p(t)$, as an analytic manifold with one chart; by Lemma 91 it has no nonzero bounded linear functional. Let the fiber be $H = D_k \times E_k$ with two global trivializations differing by the shear $g(x) = \begin{pmatrix} \text{id} & u(x) \\ 0 & \text{id} \end{pmatrix}$, where $u(x) = M_x \oplus I_r$ is the bounded affine family of Appendix H, and take the trivial target bundle with fiber G . The transitions $g(x)$ and $g(x)^{-1} = \begin{pmatrix} \text{id} & -u(x) \\ 0 & \text{id} \end{pmatrix}$ are affine analytic. By (5.1), compressing the induced pullback recovers $Q_{E_k, D_k, G}^K(u(x))$, and evaluation at $\det_k$ followed by coefficient extraction gives the map (H.3), which is not analytic at zero. Hence the alternating-map bundle is not analytic, although the base and all fibers have finite algebraic dimension. Over a complete field this cannot happen, by Lemma 11.

6.3. Universal targets. By Proposition 39, the class $\mathsf{T}_k(K)$ of Definition 37 determines a canonical analytic domain, $\text{Vec}_K^{\text{op}} \times \mathsf{T}_k(K)$, whose objects can be adjoined to every other analytic domain. The search for large analytic domains therefore leads to the question of which targets are universal. We collect what is known.

If $k! \neq 0$ in K , then $\mathsf{T}_k(K)$ contains every normed space (Proposition 9). If K has positive characteristic, Main Theorem (2) places every nonarchimedean, spherically complete space in $\mathsf{T}_k(K)$ for every k . Since $\mathsf{T}_k(K)$ is closed under bounded linear isomorphisms, retracts and finite products (Proposition 39), it then contains every space that admits an equivalent nonarchimedean, spherically complete norm, and every bounded linear retract of such a space. If K has characteristic $p > 0$ and $k \geq p$, then $\mathsf{T}_k(K)$ does not contain the Banach target of Theorem 28, which admits no equivalent nonarchimedean norm.

Remark 46 and Proposition 47 below treat finite-dimensional targets and targets over discretely valued fields. In terms of universal targets, they give the following. Over a complete field, a nonzero finite-dimensional space lies in $\mathsf{T}_k(K)$ exactly when the scalar target K does;

if K is moreover nonarchimedean and spherically complete, every finite-dimensional space lies in $\mathsf{T}_k(K)$. Over a nonarchimedean field with $|K^\times| = r^\mathbb{Z}$ for some $r \in (0, 1)$, complete or not, every Banach space that admits an equivalent nonarchimedean norm lies in $\mathsf{T}_k(K)$ for every k . Remark 94 in Appendix I shows that spherical completeness of the given norm is not necessary for membership. Its target still admits an equivalent spherically complete norm, so the example does not decide whether such a norm is necessary. Section 7 asks for an intrinsic characterization of $\mathsf{T}_k(K)$.

Remark 46 (finite-dimensional targets). Let K be a complete nontrivially normed field, let E, E' be normed K -spaces, and let F be finite-dimensional. If $F \neq 0$, then, at every $f_0 \in \mathcal{L}(E, E')$,

$$Q_{E, E', F}^K \text{ is analytic at } f_0 \iff Q_{E, E', K}^K \text{ is analytic at } f_0.$$

If $F = 0$, precomposition is constant and hence analytic.

Indeed, choose a basis $v_1, \dots, v_d$ of F with coordinate functionals $\lambda_1, \dots, \lambda_d$. These functionals are continuous because K is complete. For a vector v and a scalar-valued form a , write va for the form $x \mapsto a(x)v$. Writing $Q_F := Q_{E, E', F}^K$ and $Q_K := Q_{E, E', K}^K$, we have

$$Q_F(f)(m) = \sum_{a=1}^d v_a Q_K(f)(\lambda_a \circ m).$$

The right-hand side is a fixed continuous linear function of the operator $Q_K(f)$, so analyticity of Q_K implies that of Q_F . Conversely, choose $v \in F$ and $\lambda \in \mathcal{L}(F, K)$ with $\lambda(v) = 1$. For $a \in \text{Alt}^k(E'; K)$,

$$Q_K(f)(a) = \lambda \circ Q_F(f)(va).$$

This expresses Q_K as a fixed continuous linear function of Q_F and proves the converse. Thus finite-dimensionality of the target reduces the question to a scalar target. This reduction gives no assertion of analyticity over a nonspherically complete base.

If K is nonarchimedean and spherically complete, there is a positive conclusion for every finite-dimensional F , with any norm. The coordinate norm $\|y\|_\infty := \max_a |\lambda_a(y)|$ is equivalent to the given norm. With this norm, F is spherically complete: a closed ball in K^d is a product of coordinate balls, and spherical completeness of K gives a point in each coordinate intersection. The Main Theorem therefore applies. Returning to the original norm preserves analyticity, since it changes the norms on the alternating-map and operator spaces only by fixed comparison factors.

Proposition 47 (targets over discretely valued bases). *Let K be a nonarchimedean normed field with $|K^\times| = r^\mathbb{Z}$ for some $r \in (0, 1)$, complete or not; for instance $K = \mathbb{F}_q((t))$, or $K = \kappa((t))$ for an arbitrary field κ . Let E, E' be normed K -spaces and F a K -Banach space that admits an equivalent nonarchimedean norm. Then $Q_{E, E', F}^K$ is analytic at every point.*

The proof is given in Appendix I.

Thus, over a discretely valued base, every Banach counterexample must have a target admitting no equivalent nonarchimedean norm. Proposition 88 establishes this property for the target of Theorem 28 over every base.

7. OPEN PROBLEMS

The distinction between restrictions on objects and restrictions on analytic families leaves several different classification problems. They are substantive only in characteristic $p > 0$ and degree $k \geq p$; when $k! \neq 0$, the alternating-map bifunctor is already analytic on the whole product category. For C^∞ smoothness, no restriction is needed in any characteristic.

- (1) *Intrinsic characterization of universal targets.* Determine $T_k(K)$ in terms of the topological linear structure of its objects, rather than by restating analyticity or the multilinear lift criterion. In particular, characterize its Banach objects and the targets that belong to $T_k(K)$ in every finite degree. By §6.3, an equivalent nonarchimedean, spherically complete norm is sufficient, and spherical completeness of the given norm is not necessary; the example there does not decide whether such an equivalent norm is necessary.
- (2) *Analytic domains over complete fields and among Banach pairs.* For a prescribed complete field K , does there exist a largest full analytic domain, closed under isomorphisms, in $\text{Vec}_K^{\text{op}} \times \text{Vec}_K$? Ask the same question in $\text{Ban}_K^{\text{op}} \times \text{Ban}_K$, also when K itself is incomplete. Theorem 43 answers the unrestricted existence question negatively over $\mathbb{F}_p(t)$, using finite-dimensional but incomplete spaces. Its construction does not decide either of these restricted questions. Where no largest domain exists, seek structural criteria for compatibility between objects and descriptions of full domains maximal under inclusion.
- (3) *Extensions of the split-pair category.* Which objects can be adjoined to the full category of pairs for which $\text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$ has a bounded linear retraction? The universal-target core can always be adjoined, by Proposition 39. The objects X_k of Theorem 43, despite having analytic self-action, cannot be adjoined. A useful answer should give sufficient and necessary conditions on the spaces themselves, rather than merely requiring analyticity of all the cross-actions.
- (4) *Parameter spaces for analytic families.* Characterize normed parameter spaces P for which every operator-valued analytic family with parameters in an open subset of P induces an analytic family on alternating-map spaces. The hypotheses on the target fibers should be recorded explicitly: completeness and nonarchimedean norms enter some of the positive results. General reflection of analyticity through closed linear embeddings is a sufficient condition, but it is stronger than the property needed for this particular finite-degree construction. The ordinary ℓ^1 parameter theorem makes this distinction concrete. The answer depends on the admitted target fibers: by Corollaries 24 and 34, the space $c_0(\mathbb{N}, \mathbb{F}_q((u)))$ qualifies for complete nonarchimedean final targets but not for all complete targets.
- (5) *Countable index.* Let K be complete, not spherically complete, of characteristic $p \geq 3$, and let $k \geq p$. Is there a complete nonarchimedean F such that $Q_{E, E', F}^K$ is not analytic for $E = \ell^\infty(\mathbb{N}, K)$? For $p = 2$, the answer is yes: degree two is the theorem of [7], and the higher degrees follow by the wedging argument of [7, Remark 18], formalized under hypothesis (K2) of that paper (Remark 77). Remark 96 in Appendix I proposes a counterexample with a larger index set; that extension and the countable case remain open.
- (6) *Densely valued spherically complete bases.* Let K be spherically complete, densely valued and of characteristic p , for example $\mathbb{F}_p((t^\mathbb{Q}))$, and let $k \geq p$. Is $Q_{E, E', F}^K$ analytic for all Banach E, E' and every complete nonarchimedean F ? Main Theorem (2) gives a positive answer for spherically complete F . The counterexample of Theorem 28 has a target with no equivalent nonarchimedean norm and therefore does not answer this question.

APPENDIX A. PROOFS FOR THE LIFT CRITERION AND FINITE COORDINATES

This appendix proves the lift criterion of Section 2 (Proposition 3) from the one-variable uniqueness of power-series coefficients (Lemma 48), so the argument uses no polarization formula, no division by $k!$ and no completeness assumption. It then gives the explicit determinant lifts

of Proposition 10 from Section 3, for spaces with a finite algebraic basis whose coordinate functionals are continuous; these lifts are finite sums and are valid in every characteristic.

Lemma 48 (one-variable uniqueness). *Let K be nontrivially normed and Z a normed K -space, not necessarily complete. Let $z_0, z_1, \dots \in Z$, $\rho > 0$ and $M \geq 0$ with $\|z_n\|\rho^n \leq M$ for all n . If $\sum_n t^n z_n$ converges to 0 for every $t \in K$ with $|t| < \rho$, then $z_n = 0$ for all n .*

Proof. Otherwise let N be least with $z_N \neq 0$. For $0 < |t| < \rho/2$ the series $\sum_{n>N} t^{n-N} z_n$ converges, being t^{-N} times the convergent tail $\sum_{n>N} t^n z_n$, and $t^N(z_N + \sum_{n>N} t^{n-N} z_n) = 0$. Hence

$$\|z_N\| = \left\| \sum_{n>N} t^{n-N} z_n \right\| \leq \sum_{n>N} M \rho^{-n} |t|^{n-N} = M \rho^{-N} \frac{|t|/\rho}{1 - |t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since K is nontrivially normed, there are $t \neq 0$ with $|t|$ arbitrarily small, so $\|z_N\| = 0$, a contradiction. $\square$

Proof of Proposition 3. (1) For $y \in \mathcal{L}(E, E')$, multilinearity gives

$$Q(f_0 + y) = P(f_0 + y, \dots, f_0 + y) = \sum_{r=0}^k q_r(y, \dots, y),$$

where $q_r(y_1, \dots, y_r)$ is the sum, over the r -element subsets $S \subseteq \{1, \dots, k\}$, of P evaluated with $y_1, \dots, y_r$ in the slots of S in increasing order and f_0 in the other slots. Each q_r is bounded r -linear, with $\|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}$, so $(q_r)_{r \leq k}$ is a power series of infinite radius for Q at f_0 .

(2) We compare the given power series along each line with the polynomial obtained by expanding precomposition. The expansion is taken in the larger space of all multilinear maps, where each coefficient is visibly defined. Let $\text{incl}: \text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$ be the isometric inclusion, and put

$$Z := \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)), \quad \mathcal{I}(T) := \text{incl} \circ T.$$

Then $\mathcal{I}$ is a linear isometry into Z , hence injective. Fix $h \in \mathcal{L}(E, E')$, $h \neq 0$. For $m \in \text{Alt}^k(E'; F)$, $x \in E^k$ and $t \in K$, multilinearity of m gives the *ambient polynomial identity*

$$(A.1) \quad \mathcal{I}(Q(f_0 + th)) = \sum_{r=0}^k t^r A_r(h), \quad A_r(h)(m)(x) := \sum_{|S|=r} m(u_1^S, \dots, u_k^S),$$

where $u_i^S = hx_i$ for $i \in S$ and $u_i^S = f_0 x_i$ for $i \notin S$. Here $A_r(h) \in Z$ is bounded, with $\|A_r(h)\| \leq \binom{k}{r} \|h\|^r \|f_0\|^{k-r}$, and $A_k(h) = \mathcal{I}(Q(h))$. (For $r < k$ we do not need to know whether $A_r(h)(m)$ is alternating; this is why (A.1) is written in Z .) Let R and $M' := \sup_n \|p_n\| R^n$ be as in the definition of analyticity, and put $\rho := R/\|h\|$. For $|t| < \rho$ we have $\|th\| < R$, so $Q(f_0 + th) = \sum_n t^n p_n(h, \dots, h)$, and applying $\mathcal{I}$ and (A.1) gives $\sum_n t^n z_n = 0$ in Z , with $z_n := \mathcal{I}(p_n(h, \dots, h)) - A_n(h)$ and $A_n(h) := 0$ for $n > k$. Since $\|z_n\|\rho^n \leq M' + \max_{r \leq k} \|A_r(h)\|\rho^r$, Lemma 48 gives $z_k = 0$, that is, $\mathcal{I}(p_k(h, \dots, h)) = \mathcal{I}(Q(h))$, hence $p_k(h, \dots, h) = Q(h)$. For $h = 0$ both sides vanish since $k \geq 1$.

(3) follows from (1) and (2), and from `ContDiffAt.analyticAt` for the last clause. $\square$

Proof of Proposition 10. Let H denote the space with a finite coordinate basis, write that basis as $e_1, \dots, e_n$, and let $\varepsilon_1, \dots, \varepsilon_n \in \mathcal{L}_K(H, K)$ be its coordinate functionals. For an increasing

k -tuple $s = (s_1 < \dots < s_k)$ in $\{1, \dots, n\}$, put

$$c_s := \prod_{a=1}^k \|\varepsilon_{s_a}\| \|e_{s_a}\|, \quad C_H := k! \sum_{s_1 < \dots < s_k} c_s.$$

Here the factor $k!$ in C_H is the positive real integer counting the terms in a determinant, not its image in K . In particular, $C_H \geq 0$ even when $k! = 0$ in K .

Suppose first that $H = E$. For $f_1, \dots, f_k \in \mathcal{L}_K(E, E')$, $m \in \text{Alt}_K^k(E'; F)$ and $x \in E^k$, define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

The determinant makes this expression alternating and multilinear in x . It is linear in m and multilinear in $f_1, \dots, f_k$. Expanding the determinant over permutations gives

$$\left| \det(\varepsilon_{s_a}(x_b))_{a,b} \right| \leq k! \prod_a \|\varepsilon_{s_a}\| \prod_b \|x_b\|.$$

Together with the operator bound for m , the ordinary triangle inequality therefore yields

$$\|P(f_1, \dots, f_k)(m)(x)\| \leq C_E \|m\| \prod_a \|f_a\| \prod_b \|x_b\|.$$

Thus the displayed formula defines a bounded k -linear map into $\mathcal{L}_K(\text{Alt}_K^k(E'; F), \text{Alt}_K^k(E; F))$, with $\|P\| \leq C_E$. On the diagonal, the finite exterior-basis expansion of $x_1 \wedge \dots \wedge x_k$ gives

$$m(fx_1, \dots, fx_k) = \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k}),$$

so $P(f, \dots, f) = Q(f)$.

Now suppose that $H = E'$. Define instead

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(f_a x_b))_{a,b} m(e_{s_1}, \dots, e_{s_k}).$$

Each row of the determinant is linear in its corresponding operator f_a , and its columns make the result alternating and multilinear in x . The expression is again linear in m . The determinant estimate now reads

$$\left| \det(\varepsilon_{s_a}(f_a x_b))_{a,b} \right| \leq k! \prod_a (\|\varepsilon_{s_a}\| \|f_a\|) \prod_b \|x_b\|.$$

It follows that the same operator bound holds with $C_{E'}$ in place of C_E . Expanding $fx_1 \wedge \dots \wedge fx_k$ in the exterior basis of E' proves the diagonal identity.

Both formulas are finite sums. Strong alternation follows from equal columns in a determinant, so the proof works in every characteristic and involves no division by $k!$. Proposition 3 gives continuous polynomiality and analyticity. If $k > n$, the sums are empty and the relevant alternating-map space is zero; the same argument gives the zero lift. Degree zero, excluded from the lift criterion's statement, is immediate because Q is constant. $\square$

APPENDIX B. SPHERICALLY COMPLETE TARGETS

This appendix proves Main Theorem (2), following the strategy of §3.5. Throughout, K is a nonarchimedean nontrivially normed field, E and E' are normed K -spaces, and F is a nonarchimedean, spherically complete normed K -space; for $k \geq 1$ we keep the notation $A = \text{Alt}^k(E; F)$ and $M = \text{Mult}^k(E; F)$. We prove an extension lemma with simultaneous

bounds (Lemma 49), deduce Proposition 14 from it, and apply it once more to construct a contraction $\rho: M \rightarrow A$ that fixes every alternating map, which gives the lift described in §3.5.

B.1. Extension with simultaneous bounds. The following form of the nonarchimedean extension argument keeps a prospective extension in several operator balls at once. We allow a seminorm on the domain, since the free space used in the next step may have nonzero vectors of seminorm zero.

Lemma 49 (compatible extension). *Let U be a nonarchimedean seminormed K -space and let G be a nonarchimedean, spherically complete normed K -space. Suppose a nonempty family (T_α, r_α) , with $T_\alpha \in \mathcal{L}(U, G)$ and $r_\alpha \geq 0$, satisfies*

$$\|T_\alpha - T_\beta\| \leq \max(r_\alpha, r_\beta) \quad \text{for all } \alpha, \beta.$$

Let $D \subseteq U$ be a linear subspace and let $T_0: D \rightarrow G$ be linear with

$$\|T_0 d - T_\alpha d\| \leq r_\alpha \|d\| \quad (d \in D, \text{ all } \alpha).$$

Then T_0 extends to a bounded linear map $T: U \rightarrow G$ such that $\|T - T_\alpha\| \leq r_\alpha$ for every α . In particular, every contraction from a subspace of U to G extends to a contraction on U .

Proof. It suffices to extend across one vector $x \notin D$ while preserving every bound. The required value Tx must belong to the balls

$$(B.1) \quad \overline{B}(T_\alpha(d+x) - T_0 d, r_\alpha \|d+x\|) \quad (\alpha, d \in D).$$

We check that these balls meet pairwise. Write $c_{\alpha,d} = T_\alpha(d+x) - T_0 d$, and suppose $r_\alpha \leq r_\beta$. Then

$$c_{\alpha,d} - c_{\beta,e} = T_\alpha(d-e) - T_0(d-e) + (T_\alpha - T_\beta)(e+x).$$

Since $d-e \in D$ and $\|d-e\| \leq \max(\|d+x\|, \|e+x\|)$, the hypotheses give

$$\begin{aligned} \|c_{\alpha,d} - c_{\beta,e}\| &\leq \max(r_\alpha \|d-e\|, r_\beta \|e+x\|) \\ &\leq \max(r_\alpha \|d+x\|, r_\beta \|e+x\|). \end{aligned}$$

This is exactly the condition for the two balls to meet. The other ordering of the radii is symmetric.

By spherical completeness, choose c in all the balls (B.1). Define

$$T(d + \lambda x) = T_0 d + \lambda c.$$

The decomposition is unique because $x \notin D$. For $\lambda \neq 0$, apply the ball condition with $\lambda^{-1}d$ in place of d and multiply by $|\lambda|$; this gives $\|T(d + \lambda x) - T_\alpha(d + \lambda x)\| \leq r_\alpha \|d + \lambda x\|$. For $\lambda = 0$ the original bound applies.

Partially order the extensions satisfying all these inequalities by inclusion of their domains. Unions of chains remain linear and preserve the bounds, so Zorn's lemma gives a maximal extension. The one-vector argument forces its domain to be all of U . Fixing any one index α gives $\|Tu\| \leq \max(\|T_\alpha\|, r_\alpha)\|u\|$, so the extension is bounded. For the last assertion, take the single pair $(T_\alpha, r_\alpha) = (0, 1)$. $\square$

B.2. Spherical completeness of alternating-map spaces. Ordinary completeness alone would not let us apply the extension lemma with target A . We prove the required stronger property by presenting alternating maps as linear maps on a free space subject to linear relations. The nonarchimedean seminorm on that free space is chosen so that this presentation preserves the operator norm, even when E itself is not nonarchimedean.

Proof of Proposition 14. For $k = 0$, the space identifies isometrically with F . Suppose $k \geq 1$. Let U be the free K -vector space on the set E^k , with generators $[x]$ and seminorm

$$\left\| \sum_x c_x [x] \right\|_U = \max_x |c_x| \prod_{i=1}^k \|x_i\|.$$

The maximum is over the finite support, with empty maximum zero. This seminorm is nonarchimedean because the absolute value on K is nonarchimedean. Let $R \subseteq U$ be the linear span of the relations expressing additivity and scalar homogeneity in each slot, together with $[x]$ whenever two coordinates of x coincide.

An alternating map $a \in A$ defines a linear map vanishing on R by

$$\hat{a}\left(\sum_x c_x [x]\right) = \sum_x c_x a(x).$$

The nonarchimedean inequality in F gives

$$\left\| \hat{a}\left(\sum_x c_x [x]\right) \right\| \leq \max_x |c_x| \|a(x)\| \leq \|a\| \left\| \sum_x c_x [x] \right\|_U.$$

Conversely, a bounded linear map $T: U \rightarrow F$ vanishing on R defines an alternating map $a(x) = T([x])$ with $\|a(x)\| \leq \|T\| \prod_i \|x_i\|$. Thus these two constructions are inverse linear isometries. In particular, the equal-input relations impose strong alternation in every characteristic.

Take a nonempty family of pairwise-meeting closed balls in A , with centres a_α and radii r_α . Their corresponding linear maps satisfy $\|\hat{a}_\alpha - \hat{a}_\beta\| \leq \max(r_\alpha, r_\beta)$ and vanish on R . Apply Lemma 49 with $D = R$, $T_0 = 0$ and $G = F$. It produces a bounded linear map $T: U \rightarrow F$ that vanishes on R and satisfies $\|T - \hat{a}_\alpha\| \leq r_\alpha$ for every α . The associated alternating map belongs to every original ball. This proves spherical completeness; no closure assumption on R is needed. $\square$

B.3. The retraction and the analytic lift. We now apply the extension lemma a second time, using the alternating-map space itself as the target. This produces the projection needed to turn ordinary precomposition into a polynomial with alternating values.

Proof of Main Theorem (2). In degree zero, $\text{Alt}^0(-; F)$ is naturally F and the bifunctor acts by the pushforward morphism, so the conclusion holds. Assume $k \geq 1$ and retain $A = \text{Alt}^k(E; F)$ and $M = \text{Mult}^k(E; F)$. Their operator norms are nonarchimedean, and A is spherically complete by Proposition 14. The inclusion $j: A \rightarrow M$ is a linear isometry. Its inverse on $j(A)$ is therefore a contraction. Lemma 49 extends this inverse to a contraction $\rho: M \rightarrow A$ with $\rho j = \text{id}_A$.

For $f_1, \dots, f_k \in \mathcal{L}(E, E')$ and $m \in \text{Alt}^k(E'; F)$, define

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k)).$$

This is linear in m and multilinear in the f_i . The estimate

$$\|P(f_1, \dots, f_k)(m)\| \leq \|m\| \prod_{i=1}^k \|f_i\|$$

shows that P is a bounded k -linear map into $\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$, with $\|P\| \leq 1$. On the diagonal, $m \circ (f, \dots, f)$ is already alternating, so $P(f, \dots, f) = Q(f)$.

For completeness, expand at an arbitrary $f_0 \in \mathcal{L}(E, E')$. For $0 \leq r \leq k$, define a bounded r -linear map q_r by summing over the r -element subsets S of $\{1, \dots, k\}$: insert its r arguments

in the slots of S , in increasing order, and put f_0 in all remaining slots of P . Multilinearity gives the finite identity

$$Q(f_0 + h) = \sum_{r=0}^k q_r(h, \dots, h), \quad \|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}.$$

The binomial coefficient in the bound is an ordinary nonnegative integer counting the summands. With $q_r = 0$ for $r > k$, this is a power series of infinite radius at every f_0 . Consequently Q is continuously polynomial, hence C^ω and C^n for all $n \in \mathbb{N} \cup \{\infty\}$. Equation (1.2) makes the full bifunctor analytic on hom spaces, and Proposition 14 shows that its values are nonarchimedean and spherically complete. This proves the categorical statement. $\square$

The retraction supplies the alternating values directly, with no division by $k!$. The source spaces need no nonarchimedean hypothesis because the extension argument takes place in the free presentation and in operator spaces with nonarchimedean targets. By contrast, the Banach target in the counterexample theorem (Theorem 28) admits no equivalent nonarchimedean norm.

APPENDIX C. ANALYTIC REFLECTION BEYOND FINITE COORDINATES

This appendix supplements Section 4. We first prove the finite-coordinate reflection results, Lemma 18 and Theorem 19. We then isolate, for a general parameter space, the coefficient problem that the proof of Theorem 19 solves for K^d , and then prove the results on nonarchimedean c_0 parameters and ordinary ℓ^1 parameters (Theorems 23 and 25, Corollary 27). Finally, §C.4 gives a tensor criterion for universal analytic reflection: over a complete field, a Banach parameter space has this property exactly when the closed spans of pure tensor powers are ranges of projections whose norms grow at most exponentially (Theorem 54).

C.1. Finite-coordinate reflection. This subsection proves Lemma 18 and Theorem 19.

Proof of Lemma 18. For $n = 0$ this follows by evaluating at x_0 . Inductively, for each fixed x and sufficiently small nonzero $t \in K$, the vector

$$t^{-n} \left(f(x_0 + tx) - \sum_{r < n} t^r p_r(x, \dots, x) \right)$$

belongs to W . It tends to $p_n(x, \dots, x)$ as $t \rightarrow 0$. To see the convergence directly, choose $\rho > 0$ and M such that $\|p_r\| \rho^r \leq M$ for all r . The norm of the remaining tail is at most

$$M \left(\frac{\|x\|}{\rho} \right)^n \frac{|t| \|x\| / \rho}{1 - |t| \|x\| / \rho}$$

when $|t| \|x\| < \rho$. Nontriviality of the norm provides arbitrarily small nonzero t . Closedness now proves the induction step. $\square$

Proof of Theorem 19. Identify W with its image, use maximum-norm coordinates on K^d , and expand jf at a parameter point. Lemma 18 shows that every homogeneous polynomial $x \mapsto p_n(x, \dots, x)$ has its values in W . For $d \geq 1$ and $n \geq 1$, let $e_1, \dots, e_d$ be the coordinate vectors. For each multi-index $\alpha = (\alpha_1, \dots, \alpha_d)$ of total degree n , put

$$c_\alpha = \sum_{\substack{(i_1, \dots, i_n) \in \{1, \dots, d\}^n \\ \#\{r: i_r = j\} = \alpha_j \ (1 \leq j \leq d)}} p_n(e_{i_1}, \dots, e_{i_n}).$$

Then $p_n(x, \dots, x) = \sum_{|\alpha|=n} c_\alpha x^\alpha$. All c_α belong to W : pass to the algebraic quotient Z/W and use polynomial uniqueness over the infinite field K . This vector-valued uniqueness follows by applying scalar coordinate functionals on the finite-dimensional span of the coefficients; no topology on the quotient is needed.

Choose, for each α , the nondecreasing word $(i_1(\alpha), \dots, i_n(\alpha))$ with those multiplicities, and define the W -valued multilinear map

$$q_n(x_1, \dots, x_n) = \sum_{|\alpha|=n} c_\alpha \prod_{r=1}^n (x_r)_{i_r(\alpha)}.$$

Its diagonal agrees with that of p_n . Moreover,

$$(C.1) \quad \|q_n\| \leq \sum_{|\alpha|=n} \|c_\alpha\| \leq d^n \|p_n\|,$$

because the original expansion has exactly d^n ordered words. The degree-zero coefficient is $f(x_0)$. The bounds give a common positive radius for the q_n . Their partial sums have images under j converging to $jf(x)$, so the isometry shows that they converge to the already given vector $f(x)$ inside W . Completeness is therefore unnecessary. The case $d = 0$ is immediate, and continuous linear coordinates give the last assertion. $\square$

C.2. The exact local coefficient problem. The argument of Theorem 19 identifies what remains to be proved for a general parameter space. Here $\gamma: U \rightarrow H_{XY}$, the action A_{XY} and the closed isometric inclusion $j: W \hookrightarrow Z$ are as in Section 4.

Proposition 50. *Let γ be analytic near x_0 and choose an ambient expansion*

$$jA_{XY}\gamma(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h).$$

The induced map $A_{XY}\gamma$ is analytic at x_0 if and only if there are bounded n -linear maps q_n into W and $r > 0$ such that

$$jq_n(h, \dots, h) = p_n(h, \dots, h) \quad \text{for all } n, h, \quad \sup_n \|q_n\| r^n < \infty.$$

Proof. If the induced map is analytic at x_0 , take an expansion there in W , apply j , and compare the two expansions on every scalar line. Lemma 48 identifies their homogeneous diagonals. Conversely, shrink r to lie within the radius of the ambient expansion. The image under j of the partial sums of the proposed series converges to $jA_{XY}\gamma(x_0 + h)$, because its homogeneous diagonals agree term by term with the ambient series. Since j is an isometry, the partial sums converge inside W to $A_{XY}\gamma(x_0 + h)$. $\square$

Admissibility on an open parameter domain means that this criterion holds at every point. For an incomplete target, the existence of an expansion at a single point need not give analyticity on a neighborhood. It is therefore not necessary that the chosen maps p_n themselves take values in W away from their diagonals. Nor can the condition be replaced by ordinary derivative jets in positive characteristic: Frobenius terms have vanishing derivatives but need not vanish as homogeneous power-series terms. The relevant issue is a bounded multilinear representative for each homogeneous diagonal, with compatible growth across the degrees.

C.3. Proofs for c_0 and ℓ^1 parameters. Theorems 23 and 25 concern an arbitrary isometric inclusion $j: W \hookrightarrow Z$ and, over ℓ^1 , an arbitrary bounded multilinear map B ; Section 4 applies them to the inclusion forgetting alternation and to the ambient polynomial (4.1). The proofs below extend the sorted-word coefficient construction of Theorem 19 to infinitely many coordinates. Over $c_0(I, K)$ the nonarchimedean estimate of Remark 20 bounds the coefficients. Over $\ell^1(I, K)$ the fixed degree of the outer polynomial bounds them.

Proof of Theorem 23. The image of j is closed. Identify it with W and expand jj at a point of U . Lemma 18, followed by finite-coordinate polynomial uniqueness, puts every grouped coordinate coefficient of $p_n(x, \dots, x)$ in W . For $n \geq 1$, choose a total order on I . For a nondecreasing word $i_1 \leq \dots \leq i_n$, denote its grouped coefficient by $c_{i_1, \dots, i_n}$. It is a sum of values of p_n on coordinate vectors. The nonarchimedean inequality gives

$$\|c_{i_1, \dots, i_n}\| \leq \|p_n\|.$$

For $n \geq 1$, define

$$q_n(x_1, \dots, x_n) = \sum_{i_1 \leq \dots \leq i_n} c_{i_1, \dots, i_n}(x_1)_{i_1} \cdots (x_n)_{i_n}.$$

This sum converges in W . Indeed, outside a sufficiently large finite set of words at least one coordinate lies outside a chosen finite support approximation for the corresponding x_r . That factor is arbitrarily small, while all other factors are bounded. Thus the summands tend to zero outside finite sets, which is the summability condition in a complete nonarchimedean normed space. The same estimate gives

$$\|q_n(x_1, \dots, x_n)\| \leq \|p_n\| \prod_{r=1}^n \|x_r\|, \quad \|q_n\| \leq \|p_n\|.$$

Summation preserves multilinearity. On finitely supported vectors the diagonal is the original grouped polynomial, and continuity and density extend this equality to every vector in $c_0(I, K)$. Use $q_0 = f(x_0)$. The coefficient bounds preserve the ambient radius, and the isometry identifies the sum with f as in Theorem 19. $\square$

Lemma 51. *Let W be a closed subspace of a Banach space Z , and let $B: \ell^1(J, K)^d \rightarrow Z$ be bounded multilinear with diagonal values in W . Then its diagonal has a bounded W -valued d -linear representative C with*

$$\|C\| \leq d! \|B\|.$$

Here $d!$ denotes a real bound for a number of terms, not a scalar to be inverted in K .

Proof. For $d = 0$ the assertion is immediate. Otherwise, choose a total order on J and group the coordinate coefficients of the diagonal polynomial by nondecreasing words. Finite-coordinate polynomial uniqueness puts each grouped coefficient in W . Each group contains at most $d!$ ordered words, so its norm is at most $d! \|B\|$. Define C by the sorted-word formula used above. Its summands are absolutely summable, since

$$\sum_{j_1 \leq \dots \leq j_d} |(x_1)_{j_1} \cdots (x_d)_{j_d}| \leq \prod_{r=1}^d \|x_r\|_1.$$

The sum exists in the complete space W , is multilinear, and has the asserted bound. Diagonal equality follows first on finite supports and then by density. $\square$

Proof of Theorem 25. The case $d = 0$ is immediate. Extend B continuously to $\widehat{H}^d$, using completeness of Z . Since H is dense in its completion and W is closed, the diagonal of this extension takes values in W ; call it $\widehat{P}$. For every bounded linear $T: \ell^1(J, K) \rightarrow \widehat{H}$, Lemma 51,

applied to $B(T \cdot, \dots, T \cdot)$, gives a bounded multilinear representative for $\widehat{P}T$ of norm at most $d! \|B\| \|T\|^d$. In particular, $\widehat{P}T$ is analytic.

It remains to factor γ locally through one such T . Translate the parameter point to zero and write

$$\gamma(x) = \sum_{n \geq 0} a_n(x, \dots, x).$$

Choose $s \in K$ with $0 < |s|$ smaller than a radius of this expansion, so that $M = \sup_n |s|^n \|a_n\| < \infty$. Let $J = \coprod_{n \geq 0} I^n$, including the single empty word when $n = 0$. On the coordinate vectors of $\ell^1(J, K)$ set

$$T(e_{n, i_1, \dots, i_n}) = s^n a_n(e_{i_1}, \dots, e_{i_n}).$$

These vectors have norm at most M , so absolute summation in $\widehat{H}$ gives a bounded linear map $T: \ell^1(J, K) \rightarrow \widehat{H}$ with norm at most M .

For $\|x\|_1 < |s|$, define a vector in $\ell^1(J, K)$ by

$$g(x)_{n, i_1, \dots, i_n} = s^{-n} x_{i_1} \cdots x_{i_n}.$$

The sum of the absolute values of its degree- n coordinates is $|s|^{-n} \|x\|_1^n$. The geometric tail estimate proves that $g(x)$ belongs to $\ell^1(J, K)$ and that its series of homogeneous coordinate blocks converges there. Each block is the diagonal of a bounded n -linear map of norm at most $|s|^{-n}$, so g is analytic at zero. These are explicitly K -valued coordinates, and convergence is verified against the explicit vector $g(x)$; completeness of K is not being used.

Expanding each a_n in ℓ^1 coordinates gives

$$Tg(x) = \sum_{n \geq 0} a_n(x, \dots, x) = \gamma(x)$$

inside $\widehat{H}$. Consequently $P\gamma = (\widehat{P}T)g$ is analytic at the chosen point. Repeating the construction at every parameter point proves the theorem. The factorial estimate used the fixed outer degree d throughout, rather than the varying Taylor degree n . $\square$

Proof of Corollary 27. Choose the Banach counterexample E, F from Theorem 28. Let $J = \{h \in H : \|h\| \leq 1\}$ and define

$$q((a_h)_{h \in J}) = \sum_{h \in J} a_h h.$$

Absolute summability and completeness of H make this a bounded linear map of norm at most one. Fix $t \in K$ with $0 < |t| = \theta < 1$. For every nonzero $h \in H$, choose an integer m such that

$$\|h\| \leq |t^m| < \theta^{-1} \|h\|.$$

Then $v = t^{-m} h$ belongs to J , and the vector having the single nonzero coordinate t^m at v maps to h and has norm less than $\theta^{-1} \|h\|$. Hence q is surjective, and this uniform bound also proves that it is open. Both its domain and codomain are Banach.

Corollary 26, applied to the analytic family $a \mapsto (q(a), \text{id}_F)$, makes $Q_{E,E,F}^K \circ q$ analytic. The original map is nowhere analytic by construction.

If a local section $s: U \rightarrow \ell^1(J, K)$ were differentiable at $h_0 \in U$, differentiating $q \circ s = \text{id}_U$ would give

$$q \circ Ds(h_0) = \text{id}_H.$$

The bounded linear map $Ds(h_0)$ would therefore be a global right inverse of q . But then

$$Q_{E,E,F}^K = (Q_{E,E,F}^K \circ q) \circ Ds(h_0)$$

would be analytic, a contradiction. $\square$

C.4. A tensor criterion for analytic reflection. We now take up the intrinsic question raised at the end of Section 4: when does every analytic map into an ambient Banach space remain analytic into a closed subspace containing its values? This is stronger than the question for alternating-map families. A tensor criterion characterizes the stronger property and clarifies why the fixed-degree argument for ℓ^1 goes further.

In this subsection and the next, K is complete and nontrivially normed, and P is Banach. For $n \geq 1$, let $T_n(P)$ be the separated completed projective tensor power representing bounded n -linear maps from P^n to Banach spaces. We use the ordinary projective seminorm, defined by infima of sums of products of norms, and take its separated completion before invoking the universal property. Its canonical multilinear map has norm at most one. Define the closed subspace

$$\Delta_n(P) = \overline{\text{span}\{x^{\otimes n} : x \in P\}} \subseteq T_n(P).$$

A projection onto $\Delta_n(P)$ will mean a bounded linear map $R_n : T_n(P) \rightarrow \Delta_n(P)$ whose restriction to $\Delta_n(P)$ is the identity.

Proposition 52 (homogeneous reflection). *For a fixed $n \geq 1$, the following conditions are equivalent.*

- (1) *For every Banach space Z , closed subspace $W \subseteq Z$, and bounded n -linear map $B : P^n \rightarrow Z$ with diagonal values in W , there is a bounded W -valued n -linear map with the same diagonal.*
- (2) *The subspace $\Delta_n(P)$ is the range of a bounded projection in $T_n(P)$.*

Proof. Suppose first that a projection R_n exists. Linearize B to a bounded map $\tilde{B} : T_n(P) \rightarrow Z$. The diagonal hypothesis and closedness of W imply $\tilde{B}(\Delta_n(P)) \subseteq W$. Composing $\tilde{B}|_{\Delta_n(P)}$ with R_n and the canonical tensor map gives the required multilinear representative, of norm at most $\|B\|\|R_n\|$. Conversely, take $Z = T_n(P)$, $W = \Delta_n(P)$, and B equal to the canonical tensor map. Linearizing the representative supplied by the first condition gives a bounded map into $\Delta_n(P)$ fixing every pure power. It fixes their linear span and, by continuity, their closed span, so it is a projection. $\square$

Definition 53. The Banach parameter space P has *universal analytic reflection* if, for every Banach space Z , closed subspace $W \subseteq Z$, every open $U \subseteq P$, and every map $f : U \rightarrow W$, analyticity of the composite $U \rightarrow W \hookrightarrow Z$ implies analyticity of f .

Theorem 54 (analytic tensor criterion). *A Banach space P has universal analytic reflection if and only if there are projections $R_n : T_n(P) \rightarrow \Delta_n(P)$ for all $n \geq 1$ such that*

$$(C.2) \quad \limsup_{n \rightarrow \infty} \|R_n\|^{1/n} < \infty.$$

Proof. Assume first that the projections exist. Expand an ambient analytic map with values in W at a point. Lemma 18 puts every homogeneous diagonal in W . The construction in Proposition 52 replaces its coefficient p_n by a W -valued coefficient of norm at most $\|p_n\|\|R_n\|$ and with the same diagonal. The exponential bound preserves a positive radius of convergence. The degree-zero coefficient is already in W , and the resulting expansion represents the given map.

Conversely, form the Banach spaces with supremum norms

$$Z = c_0(\{T_n(P)\}_{n \geq 1}), \quad W = c_0(\{\Delta_n(P)\}_{n \geq 1}) \subseteq Z.$$

Here c_0 denotes families tending to zero in their respective Banach coordinate spaces; W is closed. On the open unit ball of P , define

$$f(x) = (x^{\otimes n})_{n \geq 1}.$$

Since $\|x^{\otimes n}\| \leq \|x\|^n$, this vector lies in W . As a Z -valued map it is analytic: its degree- n coefficient is the canonical tensor map placed in coordinate n , and has norm at most one. This gives an expansion at zero. To re-expand at a with $\|a\| < 1$, choose $r > 0$ with $\|a\| + r < 1$. Expanding each tensor power in $a + h$ gives translated coefficients, and the sum of all term bounds is at most $\sum_{n \geq 1} (\|a\| + r)^n < \infty$. Completeness of Z supplies the coefficients and the rearranged sum, proving analyticity throughout the unit ball. Universal reflection gives a W -valued expansion $\sum_m q_m(x, \dots, x)$ near zero with $\|q_m\| \leq C\rho^{-m}$ for some $C, \rho > 0$. Compose this expansion with the projection $\pi_n: W \rightarrow \Delta_n(P)$ and compare on scalar lines with $x \mapsto x^{\otimes n}$. Lemma 48 gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n} \quad (x \in P).$$

Linearizing $\pi_n q_n$ therefore gives a projection R_n onto $\Delta_n(P)$ with $\|R_n\| \leq \|q_n\| \leq C\rho^{-n}$, proving (C.2). $\square$

C.5. Projection bounds and the special role of fixed degree. For $P = K^d$ with the maximum norm, the coordinate construction in Theorem 19, applied to the canonical tensor map, gives $\|R_n\| \leq d^n$ for $d \geq 1$. A bounded linear change of coordinates introduces only a fixed distortion factor to the n -th power, so it does not change the exponential-growth condition.

For the ordinary sum-norm space $P = \ell^1(I, K)$, the projective tensor power is isometrically $\ell^1(I^n, K)$. Indeed, the coordinate tensor map has norm at most one, while expansion in coordinate tensors gives the reverse projective-norm estimate; completion then gives the identification. In these coordinates, $\Delta_n(P)$ consists exactly of arrays constant on each permutation orbit in I^n . One inclusion follows because every pure power is constant on such orbits. For the other, finite-coordinate polynomial interpolation puts each finite orbit sum in the span of pure powers, and finite sums of orbit sums are dense in the space of orbit-constant ℓ^1 arrays.

Choose one representative of each orbit, send its basis vector to the sum of all basis vectors in that orbit, and send the other basis vectors in the orbit to zero. This gives a projection with norm at most $n!$. Over a nonarchimedean field and for infinite I , this estimate is optimal even among projections that do not preserve the individual orbit blocks.

Proposition 55. *If K is nonarchimedean and complete, and I is infinite, then*

$$\inf\{\|R\| : R: T_n(\ell^1(I, K)) \rightarrow \Delta_n(\ell^1(I, K)) \text{ is a bounded projection}\} = n!.$$

Consequently $\ell^1(I, K)$ does not have universal analytic reflection.

Proof. The preceding construction gives the upper bound. For the lower bound, take an orbit $\mathcal{O}$ of a tuple with n distinct indices; it has $n!$ elements. For each $w \in \mathcal{O}$, let λ_w be the common coefficient on $\mathcal{O}$ of $R(e_w)$. The orbit sum $\sum_{w \in \mathcal{O}} e_w$ belongs to $\Delta_n(P)$ and is fixed by R , hence

$$\sum_{w \in \mathcal{O}} \lambda_w = 1.$$

The nonarchimedean inequality implies that some $|\lambda_w| \geq 1$. For this w , the ordinary sum norm satisfies $\|R(e_w)\|_1 \geq n!|\lambda_w| \geq n!$. Thus $\|R\| \geq n!$. Finally, $(n!)^{1/n}$ is unbounded, so Theorem 54 excludes universal analytic reflection. $\square$

Corollary 56 (Entire polynomial approximation of a nonanalytic smooth map). *Let K be a complete nonarchimedean nontrivially normed field and put $P = \ell^1(\mathbb{N}, K)$, with the ordinary sum norm. There are Banach spaces $W \subseteq Z$, with W closed, and a map f on the open unit ball of P such that f is analytic into Z and C^∞ into W , but is not analytic into W at zero. Moreover, f is a uniform limit on every smaller ball of entire W -valued polynomials.*

Proof. Take

$$Z = c_0(\{T_n(P)\}_{n \geq 1}), \quad W = c_0(\{\Delta_n(P)\}_{n \geq 1}), \quad f(x) = (x^{\otimes n})_{n \geq 1}.$$

The spaces are Banach and W is closed in Z . As in the proof of Theorem 54, the ambient expansion has coefficient norms at most one and defines an analytic map on the open unit ball. Smooth reflection through the closed isometric inclusion makes f C^∞ as a map into W .

Suppose a W -valued expansion at zero has coefficients q_n . Uniqueness on scalar lines gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n},$$

where $\pi_n: W \rightarrow \Delta_n(P)$ is the coordinate projection. Linearizing $\pi_n q_n$ gives a projection $T_n(P) \rightarrow \Delta_n(P)$ of norm at most $\|q_n\|$. Proposition 55 therefore gives $\|q_n\| \geq n!$. Since $(n!)^{1/n}$ is unbounded, these coefficients give a formal multilinear series of radius zero, contradicting the assumed analytic expansion.

For each n , choose the sorted-orbit projection $R_n: T_n(P) \rightarrow \Delta_n(P)$. Its composition with the canonical tensor map is a bounded multilinear representative of $x \mapsto x^{\otimes n}$ with values in $\Delta_n(P)$. Consequently the truncation

$$f_N(x) = (x, x^{\otimes 2}, \dots, x^{\otimes N}, 0, \dots)$$

is an entire W -valued polynomial. For $0 < r < 1$ and $\|x\| \leq r$,

$$\|f(x) - f_N(x)\|_W \leq r^{N+1},$$

which proves the asserted uniform convergence. $\square$

This example is characteristic-free within the nonarchimedean setting. Every individual homogeneous term has an analytic lift, but their best multilinear bounds grow factorially. The fixed-degree argument for alternating families avoids this obstruction by using one outer degree throughout.

There is no conflict with Corollary 26: every alternating-map family on these parameter spaces is still admissible when its final output fiber is complete. The ambient alternating action has a fixed polynomial degree. Factoring a general analytic parameter map through a free ℓ^1 space therefore incurs a single factorial bound, rather than a sequence of factorial bounds across Taylor degrees. The special alternating-family property is consequently strictly weaker than universal analytic reflection.

For comparison, in the category of nonarchimedean normed spaces with the *nonarchimedean* projective tensor norm, the corresponding tensor power of $c_0(I, K)$ is $c_0(I^n, K)$ and the same sorted-orbit projection has norm at most one. This uses the infimum of maxima in the tensor seminorm, rather than the ordinary projective seminorm used in Theorem 54. It is the tensor counterpart of the norm-preserving coefficient construction in Theorem 23; the two tensor norms must not be interchanged.

Theorem 54 thus gives a complete answer for universal Banach-valued analytic reflection. It supplies a sufficient parameter criterion for alternating families, while Proposition 55 shows that it is not a necessary one. An intrinsic characterization of the parameter spaces for which every alternating-map family is admissible remains a separate question.

APPENDIX D. THE MULTIPLIER THEOREM OVER A FINITE FIELD

This section proves the algebraic obstruction used in the Laurent-series construction: over a finite field in which $k! = 0$, coordinatewise multiplication admits no multilinear lift satisfying a polarized diagonal identity and a uniform bound on exterior support. We give a roadmap of the complete construction, introduce the exterior support estimates, and then distinguish the

diagonal identities: over a finite field equality at all scalar points need not be a polynomial identity.

Roadmap of the Banach counterexample. The proof separates into an algebraic obstruction, its realization by a normed space, and passage to the prescribed field. The analytic input is a lift criterion valid in every characteristic: if Q is analytic at any point, the degree- k term of its power series has diagonal Q . One-variable uniqueness identifies this diagonal without dividing by $k!$ (Proposition 3).

The finite-field obstruction. Let κ be finite and put $E_0 = \kappa^{\mathbb{N}}$. Theorem 65 rules out a multilinear lift of coordinatewise multiplication satisfying the polarized diagonal identity (Pol1), with values in $\Lambda_{\kappa}^k E_0$ when $k! = 0$ and every value of the lift lies in an exterior power of a subspace of uniformly bounded dimension. Ramsey's theorem first makes the coefficients depend only on the order and equality pattern of their indices. Signed sums over clusters of four indices cancel the terms not tied to the output. A staircase of clusters converts the support bound into invariance under adjacent transpositions; the polarized diagonal identity then forces $k! \chi = 1$ for a common cluster value χ , a contradiction.

The Laurent series construction. Over $K_1 = \kappa((X))$, take $E_1 = \ell^{\infty}(\mathbb{N}, K_1)$ and let B be the completion of $\Lambda_{K_1}^k E_1$ under its projective norm, defined using ordinary sums rather than maxima. The essential estimate is that the constant-coefficient map

$$\eta: B \longrightarrow \Lambda_{\kappa}^k E_0$$

is well defined and has uniformly bounded support dimension on bounded sets (Proposition 73). Evaluating a hypothetical lift on multiplication operators and on the universal wedge map, then applying η , produces exactly the finite-field lift excluded above. The coefficient map is κ -linear; no K_1 -linearity is asserted or needed.

Passage to the prescribed field. Choose $t \in K$ with $0 < |t| < 1$. The closure of $\mathbb{F}_p(t)$ in the completion $\widehat{K}$ is a spherically complete Laurent series field $K_1 \cong \mathbb{F}_p((X))$, with $|X| = |t|$. Ingleton's Hahn–Banach theorem [4] gives a contracting K_1 -linear projection $\widehat{K} \rightarrow K_1$. It makes the completed projective tensor products

$$E = E_1 \widehat{\otimes}_{\pi, K_1} \widehat{K}, \quad F = B \widehat{\otimes}_{\pi, K_1} \widehat{K}$$

contain E_1 and B isometrically, and allows a lift over $\widehat{K}$ to descend to one over K_1 . Density handles an incomplete K . These are the explicit witnesses in Theorem 28. Finally, sums of unit wedges on disjoint blocks have norms growing linearly with the number of blocks. This growth persists in F and rules out every equivalent nonarchimedean norm (Proposition 88).

D.1. Exterior powers, supports and determinant arrays. Let L be a field, V an L -vector space and $k \geq 1$. We write $\Lambda^k V = \Lambda_L^k V$ for the algebraic exterior power, in which $x_1 \wedge \cdots \wedge x_k = 0$ whenever two x_i coincide. We use the standard facts: for a subspace $W \subseteq V$ the map $\Lambda^k W \rightarrow \Lambda^k V$ is injective, and we regard $\Lambda^k W$ as a subspace of $\Lambda^k V$; if $u_1, \dots, u_d$ is a basis of a finite-dimensional space U , the wedges $u_I := u_{i_1} \wedge \cdots \wedge u_{i_k}$, $I = \{i_1 < \cdots < i_k\}$, form a basis of $\Lambda^k U$; and every $\omega \in \Lambda^k V$ lies in $\Lambda^k U$ for some finite-dimensional $U \subseteq V$.

Definition 57. The *support dimension* of $\omega \in \Lambda^k V$ is

$$\text{sdim}(\omega) := \min\{\dim W : W \subseteq V \text{ finite-dimensional, } \omega \in \Lambda^k W\}.$$

The definition gives $\text{sdim}(0) = 0$, $\text{sdim}(c\omega) = \text{sdim}(\omega)$ for $c \neq 0$, $\text{sdim}(\omega + \omega') \leq \text{sdim}(\omega) + \text{sdim}(\omega')$ (use $W + W'$), and $\text{sdim}(\omega) \leq \dim \text{span } S$ if ω is a sum of wedges of vectors from a finite set S . If $V \subseteq V'$, then sdim computed in V' is at most sdim computed in V .

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank bounds. Let S be a set and $V \subseteq L^S$ a space of functions on S . The *determinant array* is the linear map

$$\Omega: \Lambda^k V \rightarrow L^{S^k}, \quad \Omega_{y_1 \wedge \dots \wedge y_k}(c_1, \dots, c_k) := \det(y_b(c_a))_{a,b=1}^k.$$

It is well defined because the right side is multilinear and alternating in $(y_1, \dots, y_k)$ (these are the columns). For each ω , the function Ω_ω is alternating in $(c_1, \dots, c_k)$ (the rows); in particular $\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)}) = \text{sgn}(\sigma) \Omega_\omega(c_1, \dots, c_k)$.

Lemma 58. *The determinant array $\Omega: \Lambda^k V \rightarrow L^{S^k}$ is injective.*

Proof. Let $\Omega_\omega = 0$ and choose a finite-dimensional $U \subseteq V$ with $\omega \in \Lambda^k U$. The evaluations $\text{ev}_s|_U$, $s \in S$, separate the points of U (a function in U vanishing at every s is 0), so they span the dual space U^* . Choose $s_1, \dots, s_d \in S$ such that the $\text{ev}_{s_i}|_U$ form a basis of U^* , and let $u_1, \dots, u_d$ be the dual basis of U , so that $u_j(s_i) = \delta_{ij}$. For increasing k -subsets I, J of $\{1, \dots, d\}$ we get $\Omega_{u_I}(s_{j_1}, \dots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$. Writing $\omega = \sum_I c_I u_I$ therefore gives $0 = \Omega_\omega(s_{j_1}, \dots, s_{j_k}) = c_J$ for every J , so $\omega = 0$. $\square$

Lemma 59 (support bounds flattening rank). *Let $V \subseteq L^S$, $\omega \in \Lambda^k V$ and $\alpha \in \{1, \dots, k\}$. Let $R \subseteq S$ and $R' \subseteq S^{\{1, \dots, k\} \setminus \{\alpha\}}$ be finite. The $R \times R'$ matrix M with entries $M(c_\alpha, c') := \Omega_\omega(c)$, where c has c_α in slot α and the entries of c' in the other slots, has rank at most $\text{sdim}(\omega)$.*

Proof. Write $\omega = \sum_j y_{j,1} \wedge \dots \wedge y_{j,k}$ with all $y_{j,b}$ in a subspace W of dimension $\text{sdim}(\omega)$. Expanding each determinant along row α , $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot \mu_{j,b}(c')$, where the minor $\mu_{j,b}(c')$ does not involve c_α . So for fixed c' the column $c_\alpha \mapsto M(c_\alpha, c')$ is a linear combination of the restrictions $y_{j,b}|_R$. All columns lie in the restriction of W to R , a space of dimension at most $\text{sdim}(\omega)$. $\square$

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

Lemma 60 (contraction). *Let $k \geq 2$ and $\varphi = (\varphi_1, \dots, \varphi_{k-1}) \in (V^*)^{k-1}$. There is a linear map $c_\varphi: \Lambda^k V \rightarrow V$ with*

$$c_\varphi(x_1 \wedge \dots \wedge x_k) = \sum_{i=1}^k (-1)^{k+i} \det(\varphi_a(x_b))_{1 \leq a \leq k-1, b \neq i} x_i,$$

and $c_\varphi(\Lambda^k W) \subseteq W$ for every subspace W . Consequently $\text{sdim}(\omega) \geq \dim \text{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$.

Proof. The right side is multilinear in (x_b) . For $\psi \in V^*$, applying ψ to it gives, by Laplace expansion along the last row, the determinant of the $k \times k$ matrix with entries $\varphi_a(x_b)$ in the rows $a < k$ and $\psi(x_b)$ in the last row; it vanishes when two columns coincide. Since V^* separates the points of V , the right side vanishes whenever $x_a = x_b$ for some $a \neq b$, so it factors through $\Lambda^k V$. Each term is a multiple of some x_i , which gives $c_\varphi(\Lambda^k W) \subseteq W$. If $\omega \in \Lambda^k W$ with $\dim W = \text{sdim}(\omega)$, all $c_\varphi(\omega)$ lie in W . $\square$

Proposition 61 (canonical exterior support). *Let L be any field, V an L -vector space, $k \geq 2$, and $\omega \in \Lambda^k V$. Then*

$$S(\omega) := \text{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$$

is the unique smallest subspace supporting ω . In particular, $\omega \in \Lambda^k S(\omega)$ and $\text{sdim}(\omega) = \dim S(\omega)$.

Proof. Choose a finite-dimensional supporting subspace $U \subseteq V$. Lemma 60 gives $S(\omega) \subseteq U$, so $S(\omega)$ is finite-dimensional. Choose a basis $e_1, \dots, e_d$ of U whose first r vectors form a basis of $S(\omega)$, and extend its coordinate functionals algebraically to V . Write $\omega = \sum_{|I|=k} a_I e_I$ in the exterior basis of U . If $j \in I$ and $j > r$, contract against the $k-1$ coordinate functionals indexed by $I \setminus \{j\}$. The e_j -coordinate of this contraction is $\pm a_I$: no other basis wedge contributes to that coordinate. The contraction belongs to $S(\omega)$, so this coordinate is zero. Hence $a_I = 0$ whenever I meets $\{r+1, \dots, d\}$, proving $\omega \in \Lambda^k S(\omega)$. Every supporting subspace contains all the contractions by Lemma 60, and therefore contains $S(\omega)$. This proves minimality and the dimension formula. Only unit signs occur, so the argument is valid in every characteristic. $\square$

The same conclusion holds in degree one: under $\Lambda^1 V \simeq V$, the empty-tuple contraction is the identity and $S(\omega) = L\omega$. In degree zero, every element of $\Lambda^0 V = L$ is supported by the zero subspace, so its support dimension is zero; no contraction to V is needed.

For matrix calculations, choose any finite-dimensional supporting subspace U and a basis of it. Form the finite matrix whose columns are the coordinates of the contractions against $(k-1)$ -tuples of coordinate functionals, extended to V . Arbitrary functionals restrict to linear combinations of these coordinates on U ; multilinearity in the covectors therefore shows that the column span is exactly $S(\omega)$. The matrix consequently has rank $\text{sdim}(\omega)$. This uses ordinary matrix rank, and entails no choice of a rank convention for an infinite matrix.

D.2. Lifts and three diagonal identities. Fix a field L , an integer $k \geq 1$, L -vector spaces A and V , and a linear map $D: A \rightarrow \text{End}_L(V)$, $a \mapsto D_a$. A *lift of D in degree k* is a $2k$ -linear map $\Psi: A^k \times V^k \rightarrow \Lambda^k V$ over L that is alternating in the V -slots; its diagonal identity will be specified separately. For every $n \geq 1$ write $[n] := \{1, \dots, n\}$, and for $\alpha \in \mathbb{N}^m$ put $|\alpha| := \sum_i \alpha_i$. Given $m \geq 1$, a map $f: [k] \rightarrow [m]$, a family $b = (b_1, \dots, b_m) \in A^m$ and $\lambda \in L^m$, put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \text{type } f := (|f^{-1}(1)|, \dots, |f^{-1}(m)|) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^k \lambda_{f(j)} = \lambda^{\text{type } f}.$$

The *pointwise identity* is

$$(Pw) \quad \Psi(a, \dots, a; x_1, \dots, x_k) = D_a x_1 \wedge \dots \wedge D_a x_k \quad \text{for all } a \in A, x \in V^k.$$

The *polarized identity* is: for all $m \geq 1$, $b \in A^m$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in V^k$,

$$(Pol) \quad \sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k.$$

Its *multilinear instance* is: for all $b_1, \dots, b_k \in A$ and $x \in V^k$,

$$(Pol1) \quad \sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} D_{b_{\sigma(1)}} x_1 \wedge \dots \wedge D_{b_{\sigma(k)}} x_k.$$

Here $\sum_{\text{type } f = \alpha}$ runs over the maps $f: [k] \rightarrow [m]$ of type α . (Pol1) is the instance $m = k$, $\alpha = (1, \dots, 1)$ of (Pol), since the maps of type $(1, \dots, 1)$ are the permutations. For $b \in A^m$ and $x \in V^k$ put $a_\lambda := \sum_i \lambda_i b_i$ and

$$\begin{aligned} \Delta_{b,x}(\lambda) &:= \Psi(a_\lambda, \dots, a_\lambda; x) - D_{a_\lambda} x_1 \wedge \dots \wedge D_{a_\lambda} x_k = \sum_{|\alpha|=k} \lambda^\alpha \delta_\alpha(b, x), \\ \delta_\alpha(b, x) &:= \sum_{\text{type } f = \alpha} \left[\Psi(b_f; x) - D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k \right]. \end{aligned}$$

Multilinearity of Ψ in the A -slots, linearity of D and multilinearity of the wedge give the displayed expansion. Thus $\Delta_{b,x}$ is a homogeneous polynomial of degree k with coefficients in

$\Lambda^k V$. Identity (Pol) says that every coefficient $\delta_\alpha(b, x)$ is zero. Identity (Pw) says instead that $\Delta_{b,x}$ vanishes at every point of L^m ; conversely, this pointwise vanishing gives (Pw) by taking $m = 1$ and $\lambda = 1$. The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

Lemma 62 (identity principle). *Let $\mathbb{K}$ be an infinite field, Y a $\mathbb{K}$ -vector space and $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$ a polynomial in $\lambda \in \mathbb{K}^m$ with finitely many nonzero coefficients $y_\alpha \in Y$. If $g(\lambda) = 0$ for all $\lambda \in \mathbb{K}^m$, then every $y_\alpha = 0$.*

Proof. The finitely many coefficients y_α span a finite-dimensional subspace of Y . Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on m reduces to the fact that a nonzero polynomial in one variable has only finitely many roots. $\square$

Lemma 63 (homogeneous forms over a finite field). *Let q be a prime power, $L' \supseteq \mathbb{F}_q$ a field, $1 \leq e \leq q$, and $g \in L'[\lambda_1, \dots, \lambda_m]$ homogeneous of degree e with $g(\lambda) = 0$ for every $\lambda \in \mathbb{F}_q^m$. Then $g = 0$.*

Proof. Call a monomial λ^β *reduced* if every $\beta_i \leq q - 1$. The indicator function of a point $a \in \mathbb{F}_q^m$ is $\prod_i (1 - (\lambda_i - a_i)^{q-1})$, which expands into reduced monomials. So the q^m reduced monomials span the q^m -dimensional L' -space of functions $\mathbb{F}_q^m \rightarrow L'$, and they are linearly independent as functions. A monomial of degree $e \leq q$ that is not reduced is λ_i^q (and then $e = q \geq 2$); as a function on $\mathbb{F}_q^m$ it agrees with the reduced monomial λ_i , of degree $1 < e$. If $e = q$, replace each λ_i^q in g by λ_i . These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree q . The replacement therefore sends the monomials of g injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If $e < q$, no replacement is needed. $\square$

Proposition 64 (pointwise versus polarized). *Let Ψ be a lift of D in degree k over a field L .*

- (1) (Pol) $\Rightarrow$ (Pw) $\Rightarrow$ (Pol1), over every field.
- (2) If L is infinite, or if $L = \mathbb{F}_q$ and $k \leq q$, then (Pw) $\Rightarrow$ (Pol).
- (3) If $L = \mathbb{F}_q$ and $k \geq q + 1$, then (Pw) does not imply (Pol) in general.
- (4) If $k! = 0$ in L , then (Pol1) does not imply (Pw) in general.

Proof. (1) *Inclusion-exclusion.* The implication (Pol) $\Rightarrow$ (Pw) is the case $m = 1$ and $\alpha = (k)$. To prove (Pw) $\Rightarrow$ (Pol1), fix $b_1, \dots, b_k \in A$ and $x \in V^k$, and put $b_S := \sum_{i \in S} b_i$ for $S \subseteq [k]$. Multilinearity gives

$$\begin{aligned} \sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) &= \sum_S (-1)^{k-|S|} \sum_{f: [k] \rightarrow S} \Psi(b_f; x) \\ &= \sum_{f: [k] \rightarrow [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x), \end{aligned}$$

because $\sum_{S \supseteq T} (-1)^{k-|S|} = (1 - 1)^{k-|T|}$ is 1 if $T = [k]$ and 0 otherwise. The same computation applies to the multilinear map $(a_1, \dots, a_k) \mapsto D_{a_1} x_1 \wedge \dots \wedge D_{a_k} x_k$, whose diagonal is the right side of (Pw). Evaluating (Pw) at the 2^k subset sums b_S therefore gives (Pol1). The computation is valid in every characteristic and involves no division by $k!$.

(2) *The identity principles.* The coefficients of $\Delta_{b,x}$ span a finite-dimensional subspace of $\Lambda^k V$; in a basis of it each coordinate of $\Delta_{b,x}$ is a scalar form of degree k vanishing on L^m . It is zero by Lemma 62 if L is infinite, and by Lemma 63 if $L = \mathbb{F}_q$ and $k \leq q$.

(3) *Pointwise vanishing without full polarization.* Let $\dim A \geq 3$, with $b_1, b_2, b_3 \in A$ and functionals g_1, g_2, g_3 on A such that $g_i(b_j) = \delta_{ij}$, and let $y_1, \dots, y_k \in V$ be linearly independent. Put

$$\theta(a_1, \dots, a_k) := [g_1(a_1) \cdots g_1(a_q) g_2(a_{q+1}) - g_1(a_1) g_2(a_2) \cdots g_2(a_{q+1})] \cdot g_3(a_{q+2}) \cdots g_3(a_k)$$

(for $k = q + 1$ the g_3 -factor is empty and $\dim A \geq 2$ suffices), and $\Theta(a; x) := \theta(a) x_1 \wedge \cdots \wedge x_k$. Then Θ is a lift of $D = 0$. It satisfies (Pw), because $\theta(a, \dots, a) = (g_1(a)^q g_2(a) - g_1(a) g_2(a)^q) g_3(a)^{k-q-1} = 0$ since $c^q = c$ for $c \in \mathbb{F}_q$. It violates (Pol) for $\alpha = (q, 1, k - q - 1)$ and $b = (b_1, b_2, b_3)$: the first product contributes to $\sum_{\text{type } f = \alpha} \theta(b_f)$ exactly for $f = (1, \dots, 1, 2, 3, \dots, 3)$, with value 1; the second product would need $f(2) = \cdots = f(q + 1) = 2$, that is, at least $q \geq 2$ values equal to 2, which is impossible for type α . So the coefficient $\delta_\alpha(b, y)$ of Θ is $y_1 \wedge \cdots \wedge y_k \neq 0$. When $k = q + 1$, the third label and its functional can be omitted throughout, with $\alpha = (q, 1)$.

(4) *The multilinear identity without the pointwise identity.* Take $\bar{a} \in A$ and a functional g_1 with $g_1(\bar{a}) = 1$, and $y_1, \dots, y_k$ linearly independent. The lift $\Theta'(a; x) := g_1(a_1) \cdots g_1(a_k) x_1 \wedge \cdots \wedge x_k$ of $D = 0$ satisfies (Pol1), since $\sum_\sigma \Theta'(b_\sigma; x) = k! g_1(b_1) \cdots g_1(b_k) x_1 \wedge \cdots \wedge x_k = 0$ for all $b_1, \dots, b_k \in A$ and $x \in V^k$, but $\Theta'(\bar{a}, \dots, \bar{a}; y) = y_1 \wedge \cdots \wedge y_k \neq 0$. $\square$

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over $\mathbb{F}_q$ they need not agree when $k > q$. The lift constructed in Appendix E nevertheless satisfies the full identity (Pol) in every degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma 74).

D.3. Statement and proof strategy. For the proof of the theorem, assume that L is a *finite* field with $k! = 0$ in L , that is, $\text{char } L = p \leq k$; in particular $k \geq 2$. Put $V := L^{\mathbb{N}}$ and let $V_{\text{fin}} := L^{(\mathbb{N})} \subseteq V$ be the subspace of finitely supported sequences. For $u, x \in V$ we write ux for the coordinatewise product; so $D_u x = ux$ is the multiplier family.

Theorem 65 (multipliers over a finite field). *Let L be a finite field and $k \geq 1$ with $k! = 0$ in L .*

- (1) *There is no $2k$ -linear map $\Psi: V_{\text{fin}}^k \times V_{\text{fin}}^k \rightarrow \Lambda_L^k V$ over L such that*
 - (a) *Ψ is antisymmetric in the last k slots;*
 - (b) *Ψ satisfies (Pol1) for $D_u x = ux$: for all $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\text{fin}}$,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$

- (c) *for some $d \in \mathbb{N}$, $\text{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in V_{\text{fin}}^k$ with entries in $\{0, \pm 1\}$.*
- (2) *Let $E_0 = V$ or $E_0 = V_{\text{fin}}$. There is no $2k$ -linear map $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_L^k E_0$ over L that is antisymmetric in the last k slots, satisfies (Pol1) for all finitely supported u, x , and has $\text{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in E_0^k$, for some d .*

Reduction to (1). Given a map as in (2), restrict its inputs to $V_{\text{fin}}^k \times V_{\text{fin}}^k$ and compose its output with the injective map $\Lambda^k E_0 \rightarrow \Lambda^k V$. This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§D.1), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that Ψ satisfies its hypotheses, and fix the bound d in (c). Write $e_i \in V_{\text{fin}}$ for the unit vectors, $e_a := (e_{a_1}, \dots, e_{a_k})$ for $a \in \mathbb{N}^k$, and Ω for the determinant array of §D.1 with $S = \mathbb{N}$.

The proof has three steps. Ramsey's theorem first makes the coefficients of $\Omega\Psi$ depend only on the relative order and equalities among their indices. Special weights on four-point

clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Poll) say that the sum of these $k!$ equal values is 1, contradicting $k! = 0$ in L .

D.4. Order-homogeneity from Ramsey's theorem. We use Ramsey's theorem [9] in its infinite form: *for every $n \geq 1$ and every coloring of the n -element subsets of $\mathbb{N}$ with finitely many colors there is an infinite $H \subseteq \mathbb{N}$ all of whose n -element subsets have the same color.*

For a tuple $z \in \mathbb{N}^N$ with distinct values $y_1 < \dots < y_r$, its *pattern* is the map $i \mapsto j$ determined by $z_i = y_j$. In particular, repeated entries are retained in the pattern. Two tuples z and z' have the same pattern if and only if $z_i < z_j \Leftrightarrow z'_i < z'_j$ and $z_i = z_j \Leftrightarrow z'_i = z'_j$ for all i, j .

Lemma 66 (pattern homogeneity). *For every $N \geq 1$ and every function $T: \mathbb{N}^N \rightarrow L$ there is an infinite $H \subseteq \mathbb{N}$ such that $T(z) = T(z')$ whenever $z, z' \in H^N$ have the same pattern.*

Proof. Color an N -element set $s = \{s_1 < \dots < s_N\} \subseteq \mathbb{N}$ by the vector

$$(T(s_{\beta(1)}, \dots, s_{\beta(N)}))_{\beta: [N] \rightarrow [N]} \in L^{N^N}.$$

The coloring records all maps β , including those that repeat indices. It has finitely many values because L is finite; this is the only use of the finiteness of L in the obstruction proof. Ramsey's theorem gives an infinite H on which the color is constant, say equal to $(t_\beta)_\beta$. Let $z \in H^N$ have value set $\{y_1 < \dots < y_r\}$. Since H is unbounded, add $N - r$ elements of H above y_r to obtain an N -element set $s \subseteq H$. Padding above the largest value leaves the ranks unchanged: $s_j = y_j$ for $j \leq r$, and $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$, where $\beta(i)$ is the rank of z_i in $\{y_1, \dots, y_r\}$. The map β depends only on the pattern of z , and $T(z) = t_\beta$. $\square$

Coefficients. For $(a; y; c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$ put

$$T(a; y; c) := \Omega_{\Psi(e_a; e_y)}(c_1, \dots, c_k) \in L.$$

Lemma 66 with $N = 3k$ gives an infinite set $H = \{h_0 < h_1 < h_2 < \dots\}$ on which T depends only on the pattern. For $u_j, v_j \in V_{\text{fin}}$ supported in H , multilinearity of Ψ and linearity of Ω give

$$(D.1) \quad \Omega_{\Psi(u; v)}(c) = \sum_{a, y \in H^k} \prod_{j=1}^k u_j(a_j) \prod_{j=1}^k v_j(y_j) \cdot T(a; y; c).$$

Although the sum is indexed by $H^k \times H^k$, only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple c used below also lies in H^k , so pattern homogeneity applies to all $3k$ indices in (D.1).

D.5. Clusters and the cluster value. A *cluster* is a set $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$ of four consecutive elements of H ; we write $C(i) := h_{n+i-1}$ for $i = 1, \dots, 4$. Since a cluster is an interval of H , two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write $C < C'$. Distinct clusters need not be adjacent in H .

We choose two weight vectors, one for each kind of input slot:

$$\begin{aligned} s &= (s_1, s_2, s_3, s_4) := (1, 0, -1, 0) && \text{on the operator slots,} \\ w &= (w_1, w_2, w_3, w_4) := (1, -1, 1, -1) && \text{on the vector slots.} \end{aligned}$$

Over $\mathbb{Z}$, hence over every field,

$$(W) \quad \sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i < i'} s_i w_{i'} = \sum_{i > i'} s_i w_{i'} = 0, \quad s_1 w_1 = 1.$$

Indeed $\sum_i s_i = 1 - 1$, $\sum_i w_i = 1 - 1 + 1 - 1$, $s_1 w_1 = 1$, and

$$\begin{aligned} \sum_i s_i w_i &= s_1 w_1 + s_3 w_3 = 1 - 1 = 0, \\ \sum_{i < i'} s_i w_{i'} &= s_1(w_2 + w_3 + w_4) + s_3 w_4 = -1 + 1 = 0, \\ \sum_{i > i'} s_i w_{i'} &= s_3(w_1 + w_2) = -(1 - 1) = 0. \end{aligned}$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster C put

$$s_C := \sum_{i=1}^4 s_i e_{C(i)}, \quad w_C := \sum_{i=1}^4 w_i e_{C(i)}.$$

These are vectors of V_{fin} with entries in $\{0, \pm 1\}$, and $s_C w_C = e_{C(1)} - e_{C(3)}$.

The cluster value. For pairwise disjoint clusters $C_1, \dots, C_k$, one for each slot, define

$$\begin{aligned} \chi(C_1, \dots, C_k) &:= \Omega_{\Psi(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})}(C_1(1), \dots, C_k(1)) \\ &= \sum_{i, i' \in [4]^k} \prod_{j=1}^k s_{i_j} w_{i'_j} \cdot T(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)). \end{aligned}$$

Let $\tau \in S_k$ be the *order pattern* of $(C_1, \dots, C_k)$: $\tau(j)$ is the position of C_j among $C_1, \dots, C_k$. The pattern of every $3k$ -tuple in the sum is determined by (i, i') and τ : entries in different clusters compare as the clusters do, and the entries $C_j(i_j)$, $C_j(i'_j)$, $C_j(1)$ in the same cluster compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on τ . Every permutation occurs as the order pattern of some disjoint clusters in H ; we denote the resulting scalar by $\chi(\tau)$.

Lemma 67 (cancellation of output-free clusters). *Let $C_1, \dots, C_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_j C_j$ are pairwise disjoint. Put $u_j := \sum_{C \in C_j} s_C$ and $v_j := \sum_{C \in C_j} w_C$. Let $C_l \in \mathcal{C}_l$ for $l = 1, \dots, k$ and $c := (C_1(1), \dots, C_k(1))$. Then $\Omega_{\Psi(u; v)}(c) = \chi(C_1, \dots, C_k)$.*

Proof. We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (D.1). Since the clusters are disjoint, a nonzero term has, for each slot j , an operator index $a_j = C_j^{\text{op}}(i_j)$ and a vector index $y_j = C_j^{\text{vec}}(i'_j)$ with $C_j^{\text{op}}, C_j^{\text{vec}} \in C_j$, and weight $\prod_j s_{i_j} w_{i'_j}$. Group the terms according to the *cluster assignment* $(C_j^{\text{op}}, C_j^{\text{vec}})_j$.

Call an occupied cluster *output-free* if it is none of $C_1, \dots, C_k$. Suppose a group has such a cluster Γ . Disjointness of the families assigns Γ to exactly one slot j , so the only tuple entries it can contain are a_j and y_j . Disjointness of the clusters excludes every output index from Γ .

Fix the positions of all indices outside Γ and sum over the positions inside it. Since Γ consists of consecutive elements of H , every outside entry lies either below the whole cluster or above it. Moving an index within Γ cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient T therefore depends on the internal positions only as follows:

- (i) if only $a_j \in \Gamma$, not at all, and the partial sum is $(\sum_i s_i) \cdot T = 0$;
- (ii) if only $y_j \in \Gamma$, not at all, and the partial sum is $(\sum_i w_i) \cdot T = 0$;

- (iii) if both lie in Γ , only through whether $i_j < i'_j$, $i_j = i'_j$ or $i_j > i'_j$; with the corresponding values $T_<, T_=: T_>$ of T the partial sum is

$$T_< \sum_{i < i'} s_i w_{i'} + T_=: \sum_i s_i w_i + T_> \sum_{i > i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to $\mathcal{C}_j$ is C_j , which forces $C_j^{\text{op}} = C_j^{\text{vec}} = C_j$ for every j . The contribution of this one assignment is exactly $\chi(C_1, \dots, C_k)$. $\square$

D.6. The staircase and the diagonal. We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of Ψ .

Lemma 68 (staircase). *Let $\tau \in S_k$, let $\alpha \neq \beta$ be slots with $\tau(\alpha) = \ell$ and $\tau(\beta) = \ell + 1$, and let τ' be τ with these two positions exchanged, so $\tau'(\alpha) = \ell + 1$, $\tau'(\beta) = \ell$ and $\tau'(\gamma) = \tau(\gamma)$ otherwise. Then $\chi(\tau) = \chi(\tau')$.*

Proof. Let $m \geq 1$. Choose pairwise disjoint clusters, in increasing order in H : first a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then $2m$ clusters $A_1 < B_1 < A_2 < B_2 < \dots < A_m < B_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Put $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$, $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$ and $\mathcal{C}_\gamma := \{C_\gamma\}$ for the other slots, and let u, v be as in Lemma 67 and $\omega := \Psi(u; v)$. Disjointness of the clusters ensures that the entries of u and v still lie in $\{0, \pm 1\}$, so $\text{sdim}(\omega) \leq d$ by hypothesis (c) of Theorem 65(1). For this fixed m , keep u, v and hence ω fixed while varying only the output indices.

Consider the $m \times m$ matrix M with rows indexed by $c_\alpha = A_n(1)$ and columns by $c_\beta = B_{n'}(1)$ ($n, n' = 1, \dots, m$), all other output indices being fixed at $c_\gamma = C_\gamma(1)$, and entries $M_{nn'} := \Omega_\omega(c)$. By Lemma 67, applied with the output clusters $A_n \in \mathcal{C}_\alpha$, $B_{n'} \in \mathcal{C}_\beta$ and $C_\gamma \in \mathcal{C}_\gamma$, $M_{nn'}$ is the cluster value of the tuple with A_n in slot α , $B_{n'}$ in slot β , and C_γ in every other slot. The interlacing order gives $A_n < B_{n'}$ exactly when $n \leq n'$. The tuple therefore has pattern τ on and above the matrix diagonal, and pattern τ' below it.

Put $\delta := \chi(\tau) - \chi(\tau')$, let $\mathbf{1} \in L^m$ be the all-ones column, and let U be the upper unitriangular matrix with $U_{nn'} = 1$ for $n \leq n'$ and $U_{nn'} = 0$ otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau') \mathbf{1}\mathbf{1}^\top + \delta U,$$

The rows vary only slot α ; each column fixes all the other slots, including the chosen value in slot β . Thus M is a submatrix of the flattening of this one array Ω_ω along slot α . Lemma 59 gives $\text{rank } M \leq \text{sdim}(\omega) \leq d$.

If $\delta \neq 0$, the matrix δU is invertible, whereas the constant matrix $\chi(\tau') \mathbf{1}\mathbf{1}^\top$ has rank at most one. Applying rank subadditivity to $\delta U = M - \chi(\tau') \mathbf{1}\mathbf{1}^\top$ gives

$$m = \text{rank}(\delta U) \leq \text{rank } M + 1 \leq d + 1.$$

The construction is available for every $m \geq 1$; choosing $m = d + 2$ is a contradiction. Hence $\delta = 0$, as required. $\square$

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Poll); it does not use the support bound.

Lemma 69 (the diagonal). $\sum_{\tau \in S_k} \chi(\tau) = 1$.

Proof. Take clusters $C_1 < \dots < C_k$, and put $u_j := s_{C_j}$, $v_j := w_{C_j}$ and $c := (C_1(1), \dots, C_k(1))$. Apply $\omega \mapsto \Omega_\omega(c)$ to both sides of (Pol1), with the operator vectors $u_1, \dots, u_k$ and the vector arguments $v_1, \dots, v_k$.

Right side. If $\sigma \neq \text{id}$, some slot j has $\sigma(j) \neq j$. Disjoint supports then give $u_{\sigma(j)}v_j = 0$, so the corresponding wedge vanishes. Only $\sigma = \text{id}$ survives. It gives $\Omega_{(u_1v_1) \wedge \dots \wedge (u_kv_k)}(c) = \det((u_bv_b)(C_a(1)))_{a,b}$. Since $u_bv_b = e_{C_b(1)} - e_{C_b(3)}$ and $C_a(1) \neq C_b(3)$ for all a, b , this is the determinant of the identity matrix, so the right side is 1.

Left side. Fix $\sigma \in S_k$ and write $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$ and $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$. To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of Ψ in the vector slots gives $\Psi(u_\sigma; v) = \text{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$. For every $\omega \in \Lambda^k V$ the function Ω_ω is alternating in its arguments, so $\Omega_\omega(c) = \text{sgn}(\sigma)\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$; we apply this with $\omega = \Psi(u_\sigma; v_\sigma)$. Since $\text{sgn}(\sigma)^2 = 1$,

$$\begin{aligned} \Omega_{\Psi(u_\sigma; v)}(c) &= \text{sgn}(\sigma)\Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \text{sgn}(\sigma)^2\Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}). \end{aligned}$$

In the last expression slot j has operator $s_{C_{\sigma(j)}}$, vector $w_{C_{\sigma(j)}}$ and output $C_{\sigma(j)}(1)$, so it equals $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$ by definition. Its order pattern is σ , since $C_{\sigma(j)}$ is the $\sigma(j)$ -th cluster in increasing order. Thus the convention for patterns gives σ , not σ^{-1} . No cluster cancellation is needed in this calculation.

As σ runs over S_k , each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1. \quad \square$$

Proof of Theorem 65. Exchanging adjacent ranks acts on an order pattern by $\tau \mapsto (\ell \ \ell+1) \circ \tau$, $1 \leq \ell < k$. These transpositions generate S_k , and Lemma 68 makes χ invariant under each exchange. Hence χ is constant; write its value as χ_0 . Lemma 69 now gives $k!\chi_0 = 1$ in L , contradicting $k! = 0$. This proves (1), and the preceding reduction gives (2). $\square$

Remark 70 (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of L . Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma 68 only at the finitely many test inputs obtained by taking $m = d + 2$; all have entries in $\{0, \pm 1\}$. Antisymmetry and (Pol1) enter in Lemma 69, which does not use that bound. Finally, $k! = 0$ is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition 64(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at L -points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma 69 are trivial.

Remark 71 (when $k!$ is invertible). The vanishing of $k!$ is essential. If instead $k! \neq 0$ in L , the *normalized alternatization*

$$\Psi_{\text{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)}x_1) \wedge \dots \wedge (u_{\sigma(k)}x_k)$$

is alternating in x and satisfies (Pol1). Every summand is supported in the span of the k^2 vectors $u_i x_j$, so $\text{sdim}(\Psi_{\text{alt}}(u; x)) \leq k^2$. The preceding homogeneity, cancellation, staircase and

diagonal arguments apply, but now yield $\chi \equiv 1/k!$: for pairwise disjoint clusters only $\sigma = \text{id}$ survives in $\Psi_{\text{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$, and the determinant at $(C_1(1), \dots, C_k(1))$ is that of the identity. The final equation $k! \cdot (1/k!) = 1$ is then consistent. So the obstruction is exactly the vanishing of $k!$, not positive characteristic as such. For example, at $(p, k) = (5, 3)$ every step is consistent, since $3! = 6 = 1$ in $\mathbb{F}_5$ and $\chi_0 = 1$.

APPENDIX E. THE BANACH EXTERIOR TARGET OVER A LAURENT SERIES FIELD

We now realize the finite-field obstruction in a Banach-space setting. The target is the completion of an exterior power with its ordinary projective norm. Its determinant arrays admit a constant-coefficient map into the *algebraic* exterior power over the coefficient field, with support dimension controlled by the Banach norm. This estimate turns a hypothetical bounded lift into the uniformly bounded-support map ruled out by Theorem 65. The passage to the completion is the key point: constant coefficients stabilize exactly once the approximation error is less than one.

E.1. Setting. Throughout this section $k \geq 1$, κ is a field, $r \in (0, 1)$, and $K_1 := \kappa((X))$ is the Laurent series field with absolute value $|a| := r^{\text{ord } a}$ ($|0| = 0$). This field is complete, nontrivially normed and nonarchimedean, and $|c| = 1$ for every $c \in \kappa^\times$. Write $\text{coeff}_n(a) \in \kappa$ for the X^n -coefficient of $a \in K_1$. For a set S and $y \in \ell^\infty(S, K_1)$ we take the constant coefficient coordinatewise:

$$\text{coeff}_0(y) := (\text{coeff}_0(y(s)))_{s \in S} \in \kappa^S.$$

The resulting *constant-coefficient map* coeff_0 is κ -linear but not K_1 -linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$(E.1) \quad \|y\|_\infty < 1 \implies \text{coeff}_0(y) = 0, \quad \text{hence} \quad \|y - y'\|_\infty < 1 \implies \text{coeff}_0(y) = \text{coeff}_0(y').$$

Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the sup norm, a nonarchimedean K_1 -Banach space, and let $E_0 := \kappa^\mathbb{N} \subseteq E_1$ be the κ -subspace of sequences with entries in κ . Every nonzero $z \in E_0$ has $\|z\|_\infty = 1$. For $x \in E_1$ and $n \in \mathbb{Z}$ let $x_{[n]} := (\text{coeff}_n(x(i)))_{i \in \mathbb{N}} \in E_0$ be the n -th Laurent component. For $x \neq 0$ the set $\{\text{ord } x(i) : x(i) \neq 0\}$ is bounded below, because $r^{\text{ord } x(i)} \leq \|x\|_\infty$, so it has a minimum $\nu(x)$; then $\|x\|_\infty = r^{\nu(x)}$ and $x_{[n]} = 0$ for $n < \nu(x)$. Finally put

$$M_r := \max_{l \in \mathbb{N}} (l+1)r^l.$$

The maximum exists because $(l+1)r^l \rightarrow 0$. This constant will control the number of Laurent components that can contribute to a constant coefficient. For $r \leq \frac{1}{2}$ one has $(l+1)r^l \leq (l+1)2^{-l} \leq 1$, so $M_r = 1$.

E.2. The projective exterior norm. On $\Lambda := \Lambda_{K_1}^k E_1$ put

$$\|\omega\|_\pi := \inf \left\{ \sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty : \omega = \sum_j x_{j,1} \wedge \dots \wedge x_{j,k} \right\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let $\Omega^{K_1} : \Lambda \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$ be the determinant array of §D.1 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$, so that $\Omega^{K_1}(x_1 \wedge \dots \wedge x_k)(i_1, \dots, i_k) = \det(x_b(i_a))_{a,b}$; its values are bounded by the next lemma.

Lemma 72. $\|\cdot\|_\pi$ is a K_1 -norm on Λ with $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$, and Ω^{K_1} is K_1 -linear and injective with $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Let B be the completion of $(\Lambda, \|\cdot\|_\pi)$, a K_1 -Banach space, $J: B \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$ the continuous extension of Ω^{K_1} (so $\|J\| \leq 1$), and $W_B: E_1^k \rightarrow B$, $W_B(x) := x_1 \wedge \cdots \wedge x_k$. Then $W_B \in \text{Alt}_{K_1}^k(E_1; B)$, $\|W_B\| \leq 1$, and $J(W_B(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$.

Proof. For a pure wedge, each entry of the determinant array is a signed sum of products $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$. Each product has absolute value at most $\prod_b \|x_b\|_\infty$, so the nonarchimedean inequality in K_1 gives the same bound for the determinant. For any finite decomposition of ω , it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. The map Ω^{K_1} is linear and injective by Lemma 58, with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$. In particular, $\|\omega\|_\pi > 0$ whenever $\omega \neq 0$.

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since $k \geq 1$, a scalar can be absorbed into the first factor of each wedge; for $\lambda \neq 0$ this gives $\|\lambda\omega\|_\pi \leq |\lambda|\|\omega\|_\pi$, and applying the same inequality to λ^{-1} gives equality. The pure-wedge bound follows by using a single summand.

The array space $\ell^\infty(\mathbb{N}^k, K_1)$ is complete, so Ω^{K_1} extends to the contraction J on B . The wedge map into B is K_1 -multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with J holds on its values by construction. $\square$

For $k \geq 2$, B is not nonarchimedean (Proposition 88).

E.3. The support estimate. We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let $\Omega^\kappa: \Lambda_\kappa^k E_0 \rightarrow \kappa^{\mathbb{N}^k}$ be the determinant array over κ ; it is injective by Lemma 58 with $L = \kappa$ and $V = E_0$. For $y_1, \dots, y_k \in E_0$ we have $\Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k)$: the determinants are the same, with entries in $\kappa \subseteq K_1$. Support dimensions of elements of $\Lambda_\kappa^k E_0$ are computed in E_0 over κ .

Proposition 73 (support estimate). *For every $b \in B$ there is a unique $\eta(b) \in \Lambda_\kappa^k E_0$ with $\Omega^\kappa(\eta(b)) = \text{coeff}_0(Jb)$. The map η is κ -linear, and*

$$(E.2) \quad \text{sdim}(\eta(b)) \leq kM_r \|b\|_B \quad \text{for all } b \in B.$$

Proof. Step 1: one decomposable. Let $x_1, \dots, x_k \in E_1$. If some $x_b = 0$, the wedge is zero and we take $\eta = 0$. Otherwise put $\nu_b := \nu(x_b)$ and $l := -\sum_b \nu_b$, so that $\prod_b \|x_b\|_\infty = r^{-l}$. The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N} := \left\{ n \in \mathbb{Z}^k : n_b \geq \nu_b \text{ for all } b, \sum_b n_b = 0 \right\}.$$

For such a tuple, the nonnegative integers $n_b - \nu_b$ sum to l . Hence $\mathcal{N}$ is empty if $l < 0$; if $l \geq 0$, every coordinate satisfies $\nu_b \leq n_b \leq \nu_b + l$, so $\mathcal{N}$ is finite.

Fix $(i_1, \dots, i_k) \in \mathbb{N}^k$ and expand each entry as $x_b(i_a) = \sum_{n \geq \nu_b} \text{coeff}_n(x_b(i_a)) X^n$. In the determinant expansion

$$\det(x_b(i_a))_{a,b} = \sum_\sigma \text{sgn}(\sigma) \prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over $\mathcal{N}$. Exchanging this finite sum with the permutation sum gives

$$\text{coeff}_0 \det(x_b(i_a))_{a,b} = \sum_{n \in \mathcal{N}} \det(x_{b,[n_b]}(i_a))_{a,b}.$$

The same finite set $\mathcal{N}$ works for every array coordinate. Thus $\text{coeff}_0(\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)) = \Omega^\kappa(\eta)$, where

$$\eta := \sum_{n \in \mathcal{N}} x_{1,[n_1]} \wedge \cdots \wedge x_{k,[n_k]} \in \Lambda_\kappa^k E_0.$$

This is a finite sum in the algebraic exterior power. When $l < 0$ it is zero. When $l \geq 0$, all its wedge factors belong to the set $\{x_{b,[n]} : 1 \leq b \leq k, \nu_b \leq n \leq \nu_b + l\}$, which has at most $k(l+1)$ elements. Taking their span gives

$$\text{sdim}(\eta) \leq k(l+1) = k(l+1)r^l \cdot r^{-l} \leq kM_r \prod_b \|x_b\|_\infty.$$

The final bound therefore holds in both cases.

Step 2: an element of Λ . Write $\omega = \sum_j w_j$, where $w_j = x_{j,1} \wedge \cdots \wedge x_{j,k}$, and take η_j from Step 1. Additivity gives

$$\text{coeff}_0(\Omega^{K_1}(\omega)) = \sum_j \text{coeff}_0(\Omega^{K_1}(w_j)) = \Omega^\kappa\left(\sum_j \eta_j\right).$$

Injectivity of Ω^κ shows that $\eta(\omega) := \sum_j \eta_j$ is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\text{sdim}(\eta(\omega)) \leq \sum_j \text{sdim}(\eta_j) \leq kM_r \sum_j \prod_i \|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives $\text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_\pi$, without requiring a minimizing decomposition.

Step 3: completion. Choose $\omega_n \in \Lambda$ with $\|b - \omega_n\|_B \rightarrow 0$. Since J is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \leq \|b - \omega_n\|_B < 1 \quad (n \geq n_0).$$

Equation (E.1) therefore gives the *exact* identities

$$\text{coeff}_0(Jb) = \text{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \quad (n \geq n_0).$$

By injectivity of Ω^κ , the algebraic vectors $\eta(\omega_n)$ are eventually equal. Define $\eta(b)$ to be this common value. Its array is $\text{coeff}_0(Jb)$, so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in $\Lambda_\kappa^k E_0$ is involved.

For every $n \geq n_0$ the fixed vector $\eta(b)$ satisfies $\text{sdim}(\eta(b)) \leq kM_r \|\omega_n\|_\pi$. Since $\|\omega_n\|_\pi \rightarrow \|b\|_B$, this gives (E.2). Finally, the array characterization proves κ -linearity: J is K_1 -linear, coeff_0 is κ -linear, and Ω^κ is linear and injective. $\square$

E.4. The coefficient lift. The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map W_B .

Suppose that P is a bounded k -linear lift of $Q_{E_1, E_1, B}^{K_1}$ over K_1 . For $a \in E_1$ let $D_a \in \mathcal{L}(E_1, E_1)$ be coordinatewise multiplication, $D_a x = ax$. The map $a \mapsto D_a$ is K_1 -linear and $\|D_a\| \leq \|a\|_\infty$. Define

$$\begin{aligned} \Phi(a_1, \dots, a_k; x_1, \dots, x_k) &:= P(D_{a_1}, \dots, D_{a_k})(W_B)(x_1, \dots, x_k) \in B, \\ \Psi &:= \eta \circ \Phi|_{E_0^{2k}} : E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0. \end{aligned}$$

Since $\Omega^\kappa \circ \eta = \text{coeff}_0 \circ J$, the map Ψ is $\text{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$, read in $\Lambda_\kappa^k E_0$ through the injective map Ω^κ .

Lemma 74. *The map Ψ has the following properties.*

- ($\Psi 1$) Ψ is $2k$ -linear over κ .
- ($\Psi 2$) Ψ is alternating in the last k slots.
- ($\Psi 3$) $\text{sdim}(\Psi(z)) \leq d := \lfloor kM_r \|P\| \rfloor$ for all $z \in E_0^{2k}$.
- ($\Psi 4$) Ψ satisfies (**Pol**) for the multiplier family $D^\kappa: E_0 \rightarrow \text{End}_\kappa(E_0)$, $D_a^\kappa x = ax$: for all $m \geq 1$, $b_1, \dots, b_m \in E_0$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in E_0^k$,

$$\sum_{\text{type } f=\alpha} \Psi(b_f; x) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1) \wedge \dots \wedge (b_{f(k)}x_k) \quad \text{in } \Lambda_\kappa^k E_0.$$

In particular Ψ satisfies (**Pol1**).

Proof. The map Φ is K_1 -multilinear, so its restriction to E_0^{2k} is κ -multilinear. Composing with the κ -linear map η proves ($\Psi 1$). Since $P(D_{a_1}, \dots, D_{a_k})(W_B)$ belongs to $\text{Alt}_{K_1}^k(E_1; B)$, it vanishes when two of the last k arguments coincide; the same holds after applying η , proving ($\Psi 2$).

For ($\Psi 3$), all inputs from E_0 have norm at most one. Thus

$$\|\Phi(a; x)\| \leq \|P\| \prod_i \|D_{a_i}\| \|W_B\| \prod_i \|x_i\|_\infty \leq \|P\|.$$

Apply (E.2) and use that support dimension is an integer to obtain the stated bound by d .

To prove ($\Psi 4$), we first compare coefficients over the infinite field K_1 , while the values still lie in B . Fix $b_1, \dots, b_m \in E_0$ and $x \in E_0^k$. For $\lambda \in K_1^m$ put $a_\lambda := \sum_i \lambda_i b_i \in E_1$. The diagonal identity for the lift gives

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(D_{a_\lambda}, \dots, D_{a_\lambda})(W_B)(x) = Q(D_{a_\lambda})(W_B)(x) = W_B(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expanding both sides by multilinearity gives $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$ for all $\lambda \in K_1^m$, where

$$L_\alpha := \sum_{\text{type } f=\alpha} \Phi(b_f; x) \in B, \quad R_\alpha := \sum_{\text{type } f=\alpha} W_B(b_{f(1)}x_1, \dots, b_{f(k)}x_k) \in B.$$

The field K_1 is infinite even when κ is finite. Lemma 62, applied in the K_1 -vector space B , therefore yields $L_\alpha = R_\alpha$ for every α .

We can now apply η . Additivity gives $\eta(L_\alpha) = \sum_{\text{type } f=\alpha} \Psi(b_f; x)$. To identify the other side, take $y_1, \dots, y_k \in E_0$. By Lemma 72 and the determinant-array agreement preceding Proposition 73,

$$J(W_B(y)) = \Omega^{K_1}(y_1 \wedge \dots \wedge y_k) = \Omega^\kappa(y_1 \wedge \dots \wedge y_k).$$

This array has entries in κ , so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition 73 then gives $\eta(W_B(y)) = y_1 \wedge \dots \wedge y_k \in \Lambda_\kappa^k E_0$. Using $y_j = b_{f(j)}x_j$ in each summand of R_α proves

$$\eta(R_\alpha) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1) \wedge \dots \wedge (b_{f(k)}x_k),$$

which is the required identity. $\square$

The order of these operations is essential for the full identity ($\Psi 4$). Evaluating the diagonal only at κ -points would give (**Pw**) there, which for $\kappa = \mathbb{F}_q$ and $k > q$ is strictly weaker than (**Pol**) (Proposition 64(3)). The proof below uses only the consequence (**Pol1**).

E.5. No bounded lift over a Laurent series field. We now combine the support estimate (Proposition 73) and the coefficient lift (Lemma 74) with the finite-field theorem of Appendix D (Theorem 65) to prove Theorem 32, which is stated in §5.3.

Proof of Theorem 32. Suppose P exists, and form Ψ as in §E.4. By Lemma 74, $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0$ with $E_0 = \kappa^\mathbb{N}$ is $2k$ -linear over κ , alternating and hence antisymmetric in the last k slots, satisfies (Pol1), and has $\text{sdim}(\Psi(z)) \leq d$ for all inputs z . Moreover $k! = 0$ in κ , since $k \geq p = \text{char } \kappa$. This contradicts Theorem 65(2) with $L = \kappa$ and $E_0 = \kappa^\mathbb{N}$. The last sentence follows from Proposition 3. $\square$

Remark 75. Lemmas 72 and 74 and Proposition 73 hold for every field κ , and nothing in them uses a normalization of r : the constant M_r makes the argument uniform in $r \in (0, 1)$. Only the final appeal to Theorem 65 needs κ finite. We do not need, and do not discuss, the analogue of Theorem 32 for infinite κ with this particular pair (E_1, B) . For an infinite κ of characteristic p the field $\kappa((X))$ is covered by Theorem 28, with the pair obtained by ascent from its subfield $\mathbb{F}_p((X))$.

E.6. Multipliers on c_0 . We prove Proposition 33 of §5.4, with $\kappa, k, r, K_1, E_1, B, W_B, D_a$ and σ as there. The proof reruns the construction of §E.4 with test vectors restricted to the finitely supported sequences with entries in κ . These lie in E_0 and also in $c_0(\mathbb{N}, K_1)$, so a lift defined only on $c_0(\mathbb{N}, K_1)^k$ can be tested on them.

Proof of Proposition 33. Suppose $p: c_0(\mathbb{N}, K_1)^k \rightarrow \text{Alt}_{K_1}^k(E_1; B)$ is a bounded k -linear lift of σ . Let $V_{\text{fin}} = \kappa^{(\mathbb{N})}$ be the finitely supported sequences with entries in κ ; it lies in both $c_0(\mathbb{N}, K_1)$ and E_0 . Define $\Psi(a; x) = \eta(p(a_1, \dots, a_k)(x_1, \dots, x_k))$ for $a, x \in V_{\text{fin}}^k$. The proof of Lemma 74 applies verbatim. The map Ψ is $2k$ -linear over κ and alternating in the last k slots. Its inputs have supremum norm at most one, so (E.2) gives $\text{sdim } \Psi(a; x) \leq \lfloor kM_r \|p\| \rfloor$. The polarization points $a_\lambda = \sum_i \lambda_i b_i$, with $b_i \in V_{\text{fin}}$, are finitely supported and hence lie in $c_0(\mathbb{N}, K_1)$, so the identity (Pol), and in particular (Pol1), follows from the diagonal identity as before. This contradicts Theorem 65(1), since $k! = 0$ in κ .

For the second assertion, suppose that the map is analytic at a_0 , with expansion $\sum_n p_n$ there. For $h \in c_0(\mathbb{N}, K_1)$ and $t \in K_1$, multilinearity gives, in $\text{Mult}_{K_1}^k(E_1; B)$,

$$W_B \circ (u_0 + D_{a_0} + tD_h, \dots, u_0 + D_{a_0} + tD_h) = \sum_{r=0}^k t^r A_r(h), \quad A_k(h) = \sigma(h).$$

Comparing this with the given expansion on scalar lines, Lemma 48 shows, exactly as in Proposition 3(2), that p_k is a bounded lift of σ . This is impossible. The operator-valued statement follows by evaluation at W_B . $\square$

APPENDIX F. ASCENT TO AN ARBITRARY BASE FIELD

To pass from a Laurent series field to a prescribed field, we work first inside its completion. A suitable Laurent subfield is spherically complete, so Ingleson's theorem supplies a contractive scalar retraction. The ordinary projective tensor norm then gives isometric scalar extensions of the source and target, together with a retraction onto the original Banach target. Operators and alternating forms extend compatibly with these embeddings. As a result, any lift after scalar extension descends to the Laurent subfield, and nonexistence of a lift ascends. The final density argument removes the completeness assumption on the prescribed field.

F.1. The Laurent subfield. We use spherical completeness in the sense of §3.5: every nonempty family of pairwise-meeting closed balls has a common point. For the field case of the following lemma, see Schneider [13, Lemmas 1.3 and 1.6].

Lemma 76 (discrete distances). *Let Y be a complete nonarchimedean space whose distances lie in $r^{\mathbb{Z}} \cup \{0\}$ for some $r \in (0, 1)$. Then Y is spherically complete.*

Proof. Write $\mathcal{D}(a, \rho) = \{y \in Y : d(y, a) \leq \rho\}$. In a nonarchimedean space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^b), \quad \rho^b := \max\{s \in r^{\mathbb{Z}} \cup \{0\} : s \leq \rho\}.$$

We may therefore replace every radius by a value in $r^{\mathbb{Z}} \cup \{0\}$.

Let $(\mathcal{D}(a_i, \rho_i))_{i \in I}$ be a nonempty family of pairwise meeting closed balls. If its radii have a least element ρ_{i_0} , then $\mathcal{D}(a_{i_0}, \rho_{i_0})$ lies in every ball of the family, so a_{i_0} is a common point. Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose $i_1, i_2, \dots$ so that ρ_{i_n} decreases strictly to zero. The corresponding balls are nested, and

$$d(a_{i_{n'}}, a_{i_n}) \leq \rho_{i_n} \quad (n' \geq n).$$

Their centers form a Cauchy sequence. By completeness it has a limit a , and closedness puts a in every ball of the sequence. For any $i \in I$, choose n with $\rho_{i_n} \leq \rho_i$. The n -th ball then lies in $\mathcal{D}(a_i, \rho_i)$, so a belongs to every ball of the original family. $\square$

Remark 77 (the hypothesis (K2) of [7]). The construction of [7] assumes that the base field satisfies the hypothesis (K2) there: there are closed balls $\mathcal{D}(c_n, r_n)$ with $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \dots$ and $\bigcap_n \mathcal{D}(c_n, r_n) = \emptyset$. A nonarchimedean field (indeed any nonarchimedean space) satisfies (K2) if and only if it is not spherically complete. Indeed, the balls in (K2) pairwise meet, since $|c_n - c_m| \leq \max_{m \leq j < n} |c_{j+1} - c_j| \leq r_m$ for $m \leq n$. Conversely, let a family of pairwise meeting closed balls have no common point. A ball of least radius in the family would lie in all the others (as in the proof of Lemma 76), and its center would be a common point; so the radii have no least element, and every radius in the family exceeds their infimum. A sequence of balls from the family whose radii decrease strictly to that infimum is then nested, so it satisfies $|c_{n+1} - c_n| \leq r_n$, and it has empty intersection: a common point of the sequence would lie in every ball of the family, because every ball of the family contains a ball of the sequence. This remark is used only in §1, §7 and Appendix I.

Lemma 78 (Laurent subfield). *Let K' be a complete nontrivially normed field of characteristic p , and let $t \in K'$ with $0 < |t| = r < 1$.*

- (1) K' is nonarchimedean, and $|c| = 1$ for every $c \in \mathbb{F}_p^\times$.
- (2) For $a_m, \dots, a_M \in \mathbb{F}_p$, not all zero, $|\sum_{i=m}^M a_i t^i| = r^{\min\{i: a_i \neq 0\}}$.
- (3) Give $\mathbb{F}_p((X))$ the absolute value with $|X| = r$. There is a continuous isometric ring homomorphism $\text{ev}_t: \mathbb{F}_p((X)) \rightarrow K'$ such that $\text{ev}_t(\sum_{i \geq m} a_i X^i) = \sum_{i \geq m} a_i t^i$. Its image K_1 is a closed subfield of K' ; it is the closure of $\mathbb{F}_p(t)$, and it is isometrically isomorphic to $\mathbb{F}_p((X))$.
- (4) K_1 is spherically complete.

Proof. (1) Lemma 2 makes K' nonarchimedean. For $c \in \mathbb{F}_p^\times$, the identity $c^{p-1} = 1$ gives $|c| = 1$.

(2) The nonzero terms $a_i t^i$ have pairwise distinct absolute values r^i . In a nonarchimedean field, $|x + y| = |x|$ whenever $|y| < |x|$. Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely $r^{\min\{i: a_i \neq 0\}}$.

(3) Evaluation on Laurent polynomials defines a ring homomorphism $\text{ev}_t: \mathbb{F}_p[X, X^{-1}] \rightarrow K'$, and (2) shows that $|\text{ev}_t(g)| = r^{\text{ord } g} = |g|$. Laurent polynomials are dense in $\mathbb{F}_p((X))$, since truncations converge. Completeness of K' therefore gives a unique continuous extension to $\mathbb{F}_p((X))$. Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of $\sum_{i \geq m} a_i X^i$ map to the partial sums of $\sum_{i \geq m} a_i t^i$.

The image is complete as the isometric image of a complete space, hence closed in K' . It is a subfield: for $g \neq 0$, the identity $\text{ev}_t(g)\text{ev}_t(g^{-1}) = 1$ provides the inverse. This subfield contains $\mathbb{F}_p(t)$, while the Laurent polynomials $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$ are dense in it. The image is consequently exactly the closure of $\mathbb{F}_p(t)$.

(4) By (3), K_1 is complete and has value group $|K_1^\times| = r^\mathbb{Z}$. It is nonarchimedean by (1), so its distances lie in $r^\mathbb{Z} \cup \{0\}$. Lemma 76 gives spherical completeness. $\square$

F.2. Ingleton's theorem and its consequences. Throughout the rest of this section we fix fields $K_1 \subseteq K'$ with the following properties.

- (H1) K_1 , with the restricted absolute value, is nontrivially normed, complete and spherically complete.
- (H2) K' is nonarchimedean.

By multiplication, K' is a nonarchimedean normed K_1 -space. In our application $K' = \widehat{K}$ and K_1 is the field of Lemma 78.

Theorem 79 (Ingleton [4]; see also [11]). *Let K_1 be a spherically complete nonarchimedean normed field, V a K_1 -vector space with a nonarchimedean norm, $V_0 \subseteq V$ a subspace, $C \geq 0$, and $T_0: V_0 \rightarrow K_1$ linear with $|T_0 v| \leq C\|v\|$ for $v \in V_0$. Then T_0 extends to a linear map $T: V \rightarrow K_1$ with $|Tx| \leq C\|x\|$ for all $x \in V$.*

Schneider [13, Prop. 9.2] gives the more general formulation with a dominating nonarchimedean seminorm.

We will use Theorem 79 on the nonarchimedean K_1 -space K' in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain K_1 . The spaces E_1 and F_1 introduced later need not be nonarchimedean.

Corollary 80 (Ingleton projection). *There is a K_1 -linear map $\varpi: K' \rightarrow K_1$ with $\varpi|_{K_1} = \text{id}$ and $|\varpi(\lambda)| \leq |\lambda|$ for all $\lambda \in K'$.*

Proof. Theorem 79 with $V = K'$, $V_0 = K_1$, $T_0 = \text{id}$ and $C = 1$; its hypotheses hold by (H1) and (H2). $\square$

Lemma 81 (extension). *Let $\mathbb{K} \in \{K_1, K'\}$ and let $C \geq 0$.*

- (1) *Let V_0 be a dense $\mathbb{K}$ -subspace of a normed $\mathbb{K}$ -space V , let Y be a $\mathbb{K}$ -Banach space and $T_0: V_0 \rightarrow Y$ a $\mathbb{K}$ -linear map with $\|T_0 x\| \leq C\|x\|$. Then T_0 extends uniquely to a continuous map $T: V \rightarrow Y$, which is $\mathbb{K}$ -linear with $\|Tx\| \leq C\|x\|$.*
- (2) *Let $V_1, \dots, V_k$ be normed K' -spaces with dense subspaces V_i^0 , Y a K' -Banach space, and $M: \prod_i V_i^0 \rightarrow Y$ a K' -multilinear map with $\|M(d)\| \leq C \prod_i \|d_i\|$. Then M extends uniquely to a continuous K' -multilinear map $\bar{M}: \prod_i V_i \rightarrow Y$ with $\|\bar{M}\| \leq C$.*

Proof. (1) The bound makes T_0 Lipschitz, hence uniformly continuous. Density of V_0 and completeness of Y give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case $k = 0$ is immediate. For $k \geq 1$, multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \leq C\rho^{k-1} \sum_i \|d_i - d'_i\|$$

whenever $\|d_i\|, \|d'_i\| \leq \rho$ for every i , with $\rho > 0$. For any $v \in \prod_i V_i$, choose tuples $d^{(n)} \in \prod_i V_i^0$ converging to v . Their coordinates are uniformly bounded, so the estimate makes $M(d^{(n)})$ Cauchy in Y . Define $\bar{M}(v)$ to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of $\prod_i V_i^0$ proves uniqueness. $\square$

F.3. Projective base change. The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary 80 is the essential ingredient; the construction applies to arbitrary normed spaces.

For a normed K_1 -space V let $V_{K'} := V \otimes_{K_1} K'$, a K' -vector space through the second factor, and put

$$\|u\|_\pi := \inf \left\{ \sum_j \|x_j\| |\lambda_j| : u = \sum_j x_j \otimes \lambda_j \right\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded K_1 -linear $\psi: K' \rightarrow K_1$ let $T_\psi: V_{K'} \rightarrow V$ be the K_1 -linear map with $T_\psi(x \otimes \lambda) = \psi(\lambda)x$; it exists because $(x, \lambda) \mapsto \psi(\lambda)x$ is K_1 -bilinear. Let $\iota_V: V \rightarrow V_{K'}$, $x \mapsto x \otimes 1$; it is K_1 -linear, since $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$ for $c \in K_1$.

Lemma 82 (projective base change). *Let V be a normed K_1 -space; no nonarchimedean hypothesis is made on V .*

- (1) $\|T_\psi u\| \leq \|\psi\| \|u\|_\pi$ for every bounded K_1 -linear $\psi: K' \rightarrow K_1$.
- (2) $\|\cdot\|_\pi$ is a K' -norm on $V_{K'}$.
- (3) ι_V is isometric.
- (4) Let $V \widehat{\otimes}_\pi K'$ be the completion of $(V_{K'}, \|\cdot\|_\pi)$. If V is complete, then T_ϖ , with ϖ as in Corollary 80, extends to a K_1 -linear map $\Pi_V: V \widehat{\otimes}_\pi K' \rightarrow V$ with $\|\Pi_V\| \leq 1$ and $\Pi_V \circ \iota_V = \text{id}_V$.

Proof. (1) For every decomposition $u = \sum_j x_j \otimes \lambda_j$,

$$\|T_\psi u\| = \left\| \sum_j \psi(\lambda_j) x_j \right\| \leq \sum_j \|\psi\| |\lambda_j| \|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by $\mu \in K'$ gives $\|\mu u\|_\pi \leq |\mu| \|u\|_\pi$; for $\mu \neq 0$, applying the same inequality with μ^{-1} gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let $u \neq 0$. Choose a finite tensor representation and rewrite its second factors in a basis of their K_1 -span. This gives

$$u = \sum_{j=1}^n x_j \otimes \lambda_j$$

with $\lambda_1, \dots, \lambda_n$ linearly independent over K_1 . Some x_{j_0} is nonzero, since otherwise $u = 0$. On the finite-dimensional subspace $U := \text{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$, take the coordinate functional $\psi_0(\lambda_j) = \delta_{jj_0}$. It is bounded by some $C \geq 0$ by Lemma 11, using completeness of K_1 in (H1).

Apply Theorem 79 to the nonarchimedean K_1 -space K' to extend it to $\psi: K' \rightarrow K_1$ with $\|\psi\| \leq C$. Then $T_\psi u = x_{j_0}$, so (1) gives

$$0 < \|x_{j_0}\| \leq C\|u\|_\pi.$$

Thus $\|u\|_\pi > 0$. This argument detects a nonzero vector coefficient by a functional on the *scalar* factor K' ; it requires no separating continuous dual on V .

(3) The elementary tensor representation gives $\|x \otimes 1\|_\pi \leq \|x\|$. For the reverse inequality, use $\varpi(1) = 1$ and $\|\varpi\| \leq 1$ in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \leq \|x \otimes 1\|_\pi.$$

(4) When V is complete, the contraction T_ϖ on the dense tensor subspace extends by Lemma 81(1), with $\mathbb{K} = K_1$, to a K_1 -linear contraction Π_V with values in V . On the embedded copy of V it satisfies $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$. $\square$

We now fix the spaces to which the descent argument will be applied. Let E_1 be a normed K_1 -space and F_1 a K_1 -Banach space, and put

$$E := E_1 \widehat{\otimes}_\pi K', \quad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of $((E_1)_{K'}, \|\cdot\|_\pi)$ and $((F_1)_{K'}, \|\cdot\|_\pi)$. They are K' -Banach spaces; $\iota_E: E_1 \rightarrow E$ and $\iota_F: F_1 \rightarrow F$ are K_1 -linear isometries, and $\Pi_F: F \rightarrow F_1$ is K_1 -linear with $\|\Pi_F\| \leq 1$ and $\Pi_F \circ \iota_F = \text{id}_{F_1}$ (Lemma 82).

Lemma 83 (operators). *For $f \in \mathcal{L}_{K_1}(E_1, E_1)$ the map $f \otimes \text{id}$ on $(E_1)_{K'}$ is K' -linear with $\|(f \otimes \text{id})u\|_\pi \leq \|f\| \|u\|_\pi$. Its extension $f_{K'} \in \mathcal{L}_{K'}(E, E)$ satisfies $\|f_{K'}\| \leq \|f\|$ and $f_{K'} \circ \iota_E = \iota_E \circ f$, and $f \mapsto f_{K'}$ is K_1 -linear.*

Proof. For any decomposition $u = \sum_j x_j \otimes \lambda_j$, the image under $f \otimes \text{id}$ has the decomposition $\sum_j f(x_j) \otimes \lambda_j$, whose cost is at most $\|f\| \sum_j \|x_j\| \|\lambda_j\|$. Taking the infimum proves the norm bound on the algebraic tensor product. Since E is complete, Lemma 81(1), with $\mathbb{K} = K'$, gives the continuous extension $f_{K'}$ with the same bound. Its action on $x \otimes 1$ proves the embedding identity. Additivity and K_1 -homogeneity in f hold on elementary tensors, hence on their span, and then on E by continuity. $\square$

Lemma 84 (forms). *For $m \in \text{Alt}_{K_1}^k(E_1; F_1)$ there is $m_{K'} \in \text{Alt}_{K'}^k(E; F)$ with $\|m_{K'}\| \leq \|m\|$ and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F(m(x_1, \dots, x_k)) \quad (x_i \in E_1),$$

and $m \mapsto m_{K'}$ is K_1 -linear.

Proof. We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a K_1 -basis $(\lambda_\beta)_{\beta \in \mathfrak{B}}$ of K' . Every $u \in (E_1)_{K'}$ has a unique finite expansion $u = \sum_\beta u^\beta \otimes \lambda_\beta$, with $u^\beta \in E_1$, and each coordinate map $u \mapsto u^\beta$ is K_1 -linear. Define

$$m^{\text{alg}}(u_1, \dots, u_k) := \sum_{\beta_1, \dots, \beta_k} m(u_1^{\beta_1}, \dots, u_k^{\beta_k}) \otimes \lambda_{\beta_1} \cdots \lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

- (i) Linearity of the coordinate maps and multilinearity of m make m^{alg} K_1 -multilinear.
- (ii) On elementary tensors, the definition gives

$$m^{\text{alg}}(x_1 \otimes \mu_1, \dots, x_k \otimes \mu_k) = m(x_1, \dots, x_k) \otimes \mu_1 \cdots \mu_k \quad (x_i \in E_1, \mu_i \in K').$$

Indeed, writing $\mu_i = \sum_{\beta} c_{i\beta} \lambda_{\beta}$ with $c_{i\beta} \in K_1$ gives $x_i \otimes \mu_i = \sum_{\beta} (c_{i\beta} x_i) \otimes \lambda_{\beta}$. Substitution in the defining sum yields

$$\sum_{\beta_1, \dots, \beta_k} \left(\prod_i c_{i\beta_i} \right) m(x) \otimes \prod_i \lambda_{\beta_i} = m(x) \otimes \prod_i \left(\sum_{\beta} c_{i\beta} \lambda_{\beta} \right),$$

which is the asserted formula.

(iii) The map m^{alg} is K' -multilinear. By additivity in each slot, it suffices to check K' -homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

(iv) For arbitrary decompositions $u_i = \sum_j x_{ij} \otimes \mu_{ij}$, multilinearity and (ii) give

$$m^{\text{alg}}(u) = \sum_{j_1, \dots, j_k} m(x_{1j_1}, \dots, x_{kj_k}) \otimes \prod_i \mu_{ij_i}.$$

The cost of this tensor decomposition is at most $\|m\| \prod_i (\sum_j \|x_{ij}\| |\mu_{ij}|)$. Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\text{alg}}(u_1, \dots, u_k)\|_{\pi} \leq \|m\| \prod_i \|u_i\|_{\pi}.$$

(v) To prove strong alternation, suppose two distinct slots a, b both contain $u = \sum_j x_j \otimes \mu_j$. Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index j in slots a and b vanish because m has a repeated argument. For $j \neq j'$, the term using indices (j, j') in these slots pairs with the term using (j', j) . Their scalar factors agree, while the values of m are negatives of one another, so the pair sums to zero. Thus m^{alg} vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking $m^{\text{alg}}(u, \dots, u) = 0$ would not establish strong alternation for $k \geq 3$.

(vi) Regard m^{alg} as taking values in the completion F . By (iii), (iv) and Lemma 81(2), it extends to a continuous K' -multilinear map $m_{K'}: E^k \rightarrow F$ with $\|m_{K'}\| \leq \|m\|$. Strong alternation also passes to the limit: if $v_a = v_b$, approximate this common vector by one sequence in $(E_1)_{K'}$, and approximate the other entries separately. Every approximating tuple has equal entries in slots a, b , so (v) and continuity give $m_{K'}(v) = 0$.

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = m^{\text{alg}}(x_1 \otimes 1, \dots, x_k \otimes 1) = m(x) \otimes 1 = \iota_F(m(x)).$$

The defining formula is K_1 -linear in m . These linear identities pass to the continuous extensions by uniqueness, proving K_1 -linearity of $m \mapsto m_{K'}$. $\square$

Proposition 85 (descent of a lift). *Assume (H1) and (H2). Let E_1 be a normed K_1 -space, F_1 a K_1 -Banach space, and E, F as above. If $Q_{E,E,F}^{K'}$ admits a bounded k -linear lift P over K' , then $Q_{E_1,E_1,F_1}^{K_1}$ admits a bounded k -linear lift P_1 over K_1 with $\|P_1\| \leq \|P\|$. Equivalently, by Proposition 3 over K_1 and over K' : if $Q_{E_1,E_1,F_1}^{K_1}$ is analytic at no point, then $Q_{E,E,F}^{K'}$ is analytic at no point either.*

Proof. Extend the operator and form inputs to K' , apply the given lift, and retract the value to F_1 . Explicitly, for $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$, $m \in \text{Alt}_{K_1}^k(E_1; F_1)$ and $x \in E_1^k$, set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F \left(P((f_1)_{K'}, \dots, (f_k)_{K'})(m_{K'})(\iota_E x_1, \dots, \iota_E x_k) \right).$$

We check the norm bound and then the diagonal identity.

For fixed f and m , the map $P((f_i)_{K'})(m_{K'})$ is a continuous alternating K' -multilinear form on E . Precomposing each slot with the K_1 -linear isometry ι_E and postcomposing with Π_F gives a K_1 -multilinear map on E_1 , still vanishing whenever two arguments coincide. Lemmas 82–84 give

$$\|P_1(f)(m)\| \leq \|\Pi_F\| \|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \leq \|P\| \prod_i \|f_i\| \|m\|.$$

In particular, $P_1(f)(m) \in \text{Alt}_{K_1}^k(E_1; F_1)$. The form extension is K_1 -linear in m , the operator extension is K_1 -linear in each f_i , and P is K' -multilinear with values in K' -linear operators. Consequently P_1 is K_1 -linear in m and K_1 -multilinear in $(f_1, \dots, f_k)$. The displayed estimate proves boundedness at both levels and yields $\|P_1\| \leq \|P\|$.

On the diagonal, use $P(g, \dots, g) = Q^{K'}(g)$ with $g = f_{K'}$, followed by the embedding identities in Lemmas 83 and 84 and the retraction identity $\Pi_F \iota_F = \text{id}$:

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F(m_{K'}(f_{K' \iota_E x_1}, \dots, f_{K' \iota_E x_k})) \\ &= \Pi_F(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)) \\ &= \Pi_F(\iota_F(m(f x_1, \dots, f x_k))) \\ &= m(f x_1, \dots, f x_k) = Q^{K_1}(f)(m)(x). \end{aligned} \quad \square$$

Remark 86. Proposition 85 is characteristic-free and uses no hypothesis on k . It transports negative results upward only, from K_1 to K' . Ingleton's theorem is used only on K' , with no nonarchimedean norm hypothesis on F_1 . The completeness of F_1 is used essentially: it is what makes the retraction Π_F land in F_1 .

F.4. Incomplete base fields. The preceding construction takes place over the completion $\widehat{K}$ of the prescribed field. To return to K , we restrict scalars on the resulting Banach spaces. Density of K in $\widehat{K}$ forces every continuous K -multilinear map between these spaces to be $\widehat{K}$ -multilinear, so a lift cannot appear under this restriction.

Lemma 87. *Let K be a nontrivially normed field with completion $\widehat{K}$, and let E, E', F be $\widehat{K}$ -Banach spaces, regarded as K -Banach spaces by restriction of scalars (with the same norms). Then $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$, $\text{Alt}_K^k(E; F) = \text{Alt}_{\widehat{K}}^k(E; F)$ and $\text{Alt}_K^k(E'; F) = \text{Alt}_{\widehat{K}}^k(E'; F)$ with the same norms, $Q_{E, E', F}^K = Q_{E, E', F}^{\widehat{K}}$, and every bounded k -linear lift of $Q_{E, E', F}^K$ over K is a bounded k -linear lift of $Q_{E, E', F}^{\widehat{K}}$ over $\widehat{K}$.*

Proof. Let g be a continuous K -linear map between normed spaces that already carry compatible $\widehat{K}$ -scalar actions. For $\lambda \in \widehat{K}$, choose $\lambda_n \in K$ with $\lambda_n \rightarrow \lambda$. Since $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda| \|x\|$, continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus g is $\widehat{K}$ -linear. Applying the same argument with all but one slot fixed shows that every continuous K -multilinear map is $\widehat{K}$ -multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are $\widehat{K}$ -Banach spaces, and $Q_{E, E', F}^K$ and $Q_{E, E', F}^{\widehat{K}}$ agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded k -linear lift P between these map spaces. It becomes $\widehat{K}$ -multilinear with the same norm and the same diagonal identity, and hence is a lift over $\widehat{K}$. $\square$

APPENDIX G. PROOF OF THE COUNTEREXAMPLE THEOREM

Appendices D–F give the obstruction to a bounded lift over a Laurent series field and its transfer to a prescribed nontrivially normed field of characteristic p . We first establish the remaining property of the target, that it admits no equivalent nonarchimedean norm (Proposition 88), and then assemble the proof of Theorem 28. Corollary 29 was already deduced from Theorem 28 in §5.1.

G.1. The target has no equivalent nonarchimedean norm.

Proposition 88. *Let κ be any field, $k \geq 2$, $r \in (0, 1)$, and let $K_1 = \kappa((X))$, E_1 , E_0 and B be as in Appendix E. Let $e_i \in E_0$ be the unit vectors and $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in B$ ($j \geq 0$), wedges on pairwise disjoint blocks of indices. Then $\|\omega_j\|_B = 1$ and*

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq N \quad (N \geq 1).$$

Hence the norm of B is not nonarchimedean, and B admits no equivalent nonarchimedean norm. The same holds for $F = B \widehat{\otimes}_\pi K'$ as in Appendix F, for any $K' \supseteq K_1$ satisfying (H1) and (H2), regarded as a normed space over K' or over any subfield of K' .

Proof. Each ω_j is a wedge of unit vectors, so $\|\omega_j\|_B \leq 1$ by the defining projective norm. Evaluation of its determinant array at the block indices gives $\det I = 1$, hence $\|\omega_j\|_B \geq \|J\omega_j\|_\infty \geq 1$. The triangle inequality now gives the upper bound N .

For the lower bound, write ξ_N for the sum of the same block wedges in $\Lambda_\kappa^k E_0$, and put $\omega := \sum_{j < N} \omega_j \in B$. Its array $J\omega$ is the constant Laurent array $\Omega^\kappa(\xi_N)$. Thus $\text{coeff}_0(J\omega) = \Omega^\kappa(\xi_N)$, and uniqueness in Proposition 73 gives $\eta(\omega) = \xi_N$.

We apply Lemma 60 to ξ_N . Fix one of the first N blocks and an index i_0 in it. Take for φ the $k-1$ coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here $k \geq 2$ ensures that the minors have positive size. On the chosen block, the coefficient of e_{i_0} is ± 1 , while every other cofactor contains the zero column obtained by evaluating the functionals at e_{i_0} . Consequently $c_\varphi(\xi_N) = \pm e_{i_0}$. These contractions span all kN unit vectors in the first N blocks, so

$$kN \leq \text{sdim}(\xi_N) = \text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_B$$

by (E.2). This proves the lower bound.

If $\|\cdot\|_u$ were an equivalent nonarchimedean norm, choose positive constants c, c' with $\|\cdot\|_B \leq c\|\cdot\|_u \leq c'\|\cdot\|_B$. Then

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq c \max_{j < N} \|\omega_j\|_u \leq c'$$

for every $N \geq 1$, a contradiction. In particular, the original norm is not nonarchimedean. This argument uses only the nonarchimedean inequality for $\|\cdot\|_u$, not its scalar homogeneity.

Finally, $\iota_F: B \rightarrow F$ is a K_1 -linear isometry by Lemma 82(3). The images $\iota_F \omega_j$ satisfy the same unit-norm and sum-growth estimates, so the argument applies to F as well. Restriction to a subfield leaves these estimates unchanged. $\square$

If $r \leq \frac{1}{2}$, then $M_r = 1$ and $\|\sum_{j < N} \omega_j\|_B = N$ exactly.

By Proposition 47 in §6.3, over a discretely valued base the target of every Banach counterexample must admit no equivalent nonarchimedean norm; Proposition 88 shows that this necessary condition holds for our target.

Remark 89 (necessary, but not sufficient). Completeness of the target is essential in Proposition 47: it is used through Lemma 76. Even for complete targets, the absence of an equivalent nonarchimedean norm is not sufficient for a counterexample. Take the degree-3 space B of §E.2 over $K_1 = \mathbb{F}_5((X))$. It admits no equivalent nonarchimedean norm by Proposition 88, which requires only $k \geq 2$, but $Q_{E_1, E_1, B}^{K_1}$ is analytic by Corollary 29(1), since $3! = 1$ in $\mathbb{F}_5$.

The factorial hypothesis is also essential to the finite-field obstruction. Applying the construction of §E.4 to this bounded lift gives a map Ψ with all the properties of Lemma 74, which is valid in every degree. Thus Ψ satisfies every hypothesis of Theorem 65(2), with $L = \mathbb{F}_5$, except $k! = 0$; compare Remark 71.

G.2. Proof of Theorem 28.

Proof. Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \geq p$.

- (1) *Choose a Laurent subfield.* By Lemma 2, K is nonarchimedean. Its completion $\widehat{K}$ is a complete, nontrivially normed, nonarchimedean field of characteristic p . Choose $t \in K$ with $0 < |t| = r < 1$, and let K_1 be the closure of $\mathbb{F}_p(t)$ in $\widehat{K}$. Lemma 78 identifies K_1 isometrically with $\mathbb{F}_p((X))$, with $|X| = r$, by ev_t . In particular, K_1 is complete and spherically complete.
- (2) *Obtain the obstruction over K_1 .* Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda_{K_1}^k E_1$ under the projective exterior norm. Transport of scalars along ev_t preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem 32, with $\kappa = \mathbb{F}_p$, therefore shows that $Q_{E_1, E_1, B}^{K_1}$ has no bounded k -linear lift.
- (3) *Ascend to $\widehat{K}$.* The inclusion $K_1 \subseteq \widehat{K}$ satisfies (H1) and (H2), and B is complete. Proposition 85, with $K' = \widehat{K}$, gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \quad F := B \widehat{\otimes}_\pi \widehat{K}$$

for which $Q_{E, E, F}^{\widehat{K}}$ has no bounded k -linear lift.

- (4) *Descend to K .* Regard the same E and F as K -Banach spaces. By Lemma 87, a bounded lift of $Q_{E, E, F}^K$ would extend to a bounded lift of $Q_{E, E, F}^{\widehat{K}}$, which is impossible. Proposition 3 applies over the possibly incomplete field K : $Q_{E, E, F}^K$ is analytic at no point, and $\text{ContDiffAt } K \omega$ fails at every point. Reindexing as in §2 gives the same conclusions, with the same spaces E, F , for every finite index set with k elements.
- (5) *Exclude an equivalent nonarchimedean target norm.* Since $k \geq p \geq 2$, Proposition 88 applies to B and F , after the scalar identification in step (2). Its conclusion persists when F is regarded over K .

□

Remark 90. If $K_1 = \widehat{K}$, as when $K = \mathbb{F}_p((u))$ and $t = u$, the completed tensor products identify isometrically with E_1 and B : use Lemma 82(3) and $x \otimes \lambda = \lambda x \otimes 1$. Thus E is nonarchimedean in this case. For $K = \mathbb{F}_q((u))$, $q = p^n$, with its u -adic absolute value, one can instead apply Theorem 32 directly with $\kappa = \mathbb{F}_q$ and $X = u$, obtaining $E = \ell^\infty(\mathbb{N}, K)$ and $F = B$. The proof over a general base does not require, or assert, that E is nonarchimedean.

APPENDIX H. CONSTRUCTION FOR THE NO-LARGEST-DOMAIN THEOREM

This appendix proves Theorem 43, following the outline in Section 6. Throughout, p is prime and $K = \mathbb{F}_p(t)$ carries its t -adic absolute value. We treat the degree $k = p$ first and then pass

to an arbitrary degree $k = p + r \geq p$. The objects E, D, G, M_x, E_k and D_k and the coordinate map (H.3) used in Remark 45 are constructed here.

H.1. The rigid source and the split pair. We use the following elementary completion principle. Let K be dense in a complete valued extension L , let (E_i) be a finite family of dense K -subspaces of Banach L -spaces H_i , and let Z be Banach over L . A bounded K -multilinear map $\prod_i E_i \rightarrow Z$ extends uniquely to a bounded L -multilinear map $\prod_i H_i \rightarrow Z$, with the same norm. Indeed, approximate each input by vectors in E_i . A telescoping expansion and the multilinear bound make the resulting values Cauchy, independently of the approximations. The bound persists in the limit, and density of K in L gives L -linearity in each slot. In the applications below the extended codomain is the complete space L or A , while membership of values on the original inputs in the smaller spaces is retained.

Put $L = \mathbb{F}_p((t))$, the completion of K , and consider the normed L -algebra

$$A = L[\epsilon]/(\epsilon^p), \quad e_i = \epsilon^i \quad (0 \leq i < p),$$

with the maximum norm in this basis. Multiplication satisfies $\|xy\| \leq \|x\| \|y\|$. Choose elements $a_0, \dots, a_{p-1} \in L$, together with the finitely many auxiliary scalars τ_w introduced below, algebraically independent over K . Such a choice is possible because K is countable and L is uncountable: the elements of L algebraic over any finitely generated extension of K form a countable set. Set

$$a = \sum_{i=0}^{p-1} a_i e_i, \quad E = K^p + Ka \subseteq A.$$

Every subspace used below carries the norm induced by A or L . In particular, E has K -dimension $p + 1$ and completion A .

Lemma 91. *Every bounded endomorphism of E is multiplication by a scalar in K , and every bounded linear map $E \rightarrow K$ is zero.*

Proof. A bounded K -linear map extends to completions, and the extension is L -linear by density of K in L . If it preserves E , its columns, the images of the e_i , give a matrix $M + ac^T$, with $M \in M_p(K)$ and $c \in K^p$. Its value at a lies in $K^p + Ka$. Comparison of homogeneous degrees in the algebraically independent a_i shows first that $a(c^T a) = 0$, hence $c = 0$, and then that $M = sI$ for some $s \in K$.

Similarly, the extension of a bounded map $E \rightarrow K$ is an L -linear functional on A . Its coefficients belong to K , since its values on the e_i do. Its value on a also lies in K , so algebraic independence makes all its coefficients zero. $\square$

Multiplication by a has lower triangular Toeplitz matrix

$$(M_a)_{ij} = \begin{cases} a_{i-j}, & i \geq j, \\ 0, & i < j. \end{cases}$$

Define

$$D_0 = \text{span}_K\{e_i, \epsilon^i a : 0 \leq i < p\} + Ka^2, \quad \mathcal{W} = \{e_i : 0 \leq i < p\} \cup \{e_i + e_j : 0 \leq i < j < p\},$$

and put

$$D = D_0 + \sum_{w \in \mathcal{W}} K\tau_w w, \quad G = \text{span}_K\{\det(d_1, \dots, d_p) : d_i \in D\} \subseteq L.$$

The determinant is taken in the basis $(e_0, \dots, e_{p-1})$. These are finite-dimensional K -spaces, with completions A and L : D contains K^p , and G contains 1. By multilinearity, G is spanned

by determinants of the displayed finite generating set for D . We may therefore identify G with a finite-dimensional subspace of $K[a_0, \dots, a_{p-1}, (\tau_w)_w]$. Every element has degree at most one in each individual τ_w , since a nonzero determinant cannot repeat the generator $\tau_w w$. Algebraic independence gives

$$(H.1) \quad \tau_w^2 G \cap G = \{0\} \quad (w \in \mathcal{W}).$$

Lemma 92. *Every bounded p -linear map $D^p \rightarrow G$ is alternating. Thus (D, G) is split in degree p .*

Proof. Compose such a map m with $G \hookrightarrow L$ and extend to a bounded L -multilinear map on A^p . Fix two argument positions. Put the same $w \in \mathcal{W}$ in those positions and arbitrary standard basis vectors in the others, and call the value $c \in G$. Replacing both copies of w by $\tau_w w \in D$ shows that $\tau_w^2 c \in G$, so (H.1) gives $c = 0$. Taking $w = e_i$ gives vanishing on repeated basis vectors. Taking $w = e_i + e_j$ then gives the signed interchange identity in the two positions. Expanding two equal arbitrary arguments in the basis proves that the completed map, and hence m , is alternating, including when $p = 2$. The inclusion of alternating maps into multilinear maps is therefore an isometric bijection, whose inverse is the desired retraction. $\square$

H.2. A missing coefficient. The coefficient space of alternating maps on E is

$$C = \{\lambda \in G : a_i \lambda \in G \text{ for every } i\}.$$

Indeed, an alternating map extends to $\lambda \det$ on the p -dimensional completion A . Determinants of tuples from E span $\text{span}_K\{1, a_0, \dots, a_{p-1}\}$: a determinant can contain at most one copy of a , and replacing one basis vector by a realizes each a_i up to sign. Consequently

$$(H.2) \quad \text{Alt}^p(E; G) = C \det.$$

This identification is isometric, since the determinant bound gives one inequality and evaluation on the standard basis gives the reverse inequality.

Algebraically set every τ_w to zero. The image of G is

$$G_0 = \text{span}_K\{\det(d_1, \dots, d_p) : d_i \in D_0\}.$$

Put $C_0 = \{\lambda \in G_0 : a_i \lambda \in G_0 \text{ for every } i\}$. Specialization sends C into C_0 ; no continuity of specialization is asserted. The displayed generators of D_0 are homogeneous polynomial vectors of degrees zero, one, and two. Their determinants are homogeneous, so G_0 is graded, and the defining conditions show that C_0 is graded as well.

Lemma 93. *The homogeneous degree-one part of C_0 is zero.*

Proof. The degree-two part of G_0 is spanned by the 2×2 minors of M_a and the coordinates of a^2 . These correspond respectively to choosing two degree-one columns, or one degree-two column, in a determinant of generators. On homogeneous quadratic polynomials define the K -linear map

$$\Phi(a_i a_j) = \begin{cases} T^{i+j}, & i + j \geq p - 1, \\ 0, & i + j < p - 1. \end{cases}$$

Every 2×2 minor of M_a is killed by Φ . When both product terms are nonzero, their index sums coincide. If only one is nonzero, write the rows as $r < s$ and columns as $i < j$. The surviving term has index sum $r + s - i - j$, and the missing term implies $r < j$. Hence $r + s - i - j \leq s - i - 1 \leq p - 2$, so its image is again zero.

The coefficient of ϵ^r in a^2 is $\sum_{i+j=r} a_i a_j$. Its image is zero for $r < p-1$; for $r = p-1$ there are exactly p summands and its image is $pT^{p-1} = 0$. Thus Φ kills the entire degree-two part of G_0 . If $\ell = \sum_j c_j a_j$ lies in the degree-one part of C_0 , then $a_{p-1}\ell \in G_0$, whereas

$$\Phi(a_{p-1}\ell) = \sum_j c_j T^{p-1+j}.$$

The powers are distinct, so all c_j vanish. $\square$

H.3. The nonanalytic cross-action. Set $X = (E, G)$ and $Y = (D, G)$. Lemma 91 identifies $\text{End}_K(E)$ isometrically with K , and the self-action of X is

$$(s, v) \longmapsto s^p v_*.$$

This is a bounded polynomial, so the singleton full subcategory on X is analytic. The singleton on Y is analytic by Lemmas 92 and 42.

For $x \in E$, multiplication by x in A restricts to a bounded map $M_x: E \rightarrow D$. Indeed, writing $x = b + ha$ with $b \in K^p$ and $h \in K$, its values on the generators e_j, a lie in

$$K^p + \sum_i K \epsilon^i a + K a^2 \subseteq D.$$

Submultiplicativity gives a bounded linear family $x \mapsto M_x$ from E to $\mathcal{L}(E, D)$. Its matrix is lower triangular with diagonal entry x_0 , so $\det M_x = x_0^p$. In the cross-action from Y to X , fix the target morphism to id_G , evaluate at $\det \in \text{Alt}^p(D; G)$, and use (H.2). The resulting map is

$$(H.3) \quad f: E \longrightarrow C, \quad f(x) = x_0^p.$$

Its values lie in C because it was obtained by pulling back an alternating G -valued map.

Suppose that f were analytic at zero. Since $f(sx) = s^p f(x)$, uniqueness of power-series coefficients on scalar lines gives a bounded p -linear map $B: E^p \rightarrow C$ with diagonal f . This is the homogeneous coefficient argument used in Proposition 3; it requires no factorial division or completeness. Compose B with $C \hookrightarrow L$ and extend to an L -multilinear map on A^p . Put

$$b_i = B(e_0, \dots, e_0, e_i) \in C.$$

There are $p-1$ initial copies of e_0 , and $b_0 = f(e_0) = 1$. Evaluation at a in the last position gives $\sum_i a_i b_i \in C$. Specializing all τ_w to zero yields elements $d_i \in C_0$ with

$$d_0 = 1, \quad \sum_i a_i d_i \in C_0.$$

The degree-one part of this last expression is $\sum_i a_i (d_i)_0$, where $(d_i)_0 \in K$ is the constant term. Its a_0 -coefficient equals one. This contradicts Lemma 93. Thus (H.3), and therefore the cross-action from Y to X , is not analytic.

H.4. Higher degrees and the categorical conclusion. For $k = p + r$, let

$$E_k = E \oplus K^r, \quad D_k = D \oplus K^r$$

with maximum norms. Their completions have dimension k over L . Determinants of tuples from E_k still span $\text{span}_K\{1, a_0, \dots, a_{p-1}\}$, so extraction of the top-degree coefficient identifies

$$(H.4) \quad \text{Alt}^k(E_k; G) = C \det_k$$

isometrically. In particular, the coefficient takes values in C , a fact needed to retain the preceding obstruction after increasing the degree.

By Lemma 91, the extension of every bounded endomorphism of E_k to its L -completion $A \oplus L^r$ has block form

$$\begin{pmatrix} sI_p & B \\ 0 & T \end{pmatrix}, \quad s \in K, \quad T \in M_r(K).$$

The scalar s and entries of T are bounded linear functions of the endomorphism, as is seen by evaluation on the standard basis. Its determinant is $s^p \det T$, so its action on (H.4) is multiplication by this bounded polynomial. Including postcomposition shows that the singleton full subcategory on $X_k = (E_k, G)$ is analytic.

The pair $Y_k = (D_k, G)$ is split. Given $m \in \text{Mult}^k(D_k; G)$, restrict the first p arguments to D and fix the last r arguments at the standard auxiliary basis. The resulting p -linear map is alternating by Lemma 92. Pull it back along $D_k \rightarrow D$ and take its exterior product with the standard volume form on K^r . The shuffle formula defines this exterior product without factorial division and gives a bounded linear map from $\text{Mult}^k(D_k; G)$ to $\text{Alt}^k(D_k; G)$. It fixes every alternating map: after extension to the k -dimensional completion, both maps are determined by their value on the standard basis, where the shuffle formula has exactly one nonzero term. This is the required retraction. For $r > 0$, we do not need or assert that every k -linear map on D_k is alternating.

Finally, restrict the cross-action along the bounded affine family

$$E \longrightarrow \mathcal{L}(E_k, D_k), \quad x \longmapsto M_x \oplus I_r.$$

Evaluation at $\det_k$ and coefficient extraction through (H.4) give exactly the nonanalytic map (H.3). Thus the cross-action from Y_k to X_k is not analytic.

A largest analytic full subcategory would have to contain both individually analytic singleton subcategories, and fullness would then require this cross-action to be analytic. This is impossible. If closure under isomorphisms is required, replace each singleton by its full isomorphism closure. Every action between two copies is transported from the self-action by bounded linear changes of coordinates, so both closures remain analytic. The same contradiction proves Theorem 43.

APPENDIX I. FURTHER COMPLEMENTS ON NONARCHIMEDEAN TARGETS

This appendix first proves Proposition 47 on targets over discretely valued fields, with the example of Remark 94. It then collects complements that the main results do not use: an explicit lift from an orthogonal basis (Remark 95), a comparison with the characteristic-2 construction of [7], and two proposed approaches that are not proved here, namely a counterexample with a nonarchimedean target over complete fields that are not spherically complete (Remark 96) and an alternative route, via an infinite-field analogue of Theorem 65, to the obstruction used in Appendix E (Remark 97). The one exception is Serre's classification of nonarchimedean Banach spaces over complete discretely valued fields, recorded at the end of Remark 95, which the proof of Corollary 35 uses.

Targets over discretely valued bases. This subsection proves Proposition 47 and records an example showing that spherical completeness of the given target norm is not necessary for analyticity.

Proof of Proposition 47. Round the norm. Let $\|\cdot\|_u$ be an equivalent nonarchimedean norm on F . For $y \neq 0$, round $\|y\|_u$ up to the least element $\|y\|_u^\sharp$ of $r^\mathbb{Z}$ above it, and put $\|0\|_u^\sharp := 0$. Then

$$\|y\|_u \leq \|y\|_u^\sharp \leq r^{-1} \|y\|_u.$$

Rounding is monotone and commutes with multiplication by elements of $r^{\mathbb{Z}}$. Since $|\lambda| \in r^{\mathbb{Z}}$ for $\lambda \in K^\times$, it preserves scalar homogeneity and the nonarchimedean inequality. Thus $\|\cdot\|_u^\sharp$ is an equivalent nonarchimedean norm, and $F' := (F, \|\cdot\|_u^\sharp)$ is complete with distances in $r^{\mathbb{Z}} \cup \{0\}$.

Apply the Main Theorem. Lemma 76 makes F' spherically complete. Part (1) in characteristic zero, and part (2) in positive characteristic, therefore give analyticity of $Q' := Q_{E,E',F'}^K$ at every point.

Return to the original norm. Equivalent target norms give the same spaces of bounded alternating maps, with equivalent operator norms. Consequently

$$\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \quad \text{and} \quad \mathcal{L}(\text{Alt}^k(E'; F'), \text{Alt}^k(E; F'))$$

are the same vector space with equivalent norms, and $Q = Q'$ under this identification. A power series for Q' is therefore a power series for Q : coefficient norms change by at most a fixed factor, and convergence is unchanged. $\square$

Remark 94 (a nonspherical target with analytic precomposition). Spherical completeness of the given target norm is not necessary for analyticity, even in degrees at least the characteristic. Let $K = \mathbb{F}_p((t))$ with $|t| = 1/2$, and let $F = c_0(\mathbb{N}, K)$, the space of sequences tending to zero. Put

$$w_j = 1 + \frac{1}{j+2}, \quad \|x\|_w = \sup_{j \geq 0} w_j |x_j|.$$

This is a nonarchimedean norm, and $\|x\|_\infty \leq \|x\|_w \leq \frac{3}{2}\|x\|_\infty$. Thus F is Banach in either norm. The usual supremum norm has nonzero values in $2^{\mathbb{Z}}$, since every nonzero null sequence attains its maximum absolute value. Lemma 76 makes it spherically complete. Main Theorem (2) therefore gives analytic precomposition in every finite degree with the supremum norm; the change of norm argument in Proposition 47 gives the same conclusion for $\|\cdot\|_w$, for all normed E, E' .

Nevertheless, the weighted norm is not spherically complete. Write e_j for the coordinate vectors and $a_n = \sum_{j < n} e_j$. The closed balls $B_n = \overline{B}_w(a_n, w_n)$ are nested: $w_{n+1} < w_n$ and $\|a_{n+1} - a_n\|_w = w_n$. If x belonged to every B_n , then, for each j and any $n > j$,

$$|x_j - 1| \leq \frac{w_n}{w_j} < 1,$$

so $|x_j| = 1$ for every j , contrary to $x_j \rightarrow 0$. Hence $\bigcap_n B_n = \emptyset$. This example separates spherical completeness of a particular norm from analyticity, which is invariant under equivalent norms. It still admits an equivalent nonarchimedean, spherically complete norm.

An explicit lift from an orthogonal basis.

Remark 95 (an explicit lift). Let K be nonarchimedean, let E have an orthogonal basis $(e_i)_{i \in I}$, and let F be complete and nonarchimedean. Here a basis means that every $x \in E$ has an unconditional expansion $x = \sum_i e_i^*(x) e_i$, with $\|\sum_i c_i e_i\| = \sup_i |c_i| \|e_i\|$. Fix a total order on I and put

$$P(f_1, \dots, f_k)(m)(x^1, \dots, x^k) := \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

For each x , the weighted coordinates $|e_i^*(x)| \|e_i\|$ tend to zero off finite subsets of I . Expanding the determinant bounds a summand by a fixed constant times a finite sum of products of these null families, one coordinate from each input. Such products tend to zero off finite subsets of the tuple space. Hence, for every $\varepsilon > 0$, only finitely many summands have norm at least ε , and the family is summable in the complete nonarchimedean space F .

Orthogonality gives $|e_s^*(x)| \|e_s\| \leq \|x\|$. The nonarchimedean determinant estimate therefore bounds every summand by $\|m\| \prod_a \|f_a\| \prod_b \|x^b\|$; the nonarchimedean inequality in F then gives $\|P\| \leq 1$. The formula is multilinear in f and linear in m , and its output is alternating in x because the determinant is alternating in its columns. Finally, finite coordinate expansions give

$$m(fx^1, \dots, fx^k) = \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k});$$

the general identity follows by continuity along finite coordinate projections. Thus $P(f, \dots, f) = Q(f)$. For $k = 0$, the formula has one empty-index term, with determinant 1.

This *sorted lift* works in every characteristic and degree, without division by $k!$ or spherical completeness. It is not used in the main proof. Over a complete discretely valued field, every nonarchimedean Banach space is topologically isomorphic to a space $c_0(I)$ of null families [10, §1, Prop. 1 and its Corollaire] (see also [13, Prop. 10.1]), and hence has such a basis for an equivalent norm.

Nonarchimedean counterexamples.

Comparison with characteristic 2. For $p = k = 2$, Theorem 28 gives a second counterexample over every nontrivially normed field of characteristic 2. Unlike the construction of [7], it applies to spherically complete bases such as $\mathbb{F}_2((t))$. The distinction lies in the target: that of [7] is nonarchimedean, whereas ours admits no equivalent nonarchimedean norm. Proposition 47, using Main Theorem (2), shows that a complete counterexample target over $\mathbb{F}_2((t))$ cannot carry an equivalent nonarchimedean norm.

The proofs also use different mechanisms. The characteristic-2 argument combines a parity identity for determinants—the determinant of an alternating matrix is the square of its Pfaffian—with the failure of Hahn–Banach over a base that is not spherically complete, using ℓ^∞ to assemble the construction. Here the finite-field theorem is combined with the constant-coefficient map of a Laurent series field, and Ingleton’s theorem is applied over a spherically complete subfield. In characteristic 2, Theorem 65 needs no signs (Remark 70).

Further approaches.

Remark 96 (nonarchimedean targets over bases that are not spherically complete; proposed extension). The following statement is a proposed extension; it has not been proved here and is not used elsewhere in this paper. Let $k \geq 2$, and let K be a complete nontrivially normed field, not spherically complete, with $k! = 0$ in K . Then K is nonarchimedean by Lemma 2. For a cardinal μ put $\beth_0(\mu) := \mu$ and $\beth_{n+1}(\mu) := 2^{\beth_n(\mu)}$, and let μ^+ be the successor cardinal of μ . Let I be a set of cardinality at least $\beth_{3k-1}(|K|)^+$, put $E := \ell^\infty(I, K)$, and let F_I be the closure in $\ell^\infty(I^k, K)$ of the span of the determinant arrays $(i_1, \dots, i_k) \mapsto \det(x_b(i_a))_{a,b}$, $x \in E^k$. Then E and F_I are nonarchimedean K -Banach spaces, and Q_{E,E,F_I}^K is analytic at no point. The proposed proof uses a partition argument with the Erdős–Rado theorem in place of Ramsey’s theorem. For $k = 2$ in characteristic 2 the countable index set $\mathbb{N}$ suffices [7].

An alternative algebraic approach.

Remark 97 (another possible approach). The following optional sketch suggests an alternative route to the input needed in Appendix E. It requires an additional infinite-field obstruction, which is not proved here. The contraction-rank identity needed for the base-change step is Proposition 61. No other argument in this paper depends on the proposed route.

The infinite-field input. The first required statement is the analogue of Theorem 65 over infinite fields L with $k! = 0$ in L , for lifts that satisfy (Pw) and have uniformly bounded support,

of an arbitrary linear family $D: A \rightarrow \text{End}_L(V)$ whose image modulo the finite-rank operators is infinite-dimensional.

Base change and minors. Let Ψ be a lift over $\mathbb{F}_q$ whose values have support dimension at most d . After the finite base change $\mathbb{F}_q \subseteq \mathbb{F}_{q^N}$, the values of the base change of Ψ at $\mathbb{F}_{q^N}$ -points have support dimension at most $d_N := N^{2k}d$, since each input has only N pieces over $\mathbb{F}_q$; choose N with $q^N > d_N + 1$. By Proposition 61, support dimension equals the rank of a finite contraction matrix. Now consider any finite set of independent scalar parameters in each of the $2k$ input slots. The resulting polynomial family has finitely many exterior-vector coefficients; the sum of their supports is one finite-dimensional space supporting every value, over every scalar extension. Choose a basis of that space to form the contraction matrix. By multilinearity, each matrix entry has degree at most one in each scalar parameter. Its $(d_N + 1)$ -minors therefore have degree at most $d_N + 1 < q^N$ in each variable. They vanish at every $\mathbb{F}_{q^N}$ -point, so they are zero polynomials and remain zero over the algebraic closure $\overline{\mathbb{F}_q}$. The resulting support bound would allow the proposed obstruction to be applied over this infinite field.

Compatibility requirements. In addition to the proposed infinite-field obstruction, the route requires the following properties of Ψ .

- (1) The full identity (Pol), not only (Pol1): for $k > q$ the identity (Pw) at $\mathbb{F}_q$ -points does not survive base change (Proposition 64(3)), whereas (Pol) gives (Pw) at every point after base change.
- (2) Alternation, as required by the unproved input above. Both alternation and antisymmetry survive scalar extension, but in characteristic 2 antisymmetry does not imply alternation.
- (3) An infinite-dimensional image modulo finite rank. For the multipliers $D_a x = ax$ on $L^\mathbb{N}$ this holds because the multipliers by the indicators of pairwise disjoint infinite sets are linearly independent modulo finite rank.

The lift of Appendix E satisfies (Pol) and is alternating (Lemma 74), and its multiplier family is the one in (3), with $L = \kappa$.

APPENDIX J. FORMALIZATION

The original operator proofs in the private repository [12] were verified at commit 5e9c3cf, with Lean 4.34.0-rc2 and Mathlib 2b73d9821d29. The library had no admitted proofs and used only the axioms `propext`, `Classical.choice` and `Quot.sound`. The repository's `THEOREM_MAP.md` records the current declaration-level scope.

The principal declarations, in the namespace `AlternatingAnalytic`, are

- `exists_banach_counterexample_full`, for Theorem 28, including incomplete base fields, the same Banach witnesses for every finite index type of cardinality k , no bounded lift, nowhere analyticity, nowhere C^ω , and no equivalent nonarchimedean norm on the target;
- `factorial_ne_zero_iff_allBanachPrecompositionAnalytic`, for both directions of Corollary 29;
- `finiteField_multiplier_theorem`, for both parts of Theorem 65, with the support bound required only on the stated finitely supported test inputs in its first part.

The supporting development covers the κ -linear coefficient map and support estimate (Proposition 73), all four properties of the residue lift (Lemma 74), the Laurent counterexample, and ascent and descent to a prescribed field. It also includes the lift criterion, the sharp polarization distinctions, the operator form of Main Theorem (2), Proposition 47, and the sorted lift of Remark 95. Spherical completeness uses the development's own class, with the definition in

§3.5. The formalization used Codex with Autoform and incorporated previously formalized ingredients with their provenance retained.

Proposition 7 reproduces the argument of [8], whose formal proof is included in the pinned Mathlib dependency. Proposition 10 is formalized for continuous finite coordinates on either the domain or codomain of the precomposed operator, in every characteristic and degree. Over a complete field, Corollary 12 follows when either of those two spaces is finite-dimensional; the value space is arbitrary. The finite-dimensional value-target reduction (Remark 46) remains a supplementary informal result. The weighted c_0 example of Remark 94 is likewise an additional informal corollary; its sufficient analyticity criterion is formalized, but the example is not a separate Lean declaration. Parts (1) and (2) now have Lean formalizations for the actual normed-space, Banach-space and spherical-target functors. The theorem map records the collected declarations and their statement-review status.

The extension formalization covers Lemma 18, Theorem 19 and Corollary 21, including point-wise, open-domain and supplied continuous-coordinate versions. The family proof constructs the ambient $(k + 1)$ -linear representative with norm at most one; it retains every characteristic and degree and arbitrary normed fibers, without completeness assumptions. Remark 20 is also formalized: in maximum-norm coordinates, a nonarchimedean ambient norm permits reflection with equal diagonals, no increase in coefficient norms and no loss of the represented ball, including infinite radius.

The c_0 -parameter results, Theorem 23 and Corollary 24, are also formalized for arbitrary index sets over a nonarchimedean field K , without completeness assumptions on K or the parameter space. Reflection through a linear isometry $j: W \hookrightarrow Z$ requires complete W and a nonarchimedean norm on Z ; Z need not be complete. The lifted series has no larger coefficient norms and represents the map on the original ball, including infinite radius; its intrinsic convergence radius is at least that of the original series. The family corollary holds in every degree when the final value-space target F' is complete and nonarchimedean, with the other fibers arbitrary normed spaces, and transports along bounded linear isomorphisms of parameter spaces.

The ordinary sum-norm ℓ^1 results are also formalized: Lemma 51, Theorem 25 and Corollary 26. The homogeneous lift has the stated real norm bound $d! \|B\|$ in every degree. Fixed-degree reflection retains the hypotheses that W is closed in a Banach space Z , while H is merely normed and K may be incomplete. The local word factorization uses ordinary K -valued ℓ^1 and a bounded synthesis map into the completion of H . For alternating morphism families on ℓ^1 or its bounded linear retracts, only the final value-space target F' is required to be complete; no completeness assumption is imposed on K or the other fibers.

The ordinary ℓ^1 quotient construction and Corollary 27, including its following non-retract consequence, are also formalized over a complete nontrivially normed field K . The construction applies to every Banach space H , uses its closed unit ball as the index set and retains the ordinary sum norm; it includes one-coordinate lifting estimates and an explicit ball-image bound. In characteristic $p > 0$ and degree $k \geq p$, the corollary supplies Banach spaces E, F and a bounded open linear surjection $q: \ell^1(J, K) \rightarrow H = \mathcal{L}(E, E)$ of norm at most one. The composite $Q_{E,E,F}^K \circ q$ is analytic everywhere, while $Q_{E,E,F}^K$ is nowhere analytic. Every local section of q on an open subset fails to be Fréchet differentiable at each point of its domain, and this same H is not a bounded linear retract of $\ell^1(I, K)$ for any index set I . The norms on H and F need not be nonarchimedean.

Proposition 52 and Theorem 54 are also formalized over a complete nontrivially normed field K and a Banach parameter space P . The tensor powers are separated completions for the ordinary sum projective seminorm, with the norm-controlled universal property for Banach

targets. The homogeneous lift preserves the diagonal and has norm at most $\|B\|\|R_n\|$. The analytic criterion characterizes reflection into every closed subspace of every Banach target, on every open subset of P , by one family of projections $R_n: T_n(P) \rightarrow \Delta_n(P)$, $n \geq 1$, with a common bound $\|R_n\| \leq A^n$ for some $A \geq 1$. The equivalent finite root-limsup condition is included, and the degree-zero Taylor coefficient is handled separately. The necessity proof uses the actual dependent c_0 Banach test space and its ambient-analytic pure-power family on the whole open unit ball.

The checked development further includes the joint action and its functorial laws, the core admissible-family laws and graph characterization of Proposition 17, Proposition 30 with its Banach counterexample at the identity, and the canonical-support identity of Proposition 61. The degree-zero and degree-one support cases and finite-ambient contraction-matrix formula are included. These proofs have passed Lean verification without admitted proofs. Statement-review results and declaration-level scope are recorded in the repository theorem map.

The open-manifold family theorem and analytic bundle instance in Main Theorem (3) are formalized. The declaration `contMDiffVectorBundle_alternating_of_finiteCoordinates` uses Mathlib's existing alternating-bundle topology, for arbitrary K , degree and normed model fibers, and supplied continuous coordinates $P \simeq K^d$, without completeness assumptions. The declaration `alternatingBundleFunctor` completes Main Theorem (3) as an actual bifunctor on analytic normable bundles. Its object and morphism formulas use the existing bundle topologies and operator-valued analytic Hom sections; identity, composition and transport between analytically compatible presentations are formalized. Corollary 31, which is Main Theorem (4), is formalized as `exists_banach_base_alternatingBundle_not_analytic`, over every nontrivially normed field of characteristic p and every $k \geq p$. The shear bundle is a Mathlib vector bundle core over the Banach base $\mathcal{L}(E, E)$ with two global charts; it and the trivial bundle with fiber F are analytic, but the actual alternating-map bundle, with Mathlib's existing topology, is not analytic. The multiplier strengthening of §5.4, the dichotomy of Corollary 35 and Remark 45 are not formalized. Main Theorem (5) is also formalized over the t -adic field $K = \mathbb{F}_p(t)$, for every prime p and every $k \geq p$. The checked statement rules out a largest full analytic domain, also with isomorphism closure and within the finite-dimensional nonarchimedean ambient category (with isomorphism closure relative to that ambient category). It includes the incompatible analytic objects, the split-destination obstruction and the analytic completed models of Remark 44. The witnesses are incomplete; this result does not settle the corresponding questions over complete fields or among Banach pairs. The general bundle construction of Proposition 5 and its consequences over general bases (Corollaries 8, 15 and 38), the c_0 and ℓ^1 bundle constructions, the remaining tensor results, the remaining analytic-domain results and finite-support matrix transport remain pending. The theorem map records their scope and status.

The earlier characteristic-2 development [7] is cited for comparison; it is not re-audited here.

The proposed alternative route of Remark 97, the proposed large-index construction of Remark 96, the open problems, and Serre's external classification cited in Remark 95 are excluded from the proved results. The repository's comparator challenge contains two short contradiction corollaries of Theorem 28, the operator form of Main Theorem (2) for every regularity order, the finite-coordinate analytic alternating-bundle theorem, a contradiction from the analytic alternating-bundle instance with no restriction on the field, model with corners, base manifold, normed fibers or finite index type, and the nonexistence of a largest full analytic domain over $\mathbb{F}_p(t)$ in degrees $k \geq p$. Their solutions use the corresponding library theorems. The Banach and spherical-completeness hypotheses are given by transparent definitions in Mathlib terms. The largest-domain predicate uses the development's actual categorical definition of joint

analyticity on every full hom space. Intentional placeholders occur only in the challenge file, not in the proof library or solution. Current challenge checks use Kim Morrison’s comparator, with the configuration recorded in the repository.

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